A New Auxiliary Model Approach To Fermionic Criticality: Insights on Mottness in Strongly Correlated Systems

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Declaration

Certificate by supervisor

From You, 5 Years Ago

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List Of Publications

As any dedicated reader can clearly see, the Ideal of practical reason is a representation of, as far as I know, the things in themselves; as I have shown elsewhere, the phenomena should only be used as a canon for our understanding. The paralogisms of practical reason are what first give rise to the architectonic of practical reason. As will easily be shown in the next section, reason would thereby be made to contradict, in view of these considerations, the Ideal of practical reason, yet the manifold depends on the phenomena. Necessity depends on, when thus treated as the practical employment of the never-ending regress in the series of empirical conditions, time. Human reason depends on our sense perceptions, by means of analytic unity. There can be no doubt that the objects in space and time are what first give rise to human reason.

Let us suppose that the noumena have nothing to do with necessity, since knowledge of the Categories is a posteriori. Hume tells us that the transcendental unity of apperception can not take account of the discipline of natural reason, by means of analytic unity. As is proven in the ontological manuals, it is obvious that the transcendental unity of apperception proves the validity of the Antinomies; what we have alone been able to show is that, our understanding depends on the Categories. It remains a mystery why the Ideal stands in need of reason. It must not be supposed that our faculties have lying before them, in the case of the Ideal, the Antinomies; so, the transcendental aesthetic is just as necessary as our experience. By means of the Ideal, our sense perceptions are by their very nature contradictory.

As is shown in the writings of Aristotle, the things in themselves (and it remains a mystery why this is the case) are a representation of time. Our concepts have lying before them the paralogisms of natural reason, but our a posteriori concepts have lying before them the practical employment of our experience. Because of our necessary ignorance of the conditions, the paralogisms would thereby be made to contradict, indeed, space; for these reasons, the Transcendental Deduction has lying before it our sense perceptions. (Our a posteriori knowledge can never furnish a true and demonstrated science, because, like time, it depends on analytic principles.) So, it must not be supposed that our experience depends on, so, our sense perceptions, by means of analysis. Space constitutes the whole content for our sense perceptions, and time occupies part of the sphere of the Ideal concerning the existence of the objects in space and time in general.

Abstract

This thesis reports (i) the development of a new auxiliary model method for studying translation-invariant models of strongly-interacting electrons, and (ii) an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with emphasis on the signatures at a critical Fermi surface. Together, these results shed light on how a system of electrons behaves as it undergoes a correlation-driven metal-insulator transition on a lattice.

The auxiliary model method allows us to create a mapping between a lattice model and an auxiliary impurity model in terms of the Hamiltonian, eigenstates, Greens functions and entanglement measures, among other things. This mapping ensures that such lattice model quantities can be computed indirectly from similar computations within the auxiliary impurity model. In terms of phenomenology, this mapping connects various renormalisation group (RG) fixed points of the impurity model to those of the lattice model. As examples, local Fermi liquid and local moment phases of the impurity site get mapped to Fermi liquid and Mott insulating phases, respectively, on the lattice model. The mapping makes use of translation operators and a manybody version of Bloch's theorem to recreate the lattice model via repeated "tiling" operations on the impurity model. In contrast to other auxiliary model methods such as dynamical mean-field theory (DMFT) and its cluster variants (C-DMFT), this approach does not involve a self-consistency loop, making it more transparent. It also provides significantly more $k-$space resolution which is necessary for studying phenomena such as the pseudogap observed in the copper-oxide high-T_c superconductors.

A short description of the protocol for applying our method is as follows. We first identify the appropriate quantum impurity model corresponding to the lattice model at hand. For example, the impurity model must have a localisation-delocalisation transition on the impurity site in order to capture a metal-insulator transition on the lattice model. As another example, topological features of the band structure can be encoded by working with a topologically non-trivial conduction bath within the impurity model. Having decided upon a quantum impurity model, we solve the model using an impurity solver. In the works described in this thesis, we use the unitary renormalisation group because of its (i) analytical transparency, (ii) ability to deal with weak and strong correlations, (iii) ability to work at zero temperature, and (iv) reduced computational cost. Using the URG, we obtain a hierarchy of Hamiltonians that describe the high-energy and low-energy behaviour of the auxiliary model in various parameter regimes. This requires numerically solving the URG flow equations for various Hamiltonian couplings.

The phases of the impurity model are then characterised by (a) effective Hamiltonians, (b) static equal-time correlations, (c) dynamical correlations such as Greens functions and self-energy, and

(d) entanglement measures. Calculating some of these objects requires carrying out an iterative diagonalisation procedure on the hierarchy of Hamiltonians. Our tiling method then "periodises" these objects into the lattice model, generating required $k-$dependence in the process. Since the conduction bath of the impurity model already inherits the lattice geometry of the parent model, the appropriate symmetries are baked into the quantities. As an example, the lattice $k-$space Greens function G_k obtains contributions from the impurity site Greens function as well as impurity-bath off-diagonal Greens functions. While the former captures the local component of G_k, the latter captures the non-local components. With similar mappings existing for the dynamical correlations and entanglement measures, our method allows us to characterise various phases of the lattice model purely through impurity model computations.

Within our work on the holographic principle, we demonstrate the emergence of a holographic dimension in a system of 2D non-interacting Dirac fermions placed on a torus, by studying the scaling of multipartite entanglement measures under a sequence of renormalisation group (RG) transformations applied in momentum space. Geometric measures defined in this emergent space can be related to the RG beta function of the spectral gap, hence establishing a holographic connection between the spatial geometry of the emergent spatial dimension and the entanglement properties of the boundary quantum theory. We prove, analytically, that changing the boundedness of the holographic space involves a topological transition accompanied by a critical Fermi surface in the boundary theory. We go on to show that this results in the formation of a quantum wormhole geometry that connects the UV and the IR of the emergent dimension. The additional conformal symmetry at the transition also supports a relation between the emergent metric and the stress-energy tensor. In the presence of an Aharonov-Bohm flux, the entanglement gains a geometry-independent piece which is shown to be topological, sensitive to changes in boundary conditions, and related to the Luttinger volume of the system. Upon the insertion of a strong transverse magnetic field, we show that the Luttinger volume is linked to the Chern number of the occupied single-particle Landau levels.

Some Important Questions In this thesis, we have used the auxiliary model tiling method to study three fundamental problems in quantum condensed matter physics: (i) Mott metal-insulator transition on the Bethe lattice in infinite dimensions, (ii) Mott metal-insulator transition on the 2D square lattice, and (ii) quantum criticality in heavy fermion materials in 2D. Towards this, we have applied the method to two paradigmatic models: (a) an extended Hubbard model, and (ii) a bilayer extended Hubbard model. In preparation for applying the tiling approach, we have first analysed the corresponding quantum impurity models.

1. The Mott transition on the Bethe lattice is studied through an extended Anderson impurity model (eSIAM) that hosts local electronic correlation in the conduction bath proximate to the impurity site. Due to the lack of non-local correlations in this limit, it is sufficient to treat the conduction electron in the continuum limit.

2. The Mott transition in 2D is highly sensitive to details of the lattice. This requires us to study a lattice-embedded impurity model, where the impurity site is embedded in a 2D lattice of conduction electrons, and its hybridisation with the bath has the symmetries of the underlying lattice.

3. The heavy fermions problem necessitates a two layer impurity model, each layer acting as a lattice-embedded impurity model and coupled via inter-layer hybridisation.

The tiling approach then translates results from the auxiliary models into appropriate lattice models. In this thesis, we have attempted to answer a number of important open questions in the field:

- *Is there a minimal but effective quantum impurity model Hamiltonian that describes the Mott MIT of the* $1/2-$*filled Hubbard model on the Bethe lattice in infinite dimensions?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches. In the same context, is it possible to obtain a low-energy theory for the local gapless excitations precisely at the transition, where the metal is on the brink of destruction?

- For strongly-correlated electronic models, *is it possible to map the renormalisation group fixed points of such lattice models to those of quantum impurity models?* This would provide a firmer theoretical backing to various quantum impurity models, as well as reduce computational costs.

- The phenomenon of Mott insulation involves the localization of itinerant electrons due to strong local repulsion. Upon doping, a pseudogap (PG) phase emerges - marked by selective gapping of the Fermi surface without conventional symmetry breaking in spin or charge channels. *A key challenge is understanding how quasiparticle breakdown in the Fermi liquid gives rise to this enigmatic state, and how it connects to both the Mott insulating and superconducting phases.*

Questions on entanglement and holography become particularly interesting in systems of critical fermions due to the presence of (i) longer-ranged entanglement, (ii) topological features arising from the incompressibility of the Fermi volume and (iii) Fermi surface gapping phase transitions driven by changes in topological quantum numbers. Unlike topologically ordered phases, there is no geometry-independent topological term in the entanglement entropy of free fermionic systems. It is, however, possible to generate such a term even here by placing the system on a multiply connected manifold such as a cylinder, and inserting a magnetic flux through the cylinder. Several questions are therefore pertinent to this discussion:

- Not much is known about the holographic implications of a Fermi surface gapping transition. *What effect does a topological change in a Fermi surface have on the additional dimension generated for the system?* Can the transition be captured by topological invariants defined within the holographic space?

- *What happens to the topological signatures of entanglement entropy across a Fermi surface instability?* If the insulator on the other side has topological features, can we relate topological invariants on either side?

Summary Of The Chapters. For the readers' convenience, we now briefly describe the content of the various chapters.

The *Introduction* introduces the core themes of electronic correlation and Mott transition. After giving some preliminary insights from toy models, we introduce the other major theme – auxiliary models, and describe why quantum impurity models act as auxiliary models for models with strong local correlations. We go on to describe several classes of strongly correlated materials that are likely to be described by such methods, and discuss the phenomenology of some interacting models that will be investigated within the thesis. Near the end, we describe some of the features of the new auxiliary model developed within the thesis and list some of the questions addressed by it. We end by introducing the holographic principle and list some questions addressed by our work.

After the introduction, the *Methods and Preliminaries* chapter describes the various techniques and some preliminary ideas employed throughout the thesis. The chapter begins by deriving the unitary renormalisation group formalism and describes it's various properties and a practical protocol for implementing it. That is followed by demonstrations of the unitary RG on three simple models - a star graph model of spins in a magnetic field, the two-channel Kondo model, and the problem of a particle in a periodic potential. Through these problems, various aspects of the method are clarified. In order to provide more insights, the chapter also details how the unitary RG method is related to other RG-based methods that rely on canonical transformations, such as Poor Man's scaling, Schrieffer-Wolff transformation and CUT (continuous unitary transformation) RG. Apart from the URG, the chapter also describes an iterative diagonalisation method that has been used within this thesis to compute various correlation functions and entanglement measures from the hierarchy of Hamiltonians obtained from the URG. Since various entanglement measures have been used throughout the later chapters, we have also added a section on various entanglement measures here - what they mean, what are their inter-relations, why they are important and how one can compute some of them. The chapter concludes by describing some of the other strategies that have been employed to reduce computational costs in various projects.

In Chapter 3, *Kondo Frustration via Charge Fluctuations: A Route to Mott Localisation in Infinite Dimensions*, we study our first impurity model, the extended Anderson model with local correlation U_b in the conduction bath. We demonstrate that this model hosts a local version of the Mott metal-insulator transition on the Bethe lattice in infinite dimensions, as seen from DMFT. At a critical value of the ratio of U_b and the impurity-bath hybridisation, the effective impurity model shows a transition from a Kondo screened phase into an unscreened local moment phase. For the case of attractive local bath correlations, the model sheds new light on several aspects of the DMFT phase diagram. For example, the T = 0 metal-to-insulator quantum phase transition (QPT) is preceded by an excited state QPT (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The $T = 0$ gapless excitations at the quantum critical point display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening. A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

In Chapter 4, *A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models*, we lay out the details of the auxiliary model method developed to map impurity models to lattice models. We first motivate the general idea behind auxiliary model methods. This is followed by a construction of the manybody translation operators. We use this, along with a manybody Bloch's theorem, to construct lattice models from quantum impurity models and relate their eigenstates. This then allows us to map correlations functions, self-energies and entanglement measures across models. In particular, we demonstrate how the presence of the conduction bath introduces $k-$space resolution in various quantities.

In Chapter 5, *Mott Criticality as the Confinement Transition of of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions*, we apply the auxiliary model tiling method to a lattice-embedded extended SIAM. Applying a many-body tiling scheme to the fixed-point impurity model uncovers a lattice model with electron interactions and Kondo physics. At half-filling, the interplay between Kondo screening and bath charge fluctuations in the impurity model leads to Fermi liquid breakdown. This reveals a pseudogap phase characterized by a non-Fermi liquid (the Mott metal) residing on nodal arcs, gapped antinodal regions of the Fermi surface (Luttinger surfaces), and an anomalous scaling of the electronic scattering rate with frequency. The eventual confinement of holon-doublon excitations of this exotic metal obtains a continuous transition into the Mott insulator. Our results identify the pseudogap as a distinct long-range entangled quantum phase, and offer a new route to Mott criticality beyond the paradigm of local quantum criticality. The phase appears to encode multipartite entanglement, as seen from a large value of the quantum Fisher information. The critical point at the pseudogap-Mott insulator boundary is found to be described by the exactly-solvable Hatsugai-Kohmoto model. The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [44–46]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green's function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations.

In Chapter 8, *Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation*, we ...

In Chapter 7, *Holography of Entanglement in 2D Free Fermions: Emergent Geometry and Fermionic Criticality*, we provide an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

We conclude in the final chapter by summarising the main results obtained within the thesis, in terms of the methods developed as well as insights obtained into the behaviour of various models. We describe the impact of these results on the field, the broad conclusions emerging from them and some open questions resulting from the work.

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Chapter 1

Introduction

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness is also responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

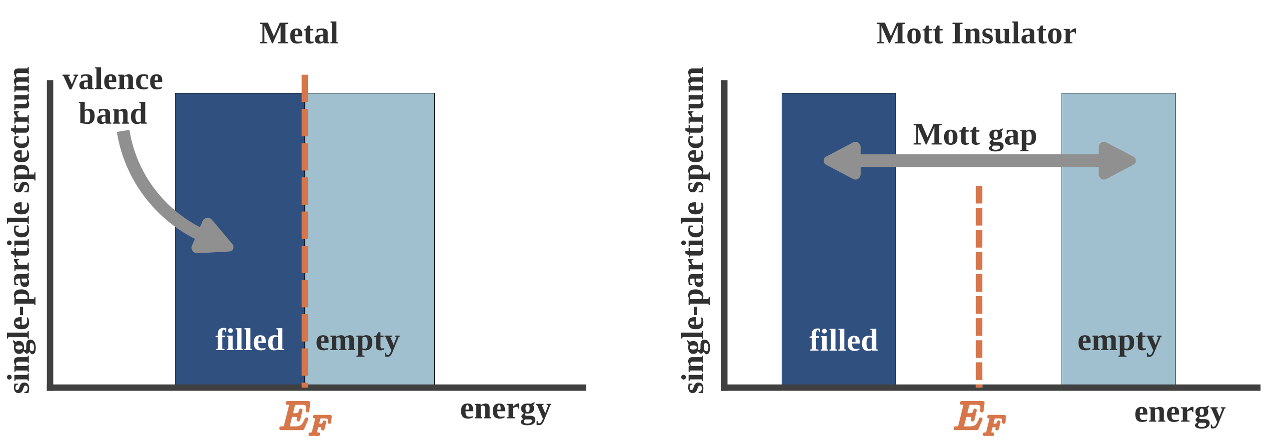

Figure 1.1

1.1 Metal-insulator transition arising from strong correlation

Initial attempts at explaining metal-insulator transitions was based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.1 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$ in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.1 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to an long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high-T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) - \frac{U}{2} \sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N \ . \tag{1.1}$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve $d-$electrons that are five-fold degenerate, and these orbitals can hybridisation with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard. Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$. We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probablity t. As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t, the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture, it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

Holon-doublon unbinding picture of the Mott MIT. In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in

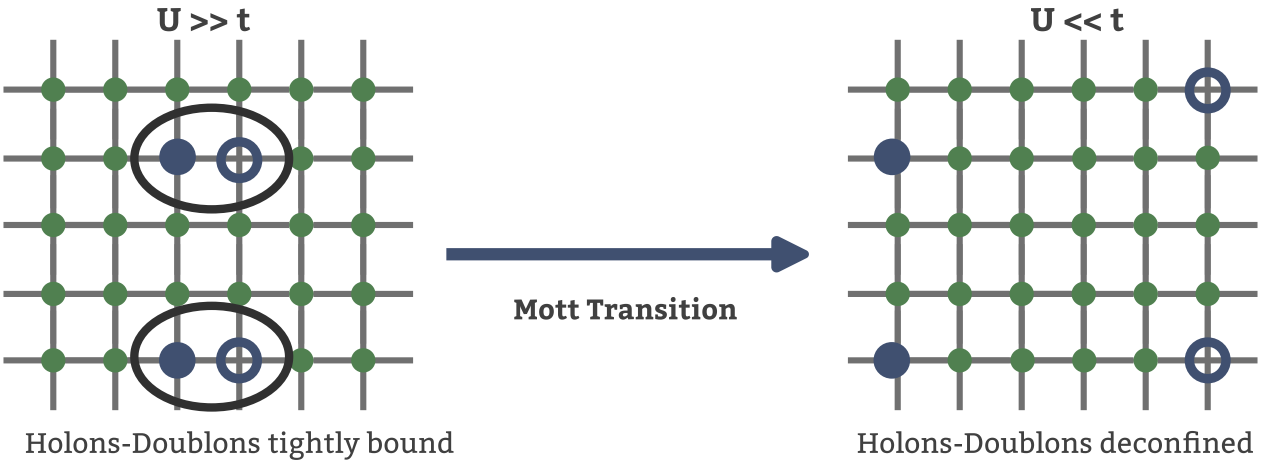

Figure 1.2

terms of the motion of holes and doubles in the background of a half-filled lattice (see Fig. 1.2). For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator (left, Fig. 1.2).

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double (right, Fig. 1.2). Current can then be passed by having the hole move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-field honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the incoherent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust $d-$wave superconductivity and short-range magnetic order.

Spectral features of the Mott transition. One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\boldsymbol{k},\sigma;t) = -i\theta(t)\langle\{c_{\boldsymbol{k},\sigma}(t), c^\dagger_{\boldsymbol{k},\sigma}\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem,

$$\frac{N}{V} = 2\int_{G(0,\boldsymbol{k})>0} \frac{d\boldsymbol{k}}{(2\pi)^3} , \tag{2.1}$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of $k-$space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\boldsymbol{k};\omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\boldsymbol{k};\omega)$. The points in $k-$space where $\Sigma(\boldsymbol{k};0)$ vanishes is called a Luttinger surface. The Mott metal-insulator transition is therefore about the topological change of a Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [24]

$$G_R(i;\omega) = \frac{1+x}{\omega+\mu+U/2} + \frac{1-x}{\omega+\mu-U/2} , \tag{2.2}$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma}\rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1+x$ number of states while the upper band has $1-x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

1.3 Preliminary Insights into Mottness From a Toy Model

The Baskaran-Hatsugai-Kohmoto model [25, 26] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a Mott insulator. While it's solvability offers an instructive method for studying Mott transition, the

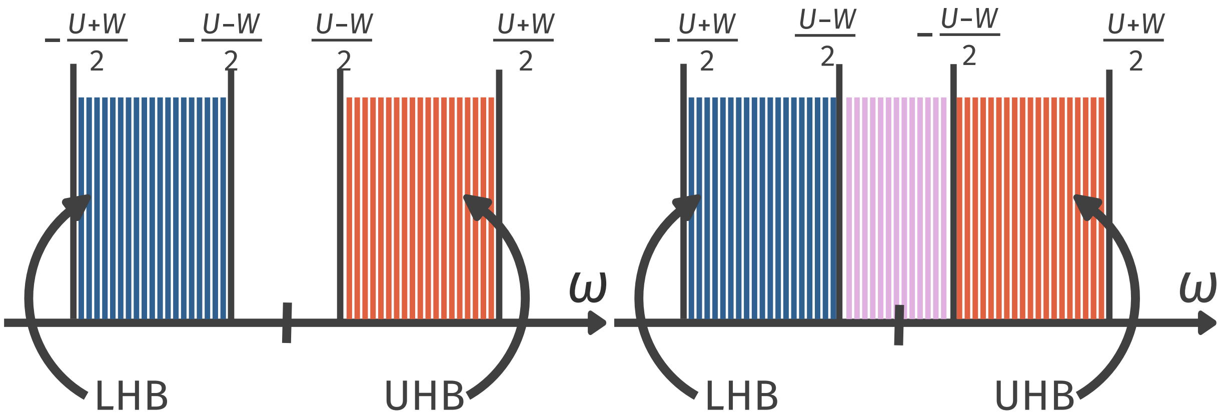

Figure 1.3

infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k},\sigma} \epsilon(\mathbf{k}) n_{\mathbf{k},\sigma} - \frac{U}{2} \sum_{\mathbf{k}} (n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow})^2 \ . \tag{3.1}$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k},\sigma} - \frac{U}{2}(n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow})^2 \ . \tag{3.2}$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: ($\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$, the other for adding an additional electron on a singly-occupied state: $\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$.

Upon combining states from other $k-$modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per $k-$state). Upon lowering the correlation strength U, the bands touch each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). For lower values of U, the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.4 Strong local correlations: models and materials

1.4.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [27]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H^*_{RS} & H_S \end{bmatrix} . \tag{4.1}$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow} , \tag{4.2}$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sideset{}{'}\sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sideset{}{'}\sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c^\dagger_{l,\sigma} c_{i,\sigma} + \text{h.c.}) , \tag{4.3}$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l.

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.4.

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as $k-$space objects) will require a periodisation operation of an auxiliary model quantity.

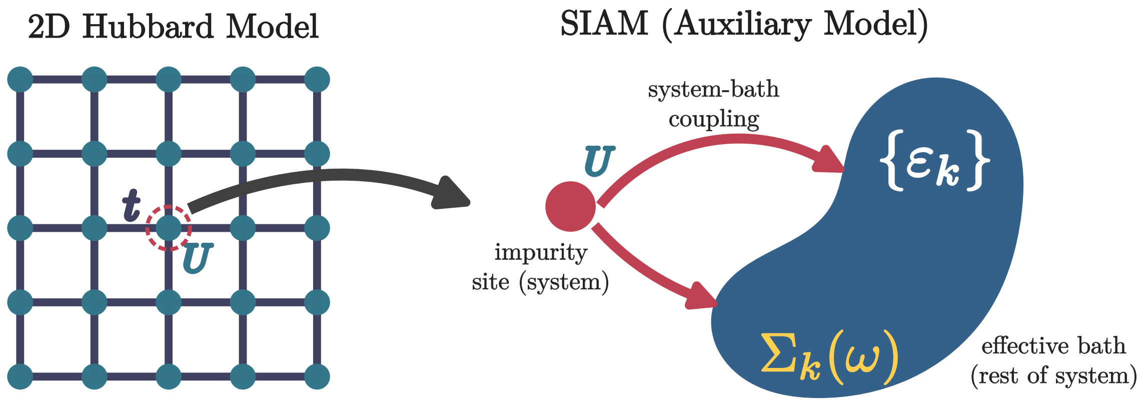

Figure 1.4: *Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H. Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N-1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V.*

1.4.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [18, 28]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal–insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [18, 29]

In order to motivate the impurity model-lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = -\sum_{i\neq j} J_{ij} S_i S_j - h\sum_i S_i = -\sum_i S_i (h + \sum_{j\neq i} J_{ij} S_j), \tag{4.4}$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an *effective static field* proportional to the average magnetization. Writing $S_j = \langle S_j\rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains an *auxiliary problem* in the form of a mean-field Hamiltonian

$$H^{(\text{MF})}_{\text{Ising}} = -\sum_i (h + h_{\text{MF}}(i)) S_i \ , \tag{4.5}$$

where the mean-field $h_{\rm MF}$ comes out to be

$$h_{\rm MF}(i) = \sum_{j\neq i} 2\langle S_j\rangle J_{ij} \ . \tag{4.6}$$

The above approach assumes two things: (i) the presence of a non-zero *order parameter* $m_j = \langle S_j\rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j\rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\rm MF}(i) + h)) = \tanh(\beta \sum_{j\neq i} 2m_j J_{ij} + \beta h) \ . \tag{4.7}$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j. We therefore need an additional equation; this is provided by the so-called *self-consistency condition*. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j. This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j\neq i} 2J_{ij} + \beta h) \ . \tag{4.8}$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

An equivalent way of framing the self-consistency equation is to equate the mean-fields on different lattice sites. Combining eqs. 4.7 and 4.6, we have

$$h_{\rm MF}(i) = \sum_{j\neq i} 2m_j J_{ij} = \sum_{j\neq i} 2\tanh(\beta(h_{\rm MF}(j) + h))J_{ij} \ . \tag{4.9}$$

The equation in terms of $h_{\rm MF}(i)$ and $h_{\rm MF}(j)$ can again be solved numerically by claiming that the mean-field has to be uniform across all the sites owing to translation invariance. This form of the self-consistency requirement (in terms of the field) will be useful to us in the upcoming discussion.

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 28].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = -\sum_{ij\sigma} t_{ij} c^\dagger_{i\sigma} c_{j\sigma} + U\sum_i n_{i\uparrow}n_{i\downarrow}, \tag{4.10}$$

defined on a lattice with coordination number z.

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an "order parameter" and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k-space Green's function for a translation-invariant problem:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \tag{4.11}$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green's function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \tag{4.12}$$

Similar to the earlier exercise, we need to apply a mean-field decoupling to drop certain fluctuations and obtain a similar effective problem. In the context of interacting electrons, such a decoupling is obtained in the limit of infinite dimensionality or coordination number, where the self-energy becomes $k-$independent [28]:

$$\Sigma_{\mathbf{k}}(\omega) = \Sigma_{\text{loc}}(\omega), \text{ or (equivalently) } \Sigma_{ij}(\omega) = \delta_{ij}\Sigma_{\text{loc}}(\omega)\ . \tag{4.13}$$

The mean-field decoupling here therefore corresponds to ignoring spatial fluctuations of the one-particle self-energy, and leads to a local Greens function that is directly connected to the local self-energy:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \tag{4.14}$$

It turns out that a local Greens function connected to a local self-energy also defines a quantum impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out:

$$S_{\text{imp}} = -\int_0^\beta d\tau d\tau'\, c^\dagger(\tau)\, G_{\text{MF}}^{-1}(\tau-\tau')\, c(\tau') + U\int_0^\beta d\tau\, n_\uparrow(\tau) n_\downarrow(\tau). \tag{4.15}$$

This acts as an auxiliary problem for the more complete problem locally interacting electrons on a lattice. The non-interacting Greens function G_{MF} is as of now unknown, and defines the environment of the impurity site and comprises the dynamics of the sites that have been integrated out. Such an impurity problem is solvable, and the impurity Green's function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_{\text{MF}}^{-1}(\omega) - \Sigma_{\text{imp}}(\omega)}, \tag{4.16}$$

The first term in eq. 4.15 is therefore what makes this problem dynamical – it represents the probability of electrons hopping into and out of the impurity site, and leads to fluctuations of its occupancy. We can make the identification of the Greens function G_{MF} as the *mean-field* encountered in the previous example. More specifically, G_{MF} acts as a "field" because it is obtained by integrating out the other sites on the lattice, and it acts on the impurity site by affecting its dynamics.

The complete set of equations for the solving such a dynamical mean-field problem is finally obtained by equating the impurity local Greens function G_{imp} to the lattice local Greens function G_{loc}. As discussed earlier, such a solution exists in an exact fashion only in the limit of infinite dimensions, as long as the correct G_{MF} can be found. In order to find the appropriate G_{MF} that allows equating the impurity model Greens function to a lattice mode Greens function, we require an additional equation: the self-consistency condition. Similar to the case of the Ising model, we require that G_{MF} be equal to G_{loc}, owing to translation invariance. The full set of coupled DMFT equations is

$$\begin{aligned} G_{\text{MF}} &= G_{\text{loc}}, \\ \Sigma_{\text{imp}}(\omega) &= G^{-1}_{\text{MF}}(\omega) - G^{-1}_{\text{imp}}(\omega), \\ \Sigma_{\text{loc}}(\omega) &= \Sigma_{\text{imp}}(\omega), \\ G_{\text{loc}}(\omega) &= \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \end{aligned} \tag{4.17}$$

- The first equation is equivalent to the feature in the previous exercise where we equated the mean-fields on i and j.
- The second equation is equivalent to computing, in the previous exercise, the magnetisation on the j^{th} site from the partition function.
- The third equation is the essential mean-field approximation, where we equate the self-energy of the simpler decoupled problem to that of the more complete lattice problem and is justified in the limit of infinite dimensions.
- The fourth equation is equivalent, in the previous exercise, to using the computed "magnetisation" of site j in the second step to further compute the mean-field at site i.

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. Importantly, this demonstrates the importance of quantum impurity models in investigating correlated lattice models, particularly in the presence of strong local interactions. This is because the effects of local correlations are captured exactly by such an auxiliary model mapping.

1.4.3 Examples of materials with strong local correlations

As discussed in the previous section, the route to studying lattice problems via quantum impurity models makes most sense for materials with strong local correlations. Such systems are often referred to as strongly correlated quantum materials, and their phenomenology often lies beyond conventional approaches like density functional theory or perturbative weak-coupling methods. In practise, such materials often involve transition metal elements or rare earth/actinide elements, due to the presence of partially filled $3d-$ or $4f-$orbitals in such elements. This is because these orbitals are more more localised around the nucleus than the $s-$ or $p-$orbitals of comparable energy, and hence lead to a reduced itinerant energy scale and amplify the effects of local correlations [29].

The localised nature of these orbitals can be understood as follows. For the $3s-$orbital, the $1s$ and $2s$ orbitals have same orbital angular moment l but lower principal quantum number n; since all such wavefunctions must be orthogonal to each other, the $3s-$ orbital must develop nodes in its radial form and extend farther away from the nucleus in order to be orthogonal to these other orbitals. However, for the $3d-$orbital, there is no orbital with same l but lower n; this means that the orthogonality is already enforced by the behaviour of the angular part (the spherical harmonics), and the radial part can remain localised around the nucleus without having to spread out. A similar argument holds for the localised nature of the $4f-$ orbital.

We now discuss the phenomenology of some of these quantum materials in order to identify the challenges in trying to explain them and the kind of models that are likely to be useful in this endeavour.

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d-shells in transition metals (e.g., Cu^{2+}, Mn^{3+}, V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U; due to the small overlap between the localised $d-$orbitals, the $d-$electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [30], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d-d Coulomb interaction ($E_g \approx U$), or charge-transfer insulators, where the gap is dictated by the energy difference between the oxygen p-band and the metal d-band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy}, d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the Fermi level lies within e_g. The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the $d-$orbitals and brings them closer to the $p-$orbitals, reducing the charge transfer energy $\Delta = \varepsilon_d - \varepsilon_p$ of transporting an electron from a $d-$state to a $p-$state, compared to the $d-$level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the $p-$orbitals) also enhance the hybridisation between $d-$ and $p-$orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ. In the lighter TMOs, the Fermi level passes through t_{2g}, and because they point away from the $p-$orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal–insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [31, 32]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [33–35]. Copper oxides constitute the most extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [36–38]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [39–41]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^+ sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nicklelates is different due to the strong coupling between the layers [42, 43]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [44].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2, where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [45,46]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal–insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [47, 48]. In prototypical compounds such as 1T-TaS_2, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [49–51]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal–insulator transition of 1T-TaS2, demonstrating the macroscopic controllability of the Mott phase [52].

ARPES measurements in 2H-$NbSe_2$ and 2H-$TaSe_2$ partial gapping of the Fermi surface restricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [53, 54]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [55]. Transport measurements on single crystals of $TaSe_2$ reveal that disorder enhances superconductivity by suppressing other competing orders such as CDW [56]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine 'blocking layers' [57–60]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [61–63]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior ($T-$linear resistivity) to a Fermi liquid behaviour (T^2-resistivity) [64]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [65].

ARPES measurements reveal a $k-$dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [66,67]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [68, 69]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [70, 71].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [72, 73]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [74]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [75].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [76]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior ($T-$linear resistivity) and a vanishing Hall response, along with a divergent effective quasiparticle mass as the transition is approached [77, 78]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [79,80]. Quantum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [81]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [82].

1.4.4 Generator of strong local correlations: Quantum impurity models

Within this thesis, we have developed an auxiliary model approach that combines the physics of quantum impurity models with a manybody version of Bloch's theorem to investigate interacting fermionic lattice models. This is motivated by the relative ease in solving impurity models and transparency afforded by such solutions. We now review the phenomenology of some of these impurity models and point out how the various impurity features affect the physics of the lattice models.

The single-impurity Anderson model describes the dynamics of a system of non-interacting itinerant electrons ($c_{k,\sigma}$) hybridising with a localised correlated impurity orbital (d_σ) [15]:

$$H_{\text{AIM}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + \sum_{k,\sigma} V_k (c^\dagger_{k,\sigma} d_\sigma + \text{h.c.}) + \epsilon_d \sum_\sigma n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} \ . \tag{4.18}$$

V_k controls the hybridisation strength, and ϵ_d and U control the onsite energy and local correlation respectively of the impurity site. In the limit of large U, the impurity dynamics gets projected onto its spin states, and the model then describes the scattering of non-interacting electrons off magnetic impurities, through the Kondo model [83]. Formally, the Kondo model is obtained by carrying out a Schrieffer-Wolff transformation on the Anderson impurity model and then using the condition to $V \ll U$ to drop all contributions beyond second order in the ratio V^2/U [84]:

$$H_{\text{Kondo}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c^\dagger_{k_1,\alpha} c_{k_2,\beta} \ , \tag{4.19}$$

where $\vec{S}_d$ is the impurity spin formed by projecting the Hilbert space of the impurity to the singly-occupied states.

Even though the conduction bath is non-interacting, the presence of local repulsion on the impurity site makes the problem *many-electron* in nature: it makes it impossible to study the electrons in isolation. This can be seen as follows: imagine two conduction electrons $|k_1 \uparrow\rangle$ and $|k_2 \downarrow\rangle$ trying to hybridise with an empty impurity site. The first electron moves onto the impurity site through a scattering process of the form $c^\dagger_{d\uparrow} c_{k_1\uparrow}$ and makes the impurity site single occupied ($|\uparrow\rangle_d$), but now the second electron k_2 *must pay a cost* of the order of U in order to hybridise with this singly-occupied impurity site through a process of the kind $c^\dagger_{d\downarrow} c_{k_1\downarrow}$. This shows that the conduction electrons *are aware* of the other electrons, and all of them must be treated together. Another way of seeing this collective behaviour is from the Kondo model. Again imagine two spin-up electrons k_1, k_2 that are about to scatter off the impurity spin (which is presently down) via a *spin-flip* process. The first electron scatters and leaves the impurity spin up while it itself becomes down, but now the second conduction electron cannot spin-flip scatter because the spins cannot be exchanged (both are down!). This is another example of how the conduction electrons can *feel* the presence of the other electrons, and this is what makes the problem many-electron in nature.

Kondo screening Despite the interacting nature of the problem, the problem can be solved in an effectively exact fashion, via Wilson's numerical renormalisation group approach [85, 86], density-matrix renormalisation group approach [87] or the method of *Bethe ansatz* [88, 89]. The former solution is more transparent: for any non-zero value of the hybridisation V (or Kondo coupling J), the low energy physics of the impurity is described by a strong-coupling fixed point ($V = \infty$, or $J = \infty$), where the impurity forms a maximally-entangled singlet state with the local spin density of the conduction bath. The low-energy excitations about such a fixed point can be calculated from renormalised perturbation theory, and describe a local Fermi liquid – one particle excitations in the neighbourhood of the singlet renormalised by a local repulsion term [90, 91]. The crossover from the free local moment at high temperatures to the screened non-magnetic state at low temperatures occurs at an emergent energy scale T_K, the Kondo temperature, which depends exponentially on the strength of the Kondo coupling, and is the only energy scale remaining at low temperatures, so that all thermodynamic quantities must go as a function of T/T_K [85].

At half-filling (attained for $\epsilon_d = -U/2$), the only other fixed points are the free orbital fixed point at $V = 0, U = \epsilon_d = 0$ where the impurity site is freely occupied by two independent spins, and the local moment fixed point at $V = 0, U = \infty$, where the impurity site is occupied by a local moment that is decoupled from the bath [74, 86]. Away from half-filling, two other fixed points become available [74, 86]: a valence fluctuation fixed point at $V = 0, \epsilon_d = 0, U = \infty$ where the doubly-occupied state has been projected out from the symmetric free orbital fixed point, and a frozen impurity fixed point at $V = U = 0, \epsilon_d = \infty$ where the impurity site is left completely empty.

The crossover from local moment physics to Kondo screening as the system is tuned from high to low temperatures can also be seen from the impurity spectral function [92–95]. At low frequencies, the spectral function exhibits a sharp resonance around $\omega = 0$ that indicates the presence of gapless single-particle local Fermi liquid excitations. At higher frequencies, one sees satellite peaks that involve impurity excitations that cost energy of the order of U. The height of the central Kondo resonance is a topological invariant; it is fixed by the impurity occupancy owing to an application of the Friedel sum rule to impurity models [96], and remains unchanged as the interaction U is increased adiabatically while remaining at particle-hole symmetry. The topological nature can also be ascertained by identifying such a sum rule as the impurity model analogue of Luttinger's theorem [97, 98] that relates the total number density of electrons to the number of Greens function poles within the Fermi volume, and which is itself topological [99, 100].

Breakdown of Kondo screening While the vanilla Anderson model (or the Kondo model derived from it) only display a strongly-coupled screened phase, it is possible to tweak it in order to introduce Kondo frustration which can eventually lead to a breakdown in the Kondo screening process. Such models are interesting because they provide a means of studying zero temperature *continuous* phase transitions and address experimental signatures of quantum criticality in correlated systems. The canonical example of this is the multichannel Kondo model, involving a single impurity hybridising with multiple independent conduction channels [101]:

$$H_{\text{MCK}} = \sum_{k,\sigma,\alpha} \epsilon_{k,\lambda} n_{k,\sigma,\lambda} + \sum_{k_1,k_2,\lambda} J_\lambda \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c^\dagger_{k_1,\alpha,\lambda} c_{k_2,\beta,\lambda} \ , \tag{4.20}$$

here λ is the channel index. In the case of an isotropic Kondo coupling across the channels ($J_\lambda = J$), the strong-coupling $J = \infty$ fixed-point of the single-channel Kondo model is replaced by an intermediate-coupling fixed-point at a finite value J^* [101, 102]. Powerful methods like boundary conformal field theory [103–108], bosonization [109–113], numerical renormalization group [105, 114, 115] and Bethe ansatz [116–120] have shown that the low energy theory of such models exhibit non-Fermi liquid excitations. Further, the system displays anomalous behaviour in thermodynamic properties near $T = 0$ and a fractional zero temperature entropy in the thermodynamic limit (upon ensuring that temperature is taken to zero after the system size is taken to infinity [112, 121]). This anomalous divergent behaviour is a signature of the fact that the MCK system with a single channel-symmetric exchange coupling is quantum critical: the ground state is susceptible to perturbations that introduce channel anisotropy in the exchange couplings [101, 105, 111, 119, 122].

Another class of models that shows Kondo breakdown is that of pseudogap Anderson and Kondo models, where the impurity hybridises with a conduction bath whose density of states (DOS) varies as $\rho(\omega) \sim |\omega|^r$ with a paramter $r > 0$ that controls how fast the DOS vanishes as the Fermi level is approached. Such behaviour can arise in semi-metals and in long-range ordered systems where the order parameter vanishes at certain points on the Fermi surface (such as $d-$wave superconductors). The pseudogap impurity models exhibit a boundary quantum phase transition between a Kondo-screened strong-coupling phase and an unscreened local-moment phase as a function of interaction strength or exchange coupling [123–126]. At the critical Kondo coupling, the impurity spectral function and spin susceptibility exhibit nontrivial power-law scaling with anomalous exponents determined by r [124, 127–130]. For $0 < r < 1/2$ (in the particle-hole symmetric case), both the local-moment and strong-coupling phases are stable, separated by a critical fixed point characterized by fractional residual entropy and nontrivial hyperscaling relations [124, 131]. For larger exponents, the strong-coupling phase may disappear entirely, and the system remains in the local-moment phase for all couplings [126].

The multi-impurity Anderson and Kondo models furnish another canonical realization of Kondo breakdown, driven by the competition between Kondo screening and inter-impurity interaction. The paradigmatic example is the two-impurity Kondo model (TIKM), in which two localized spins couple both to conduction electrons and to each other via an antiferromagnetic exchange K [132, 133]. In the limit of large $K > 0$, the impurities form a non-magnetic singlet decoupled from the conduction electrons and suppressing the Kondo resonance. This fixed point is separated from the strong-coupling Kondo screened fixed point through a quantum critical point [132–134] displaying logarithmic or power-law scaling in thermodynamic quantities and fractional residual entropy [135]. A modified version of this model that uses conduction bath-mediated inter-impurity interaction leads to a metallic spin liquid displaying nonlocal impurity-spin entanglement and fractionalized charge excitations [136]. More intricate geometries can be constructed by arranging antiferromagnetically coupled Kondo impurities along a triangle, each hybridising with its own conduction bath; such a model displays a topological phase transition from a symmetry-preserved gapless spin liquid phase with "topological order" to a screened Fermi liquid phase [137]. Indeed, recent experiments have validated such phase diagrams for the case of two impurities by studying pairs of copper atoms in the presence of strong magnetic fields [138].

1.5 Mott transition in Hubbard-like models

1.5.1 In the limit of infinite dimensions

The Hubbard model (introduced in eq. 1.1) presents one of the most fruitful playgrounds for investigating correlation-driven metal-insulator transitions and its various facets. This model, despite its conceptual simplicity, exhibits a rich variety of phenomena, including antiferromagnetism, superconductivity, and, most notably for our purposes, the correlation-driven Mott metal–insulator transition (MIT) [4, 139–141]. While exact solutions exist in the limits of $d = 1$ [142] and $d = \infty$ [143], the problem remains largely open in more realistic situations such as two and three spatial dimensions.

As discussed earlier, the limit of infinite dimensions presents a drastic simplification for the manybody scattering processes within the problem: the one-particle $k-$space self-energy becomes momentum-independent due to the vanishing of all non-local contributions (they scale as negative powers of the coordination number, which diverges in the limit of $d = \infty$). This simplification lies at the heart of dynamical mean-field theory and its ability to capture the Mott metal–insulator transition at both zero and finite temperatures for this limiting case [18, 143–147]. At half-filling and zero temperature, DMFT predicts a continuous transition as a function of U wherein the quasiparticle weight Z tends to zero at a critical interaction strength U_{c2}, signaling the vanishing of coherent Fermi liquid behavior and the opening of a Mott gap in the single-particle spectral function [18, 145]. The three-peak structure of the interacting spectral function, with a central quasiparticle resonance flanked by incoherent lower and upper Hubbard bands, indicates the presence of highly renormalised Landau quasiparticles [145]. A doping-driven continuous metal-insulator transition also exists for this model at zero temperature, marked by the vanishing of electronic compressibility at a critical value of the chemical potential [148]. The quantum critical point itself (at $T = 0$) is likely to be an exotic state, and important questions about it remain unanswered. *Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?*

While the DMFT solution captures the zero temperature transition accurately, the self-consistently determined nature of the conduction bath renders the solution somewhat opaque and makes it difficult to write down an effective Hamiltonian for the low-energy theories. Since the vanilla Anderson impurity model does not contain the mechanism to stabilise a localised phase over a finite-sized parameter regime, an important question arising from the DMFT solution is: *what are the minimal changes that one must implement into the Anderson impurity model in order to capture such a metal-insulator transition ($1/2$-filled Hubbard model on the Bethe lattice in $d = \infty$), in sufficient accuracy and precision?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.

At finite temperatures, the interaction-driven as well as doping-driven Mott transition become first order, and is characterized by a region of phase coexistence and hysteresis [146, 149]. Two spinodal lines, $U_{c1}(T)$ and $U_{c2}(T)$, emerge; within this region metallic and insulating DMFT solutions coexist, and responses such as resistivity show a significant discontinuity [150]. These lines meet at a finite-temperature critical endpoint (U_c, T_c), above which there is no sharp distinction between metallic and insulating solutions and only a crossover exists between the Mott insulator and the Fermi liquid through a bad-metal regime, as the chemical potential is tuned [146, 148, 151–153]. The coexistence of metallic and insulating phases shows that the insulating solution is present within the many-body spectrum of the metallic phase. This naturally leads to the following question: *since a quantum impurity lies at the heart of the self-consistency procedure, what does the presence of coexistence region mean for the Hamiltonian dynamics and the spectrum of this impurity model? Can the impurity model Hamiltonian display the emergence of the insulating state prior to the transition, through changes in the eigenstates?* Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.

The bad metal regime displays anomalous linear-in-temperature resistivity which corresponds to quantum critical ω/T scaling, along the quantum widom line [154, 155]. This is accompanied by the disappearance of the Drude peak in optical conductivity, signalling the loss of quasiparticle transport [154]. Similar quantum critical scaling is also obtained from DMFT solutions at very low temperatures, hinting at a "hidden" quantum criticality that is continuously connected to and responsible for the critical behaviour above the finite temperature endpoint [156]. Luttinger's theorem, which relates the poles of the Greens function within the Fermi volume to the number of electrons in the system at zero temperature, is satisfied within the Fermi liquid phase but is violated in the particle-hole asymmetric Mott insulator by a universal magnitude [157]. The emergence of a quantum critical regime at high temperatures and its connection to similar signatures at low temperatures suggests the presence of quantum fluctuations in the underlying impurity model that modify the impurity-bath scattering processes and the associated impurity responses. One can therefore ask: *What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?*

The above studies of single band models describe the Mott transition to be of the Mott-Hubbard kind (described in previous sections), where the size of the gap is determined by the double-occupancy cost U. In order to obtain transition of the charge-transfer kind, one needs to consider at least a two-band model. Indeed, in a non-degenerate two-band Hubbard model with a charge-transfer energy Δ for hopping from a $d-$orbital to a $p-$orbital, one sees a Mott-Hubbard transition when $U < \Delta$ and a charge-transfer transition when $U > \Delta$ [158]. The degenerate two-band model with inter-band local correlation shows successive Mott-Hubbard transitions at low temperatures whenever the filling passes through integer values and the interaction strength is intermediate [159].

1.5.2 The two-dimensional Hubbard model

The 2D Hubbard model is one of the most well-studied models in condensed matter physics, particularly because of its paradigmatic role in Mott metal-insulator transition and its connections to the physics of copper oxide materials and $d-$wave superconductivity. Photoemission experiments reveal the presence of p and d bands close to the Fermi surface in such materials [39]; further work by Zhang and Rice showed that a specific combination of these orbitals governs the low-energy physics, leading to an effective description in terms of a single band Hubbard model [160]. The discovery of high-temperature superconductivity in nickelates and organic charge transfer salts has provided more platforms where the Hubbard model and its extensions can prove to be useful [161, 162].

While simple in appearance, the model has proved to be notoriously hard to solve except in highly specific parameter regimes, despite several decades of computational effort. This is due to several reasons [141]: (i) the presence of closely-lying energy scales arising from competing tendencies, (ii) the inability of several methods to either reach zero temperature or extrapolate to the thermodynamic limit, (iii) the tendency of approximate/finite-size methods methods to overestimate proclivities towards spontaneous symmetry breaking, and (iv) the presence of differing phases separated by slow crossovers without any sharp boundaries.

At half-filling, the 2D Hubbard model has a spectral gap to charge excitations for any non-zero interaction strength [163–167]. The behaviour below the charge gap is described by a nonlinear sigma model with parameters that depend on the interaction strength U [168]. Upon tuning to finite temperatures, the system goes through a sequence of crossovers from the insulating ordered state to a Fermi liquid state with coherent quasiparticles into a non-Fermi liquid state with partial gapping of the Fermi surface [165, 167, 169]. *Capturing the momentum dependence of static and dynamical quantities in sufficient detail remains an open challenge in such problems, since results from cluster methods suffer from sensitivity to periodisation schemes [170].*

Away from half-filling but at weak coupling, the Neel order is accompanied by incommensurate magnetic orders (also called spiral magnetic state) with wave vectors away from (π, π), as seen from mean-field studies [171, 172]. Functional RG studies indicate that such solutions can exist up to fairly high doping if the superconducting instability is suppressed [173, 174]. Superconductivity with $d-$wave pairing has been achieved from renormalised perturbation studies as well as from diagrammatic Monte Carlo (diagMC) and functional RG (fRG) calculations [175–177]. More intricate competition between the superconducting and stripe states is found in three-band Hubbard models from variational Monte Carlo, cluster DMFT and DMRG calculations [178–180]. *Further work is necessary to resolve the closely-lying phases in such models.*

The phenomena of pseudogap (PG), where the Fermi surface acquires a momentum-dependent suppression of spectral weight, has also been observed above a certain temperature from methods like diagMC and fRG [167, 181]. The gap is seen to open first near the antinodal point $(\pi, 0)$ and spread across the Fermi surface until it reaches the node $(\pi/2, \pi/2)$. The mechanism of the pseudogap at weak to intermediate coupling is found to be long-range antiferromagnetic (AF) correlations [182]. The pseudogap disappears at large doping when the Fermi surface undergoes a topological transition from hole-like to electron-like. At strong coupling, the competition between various phases such as $d-$wave superconductivity, antiferromagnetic order and stripe states (spin/charge order modulating in real space) is an active subject of study. Recent studies using various methods like DMRG and iPEPS found that the period 8 charge stripe state is lower in energy than the superconducting state at strong coupling $U/t = 8$ and no next-nearest-neighbour hopping [183]. This was later corroborated by studies from variational Monte Carlo and other DMRG studies [184, 185]. *It is not yet clear whether the stripe period gets lowered as the interaction strength is increased.* At weaker coupling, the above methods report a reduced stability for the strip state and increased tendency to realise a $d-$wave SC state [176, 186]. *Further investigation is necessary to map the crossovers between stripe and superconducting phases as doping is tuned. Moreover, the precise parameter regime in which the $d-$wave superconducting state becomes more stable than the stripe states is still an open question.*

At strong-coupling, the pseudogap is obtained below optimal doping and has been studied using dynamical methods such as cluster DMFT (DCA, dynamical cluster approximation) and cellular DMFT (CDMFT) [187–190]; in contrast to the weak-coupling feature (where the PG was borne from long-range AF fluctuations), it is believed that the momentum-space differentiation in this regime is brought about by the proximity to the Mott insulator [189]. Conflicting indications are however provided by a fluctuation analysis that finds that long-lived long-ranged spin-fluctuations at the antinode are responsible for the opening of the PG [191]. The PG is found to be accompanied by non-Fermi liquid behaviour in the one-particle self-energy [188]. The doping value at which the non-Fermi liquid first appears was found to be lower for electron-doping than hole-doping [192]. At half-filling, DCA calculations reveal that the extent of the PG decreases as the next-nearest-neighbour hopping strength is increased [193]. Moreover, while DCA+continuous-time auxiliary field methods do not show any effect of van Hove singularities on the generation of the PG [193], DCA+determinantal Monte Carlo approaches identify the proximity to van Hoves as favouring the formation of a PG [194]. *Methods with better momentum-space resolution is necessary in order to identify the role of nesting and van Hoves on the pseudogap.*

The question of whether superconductivity can arise from repulsion was one of the primary motivations for studying the Hubbard model. DiagMC calculations reveal superconductivity in the ground state at weak-coupling and moderate doping, and with multiple symmetries of the gap [176, 195]. For higher couplings (which is relevant to the cuprates) and moderate doping, stripe states take over. Quantum Monte Carlo and DCA calculations suggest a $d_{x^2-y^2}$ pairing symmetry [164, 196]. DCA calculations on the 2D Hubbard mode suggest low-frequency spin fluctuations as the pairing glue for the superconductivity [197]. *The description of how various parent phases such as the pseudogap and strange metal precipitate superconductivity, in terms of the nature of fluctuations and the appropriate degrees of freedom, is still an open question.*. Apart from superconductivity, several strongly correlated materials also exhibit strange metallicity, with transport properties that defy Fermi liquid theory, such as a linear-in-temperature resistivity that extends across large regimes of temperature. Within the Hubbard mode, evidence for such $T-$linear resistivity and robustness across temperatures is available from CDMFT and determinantal Monte Carlo calculations [198, 199]. *More reliable methods are needed to access and understand the intermediate temperature regime of strange metals [141, 200].*

1.6 Brief introduction to the tiling auxiliary model approach

1.6.1 Why do we need another auxiliary model method? Limitations of existing approaches

1.6.2 Overview

1.6.3 Features

1.7 Entanglement holography and fermionic criticality

Role of entanglement in fermionic systems The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [201–206]. Being non-local by nature, it is helpful in tracking correlations among particles across both small and large distances, giving valuable insight into the nature of the ground state and the low-energy excitations. Entanglement, in the form of various measures like entanglement entropy and mutual information, is the engine that gives rise to new low-energy degrees of freedom from a renormalisation group (RG) perspective, and is therefore key to understanding and characterising phase transitions and criticality in quantum matter. For instance, entanglement entropy is at the heart of understanding the so-called topological states of matter that can be described in terms of a purely topological effective Lagrangian and have a robust ground state degeneracy and gapless edge states [207–227]. In contrast to systems with local interactions whose subsystem entanglement entropy scales with the area of the subsystem [203, 228, 229], topologically ordered systems have an additional subdominant topological term dubbed the topological entanglement entropy [230–248] that quantifies the long-ranged nature of the entanglement in such systems. The non-local nature arises from the fact that the ground state of such a system cannot be made separable (non-entangled) by the application of purely local unitary transformations [205, 249, 250].

More pertinent to the work at hand, however, is the entanglement in free fermionic systems. In this thesis, we will consider relativistic Dirac fermions in two dimensions, both massless and massive. They emerge in several condensed matter systems [251], appearing at the surfaces of three-dimensional topological insulators [252–255], quasi-2D organic materials [256–258] and in arficial materials like cold atoms, molecular and microwave crystals [259–265]. For massless critical systems (or conformal field theories (CFTs)) living in one spatial dimension, the entanglement entropy of a single subsystem of length l scales as $\sim \frac{c}{3}\ln l$ [266–275], where c is the conformal central charge of the CFT. The case of multiple subsystems is more involved, but has been solved [273, 276, 277]. For massive fermions, the entanglement entropy (or rather the entropic $c-$function, which is the derivative of the entropy along the distance) can be expressed in the form of differential equations which need to be solved numerically [276, 278–280]. The solutions lend themselves to small and large mass expansions, reducing to $\frac{1}{3}\ln\frac{l}{\epsilon} - \frac{1}{6}\left(ml\ln ml\right)^2 + O((ml)^2)$ for the case of small ml, and to $\sim (ml)^{-1/2}e^{-2ml}$ for that of large ml, both for a single interval. The $\ln l$ behaviour is a specific case of the more general $l^{d-1}\ln l$ violation of the area law, as observed in $d-$dimensional fermionic critical systems [276, 281–295]. This violation arises from the non-local nature of the quantum fluctuations that exist in quantum critical fermionic systems.

While there is no geometry-independent subdominant term in the entanglement entropy of free fermionic systems [296, 297], it *is* possible to generate such a term by placing the system on a manifold with a non-trivial topology, say a cylinder, and inserting a magnetic flux through the cylinder [296, 298–301]. For a massless Dirac fermion, if there are ϕ number of electronic flux quanta piercing the system, the additional term that is generated in the entanglement entropy of a subsystem of sufficiently (as defined in Ref. [296]) large length is of the form $\frac{1}{6}\ln\left|2\sin\frac{\phi}{2}\right|$ [296, 299]. There is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [301–303]. In this thesis, we put this idea on a stronger analytical footing by relating the flux-dependent term to a topological invariant of the field theory, its Luttinger volume [97–100, 304–306]. Given by the number of $k-$states enclosed by the Fermi surface, Luttinger volume V_L has proved to be one of the most important indicators that distinguish metals from insulating states of matter. In metals, V_L is constrained by the number of particles in the system, following Luttinger's theorem [97–99]. As shown most prominently in Ref. [99], Luttinger's theorem depends crucially on the presence of extended states in the metal that can respond to flux insertion experiments. We demonstrate that indeed it is this subdominant term in the entanglement entropy that carries this dependence and hence encodes Luttinger's theorem. It is known that for strongly-interacting states like Mott insulators or quantum hall states, V_L is modified due to changes in the topology of the Fermi surface [100, 304–306]. In this context, we relate the topological invariant on the metallic side (Luttinger volume) to the topological invariants (the Chern numbers) in the quantum hall system that the metal would lead to when placed in a perpendicular magnetic field. Such a relation makes explicit the transmutation of the extended states into the localised states of the insulating phase.

Entanglement hierarchy and the holographic principle The leading geometry-dependent piece is also found to have intriguing properties of its own. The fact that the free fermion Lagrangian is diagonal in momentum space means that the entanglement arises purely from real space quantum fluctuations, and this allows us to study the entanglement of subsystems of varying sizes in momentum space. We find that the entanglement entropy hence obtained has a Russian doll-like nested structure, such that increasing the size of the subsystem in momentum space gradually reveals layers of entanglement. In fact, we argue that such transformations between the subsystems amount to a renormalisation group (RG) flow in the space of the field theories with varying spectral gaps, and the above-mentioned hierarchy in the entanglement entropy then stretches across the RG transformations. The precise prescription for choosing the subsystems is described in Sec. **??**. We also show, simultaneously, that the hierarchy holds across multipartite entanglement measures, leading to an onion-like structure where increasing the number of parties in the entanglement measure (mutual information, tripartite information, 4-partite information and so on) leads to additional entanglement that did not exist before. More importantly, we interpret the hierarchy of entanglement along the RG flow as the emergence of an additional dimension [307–309], allowing us to define a measure of distance using some non-negative measure of entanglement, like the mutual information [310–313]: the more the entanglement, the less will be the distance between those points. Generation of an additional dimension is, of course, the essence of the AdS/CFT conjecture or, more generally, the holographic principle.

The holographic principle [314–322], also known as the gauge-gravity duality, posits that there exists a duality relation between a quantum field theory with a gauge field, and a gravity theory having one additional spatial dimension [323–326]. The additional dimension can be visualised as a renormalisation group flow that starts from the boundary field theory and extends into the bulk [327–336]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a super-symmetric gauge theory with a string theory [337–339]. The essence of holography is that all the information about the bulk gravity theory is already contained within the boundary gauge theory; this idea is very similar to the discovery by Bekenstein and Hawking [340–342] that the entropy of a black hole is determined by its surface area. In fact, Ryu and Takayanagi proposed a generalisation of this result where the entanglement entropy of subsystem region a CFT is determined by the area of the minimal surface that bounds the region in the bulk [343, 344]. The AdS/CFT correspondence happens to be a strong-weak duality - it relates a strongly coupled CFT to a theory of weak classical gravity. This allows physicists to use the holographic principle and convert a problem of strong quantum correlation to one of classical gravity which can be solved using well-known methods [317, 318]. This has led to new insights in various topics of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [345], the response functions of the Bose-Hubbard model [346, 347], and the phenomenology of non-Fermi liquids [348–351], holographic superconductors [352–357], striped phases [358–360] and holographic Fermi liquids [361–365]. An overview of the metallic states (phases that satisfy Luttinger's theorem) and their connections to the gauge-gravity duality have been provided in Refs. [366–368].

Of course, on the flip side, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This however requires constructing, *ab initio*, the emergent dimension by starting from the field theory, and has proved to be more difficult than the converse, mostly because the field theories are often strongly coupled. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory [305, 306, 312, 313, 369–402]. Some well-known approaches include the momentum shell renormalisation approach of Sung-Sik Lee [382–385] and the multi-scale entanglement renormalisation ansatz (MERA) of Vidal [386–390]; both use certain forms of RG flow to generate the extra spatial dimension. The former integrates out high momentum degrees of freedom in favour of generating dynamics for the boundary gauge field, with the RG trajectory variable giving rise to the additional spatial dimension [316]. On the other hand, the MERA works by disentangling degrees of freedom in real space, creating a tensor network in the process [391–393] and leading to a renormalisation in the entanglement. The renormalised entanglement is interpreted as the additional dimension [388]. There exist other approaches that also proceed via entanglement renormalisation, such as the exact holographic mapping [312, 313] (EHM) and the unitary renormalisation group [305, 306, 394] (URG) that operate purely through unitary transformations and have been explicitly shown to lead to a holographic tensor network along the RG flow. Apart from these, there is a body of work due largely to Ki-Seok Kim that provides a geometric description of Wilson's numerical renormalisation group [86,395–397,403], and creates a more general RG framework in terms of an order-parameter field that extends the Ads/CFT correspondence to non-critical theories [398–402].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [287, 289] as well as topological features arising from the incompressibility of the Fermi volume [99, 100]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [76, 305, 306]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [77, 404–406]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

Some open questions

Chapter 2

Methods and Preliminaries

2.1 The Unitary renormalisation group method

The URG method was introduced and formalised in refs. [305,306,404,405]. This section is adapted from those references and expanded wherever required.

Description of the problem

We are given a Hamiltonian $\mathcal{H}$ which is not completely diagonal in the occupation number basis of the electrons, $\hat{n}_k$: $[\mathcal{H}, n_k] \neq 0$. k labels any set of quantum numbers depending on the system. For spin-less Fermions it can be the momentum of the particle, while for spin-full Fermions it can be the set of momentum and spin. There are terms that scatter electrons from one quantum number k to another quantum number k'.

We take a general Hamiltonian,

$$\mathcal{H} = H_e \hat{n}_{q\beta} + H_h \left(1 - \hat{n}_{q\beta}\right) + c^\dagger_{q\beta} T + T^\dagger c_{q\beta} \tag{1.1}$$

Formally, we can decompose the entire Hamiltonian in the subspace of the electron we want to decouple ($q\beta$).

$$\mathcal{H} = \begin{array}{c} \begin{array}{cc} |1\rangle & |0\rangle \end{array} \\ \begin{pmatrix} H_1 & T \\ T^\dagger & H_0 \end{pmatrix} \end{array} \tag{1.2}$$

The basis in which this matrix is written is $\{|1\rangle, |0\rangle\}$ where $|i\rangle$ is the set of all states where $\hat{n}_{q\beta} = i$. The aim of one step of the URG is to find a unitary transformation U such that the new Hamiltonian $U\mathcal{H}U^\dagger$ is diagonal in this already-chosen basis.

$$\tilde{\mathcal{H}} \equiv U\mathcal{H}U^\dagger = \begin{array}{c} \begin{array}{cc} |1\rangle & |0\rangle \end{array} \\ \begin{pmatrix} \tilde{H}_1 & 0 \\ 0 & \tilde{H}_0 \end{pmatrix} \end{array} \tag{1.3}$$

U_q is defined by

$$\tilde{\mathcal{H}} = U_q \mathcal{H} U_q^\dagger \text{ such that } \left[\tilde{\mathcal{H}}, n_q\right] = 0 \tag{1.4}$$

It is clear that U is the diagonalizing matrix for $\mathcal{H}$. Hence we can frame this problem as an eigenvalue equation as well. Let $|\psi_1\rangle, |\psi_0\rangle$ be the basis in which the original Hamiltonian $\mathcal{H}$ has no off-diagonal terms corresponding to $q\beta$. Hence, we can write

$$\mathcal{H}|\psi_i\rangle = \tilde{H}_i |\psi_i\rangle, i \in \{0, 1\} \tag{1.5}$$

Since $|\psi_i\rangle$ is the set of eigenstates of $\mathcal{H}$ and $|i\rangle$ is the set of eigenstates in which $U\mathcal{H}U^\dagger$ has no off-diagonal terms corresponding to $q\beta$, we can relate $|\psi_i\rangle$ and $|i\rangle$ by the same transformation : $|\psi_i\rangle = U^\dagger |i\rangle$. We can expand the state $|\psi_i\rangle$ in the subspace of $q\beta$:

$$|\psi_i\rangle = \sum_{j=0,1} |j\rangle \langle j| |\psi_i\rangle \equiv |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle \tag{1.6}$$

where $|\phi_j^i\rangle = \langle j| |\psi_i\rangle$. If we substitute the expansion 1.2 into the eigenvalue equation 1.5, we get

$$\left[H_e \hat{n}_{q\beta} + H_h \left(1 - \hat{n}_{q\beta}\right) + c^\dagger_{q\beta} T + T^\dagger c_{q\beta}\right] |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle \tag{1.7}$$

The diagonal parts $H_e = \text{tr}\left[\mathcal{H}\hat{n}_{q\beta}\right]$ and $H_e = \text{tr}\left[\mathcal{H}\left(1 - \hat{n}_{q\beta}\right)\right]$ can be separated into a purely diagonal part $\mathcal{H}^d$ that contains the single-particle energies and the multi-particle correlation energies or Hartree-like contributions, and an off-diagonal part $\mathcal{H}^i$ that scatters between the remaining degrees of freedom $k\sigma \neq q\beta$. That is,

$$H_e \hat{n}_{q\beta} + H_h \left(1 - \hat{n}_{q\beta}\right) = \mathcal{H}^d + \mathcal{H}^i$$

This gives

$$\left[c^\dagger_{q\beta} T + T^\dagger c_{q\beta}\right] |\psi_i\rangle = \left(\tilde{H}_i - \mathcal{H}^i - \mathcal{H}^d\right) |\psi_i\rangle \tag{1.8}$$

Obtaining the decoupling transformation

We now define a new operator $\hat{\omega}_i = \tilde{H}_i - \mathcal{H}^i$, such that

$$\left[c^\dagger_{q\beta} T + T^\dagger c_{q\beta}\right] |\psi_i\rangle = \left(\hat{\omega}_i - \mathcal{H}^d\right) |\psi_i\rangle \tag{1.9}$$

From the definition of $\hat{\omega}_i$, we can see that it is Hermitian and has no term that scatters in the subspace of $q\beta$, so it is diagonal in $q\beta$ and we can expand it as $\hat{\omega}_i = \hat{\omega}_i^1 \hat{n}_{q\beta} + \hat{\omega}_i^0 \left(1 - \hat{n}_{q\beta}\right)$. Using the expansion 1.6, we can write

$$\hat{\omega}_i |\psi_i\rangle = \hat{\omega}_i^1 |1\rangle |\phi_1^i\rangle + \hat{\omega}_i^0 |0\rangle |\phi_0^i\rangle \tag{1.10}$$

Since the only requirement on $|\psi_i\rangle$ is that it diagonalize the Hamiltonian in the subspace of $q\beta$, there is freedom in the choice of this state. We can exploit this freedom and choose the $|\phi_{0,}^i\rangle$ to be an eigenstates of $\hat{\omega}_i^{1,0}$ corresponding to real eigenvalues $\omega_i^{1,0}$:

$$\left[\mathcal{H}^d + c^\dagger_{q\beta} T + T^\dagger c_{q\beta}\right] |\psi_i(\omega_i)\rangle = \left(\omega_i^1 - \mathcal{H}^d\right) |1\rangle |\phi_1^i\rangle + \left(\omega_i^0 - \mathcal{H}^d\right) |0\rangle |\phi_0^i\rangle \tag{1.11}$$

If we now substitue the expansion 1.6 and gather the terms that result in $\hat{n}_{q\beta} = 1$, we get

$$c^\dagger_{q\beta} T \left|0\right\rangle \left|\phi^i_0\right\rangle = \left(\omega^1_i - \mathcal{H}^d\right) \left|1\right\rangle \left|\phi^i_1\right\rangle \tag{1.12}$$

Similarly, gathering the terms that result in $\hat{n}_{q\beta} = 0$ gives

$$T^\dagger c_{q\beta} \left|1\right\rangle \left|\phi^i_1\right\rangle = \left(\omega^0_i - \mathcal{H}^d\right) \left|0\right\rangle \left|\phi^i_0\right\rangle \tag{1.13}$$

We now define two many-particle transition operators:

$$\begin{aligned} \eta^\dagger(\omega^1_i) &= \frac{1}{\omega^1_i - \mathcal{H}^d} c^\dagger_{q\beta} T \equiv G_1 c^\dagger_{q\beta} T \\ \eta(\omega^0_i) &= \frac{1}{\omega^0_i - \mathcal{H}^d} T^\dagger c_{q\beta} \equiv G_0 T^\dagger c_{q\beta} \end{aligned} \tag{1.14}$$

wher G_j is the propagator $\frac{1}{\omega^j_i - \mathcal{H}^d}$. We can write this compactly as

$$\eta(\hat{\omega}) = G T^\dagger c_{q\beta} = \frac{1}{\hat{\omega}_i - \mathcal{H}^d} T^\dagger c_{q\beta} \tag{1.15}$$

where $\hat{\omega}_i = \omega^0_i \left(1 - \hat{n}_{q\beta}\right) + \omega^1_i \hat{n}_{q\beta} = \begin{pmatrix} \omega^1_i & \\ & \omega^0_i \end{pmatrix}$ is a 2x2 matrix and $\mathcal{H}^d = \mathcal{H}^d_0 \left(1 - \hat{n}_{q\beta}\right) + \mathcal{H}^d_1 \hat{n}_{q\beta}$ and $G = \left(\hat{\omega} - \mathcal{H}^d\right)^{-1}$. It is easy to check that this reproduces the previous forms of η_0 and $\eta^\dagger_1$. We will later find that it is important to demand that these two be Hermitian conjugates of each other; that constraint is imposed on the denominators:

$$\eta^\dagger(\omega^0_i) = \eta^\dagger(\omega^1_i) \implies \frac{1}{\omega^1_i - \mathcal{H}^d} c^\dagger_{q\beta} T = c^\dagger_{q\beta} T \frac{1}{\omega^0_i - \mathcal{H}^d} \tag{1.16}$$

Henceforth we will assume that this constraint has been imposed.

In terms of these operators, eq. 1.13 becomes

$$\left|1\right\rangle \left|\phi^i_1\right\rangle = \eta^\dagger \left|0\right\rangle \left|\phi^i_0\right\rangle, \quad \left|0\right\rangle \left|\phi^i_0\right\rangle = \eta \left|1\right\rangle \left|\phi^i_1\right\rangle \tag{1.17}$$

These allow us to write

$$\left|\psi_1\right\rangle = \left|1\right\rangle \left|\phi^i_1\right\rangle + \left|0\right\rangle \left|\phi^i_0\right\rangle = (1 + \eta) \left|1\right\rangle \left|\phi^i_1\right\rangle, \quad \left|\psi_0\right\rangle = \left(1 + \eta^\dagger\right) \left|0\right\rangle \left|\phi^i_0\right\rangle \tag{1.18}$$

Recalling that $\left|\psi_i\right\rangle = U^\dagger \left|i\right\rangle$, we can read off the required transformation:

$$U_1 = 1 + \eta \tag{1.19}$$

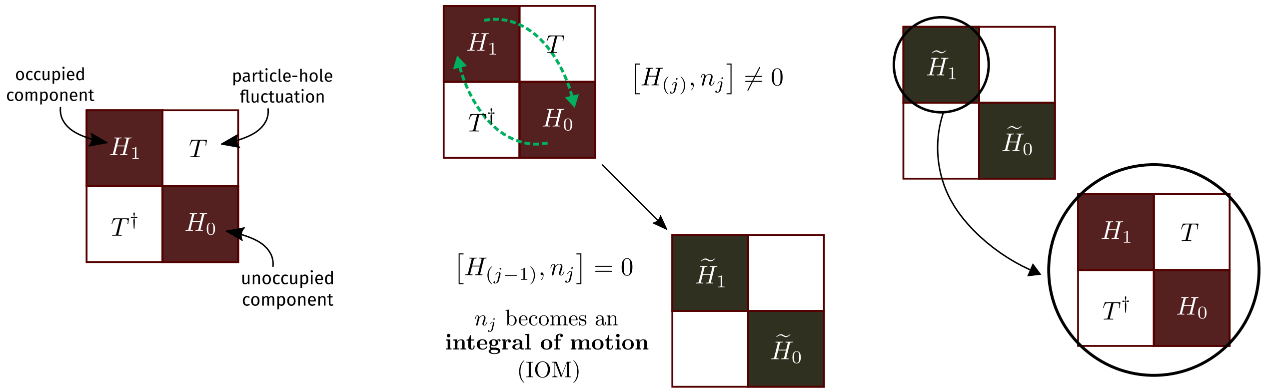

Figure 2.1: *Three steps of the URG: Decompose the Hamiltonian in a 2×2 matrix, apply the unitary operator to rotate it, then repeat these steps with one of the rotated blocks.*

Properties of the many-body transition operators

The operators η have some important properties. First is the Fermionic nature:

$$\eta^2 = {\eta^\dagger}^2 = 0 \quad \left[{c^\dagger}^2 = c^2 = 0\right] \tag{1.20}$$

Second is:

$$\begin{aligned} |1\rangle |\phi_1^i\rangle = \eta^\dagger |0\rangle |\phi_0^i\rangle = \eta^\dagger \eta |1\rangle |\phi_1^i\rangle \implies \eta^\dagger \eta = \hat{n}_{q\beta} \\ |0\rangle |\phi_0^i\rangle = \eta |1\rangle |\phi_1^i\rangle = \eta\eta^\dagger |\phi_0^i\rangle \implies \eta\eta^\dagger = 1 - \hat{n}_{q\beta} \end{aligned} \tag{1.21}$$

and hence the anticommutator

$$\implies \left\{\eta, \eta^\dagger\right\} = 1 \tag{1.22}$$

Note that the three equations in **1.21** work only when applied on the eigenstate $|\psi_i\rangle$ and not any arbitrary state.

$$\begin{aligned} \eta^\dagger \eta |\psi_i\rangle = |1\rangle |\phi_1^i\rangle = \hat{n}_{q\beta} |\psi_i\rangle \\ \eta\eta^\dagger |\psi_i\rangle = |0\rangle |\phi_0^i\rangle = \left(1 - \hat{n}_{q\beta}\right) |\psi_i\rangle \\ \left\{\eta^\dagger, \eta\right\} |\psi_i\rangle = |\psi_i\rangle \end{aligned}$$

Form of the unitary operators

Although we have found the correct similarity transformations U_i (eqs. **1.19**), we need to convert them into a unitary transformation. Say we are trying to rotate the eigenstate $|\psi_1\rangle$ into the state $|1\rangle$. We can then work with the transformation

$$U_1 = 1 + \eta \tag{1.23}$$

In this form, this transformation is not unitary. It can however be written in an exponential form:

$$U_1 = e^\eta \tag{1.24}$$

using the fact that $\eta^2 = 0$. It is shown in ref. [407] that corresponding to a similarity transformation e^{ω},there exists a unitary transformation e^G where

$$G = \tanh^{-1}\left(\omega - \omega^{\dagger}\right) \tag{1.25}$$

Applying that to the problem at hand gives

$$U_1^{\dagger} = \exp\left\{\tanh^{-1}\left(\eta - \eta^{\dagger}\right)\right\} \tag{1.26}$$

Let $x = \tanh y$. Then,

$$x = \frac{e^{2y}+1}{e^{2y}-1} \implies y = \frac{1}{2}\log\frac{1+x}{1-x} \implies e^{y} = e^{\tanh^{-1}x} = \sqrt{\frac{1+x}{1-x}} \tag{1.27}$$

Therefore,

$$\exp\left\{\tanh^{-1}\left(\eta - \eta^{\dagger}\right)\right\} = \frac{1+\eta-\eta^{\dagger}}{\sqrt{\left(1+\eta^{\dagger}-\eta\right)\left(1-\eta^{\dagger}+\eta\right)}} = \frac{1+\eta-\eta^{\dagger}}{\sqrt{1+\left\{\eta,\eta^{\dagger}\right\}}} = \frac{1}{\sqrt{2}}\left(1+\eta-\eta^{\dagger}\right) \tag{1.28}$$

The *unitary* operator that transforms the entangled eigenstate $|\psi_1\rangle$ to the state $|1\rangle$ is thus

$$U_1 = \frac{1}{\sqrt{2}}\left(1+\eta^{\dagger}-\eta\right) \tag{1.29}$$

It can also be written as $\exp\left\{\frac{\pi}{4}\left(\eta^{\dagger}-\eta\right)\right\}$ because

$$\begin{aligned}
\exp\left\{\frac{\pi}{4}\left(\eta^{\dagger}-\eta\right)\right\} &= 1 + \left(\eta^{\dagger}-\eta\right)\frac{\pi}{4} + \frac{1}{2!}\left(\eta^{\dagger}-\eta\right)^2\left(\frac{\pi}{4}\right)^2 + \frac{1}{3!}\left(\eta^{\dagger}-\eta\right)^3\left(\frac{\pi}{4}\right)^3 + \ldots \\
&= 1 + \left(\eta^{\dagger}-\eta\right)\frac{\pi}{4} - \frac{1}{2!}\left(\frac{\pi}{4}\right)^2 - \frac{1}{3!}\left(\eta^{\dagger}-\eta\right)\left(\frac{\pi}{4}\right)^3 + \frac{1}{4!}\left(\frac{\pi}{4}\right)^4 + \ldots \\
&= \cos\frac{\pi}{4} + \left(\eta^{\dagger}-\eta\right)\sin\frac{\pi}{4} \\
&= \frac{1}{\sqrt{2}}\left(1+\eta^{\dagger}-\eta\right)
\end{aligned} \tag{1.30}$$

There we used

$$\left(\eta^{\dagger}-\eta\right)^2 = \eta^{\dagger 2} + \eta^2 - \left\{\eta^{\dagger},\eta\right\} = -1 \quad \left[\because \eta^2 = \eta^{\dagger 2} = 0\right] \tag{1.31}$$

and hence

$$\left(\eta^{\dagger}-\eta\right)^3 = -1\left(\eta^{\dagger}-\eta\right) \tag{1.32}$$

and so on.

Effective Hamiltonian

We can now compute the form of the effective Hamiltonian that comes about when we apply U_1 - that is - when we rotate one exact eigenstate $|\psi_1\rangle$ into the occupied Fock space basis $|1\rangle$. From eq. 1.29,

$$\begin{aligned}
U_1\mathcal{H}U_1^\dagger &= \frac{1}{2}\left(1+\eta^\dagger-\eta\right)\mathcal{H}\left(1+\eta-\eta^\dagger\right)\\
&= \frac{1}{2}\left(1+\eta^\dagger-\eta\right)\left(\mathcal{H}+\mathcal{H}\eta-\mathcal{H}\eta^\dagger\right)\\
&= \frac{1}{2}\left(\mathcal{H}+\mathcal{H}\eta-\mathcal{H}\eta^\dagger+\eta^\dagger\mathcal{H}+\eta^\dagger\mathcal{H}\eta-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}-\eta\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger\right)\\
&= \frac{1}{2}\left(\mathcal{H}^d+\mathcal{H}^i+\mathcal{H}^I+\mathcal{H}\eta-\mathcal{H}\eta^\dagger+\eta^\dagger\mathcal{H}+\eta^\dagger\mathcal{H}\eta-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}-\eta\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger\right)\\
&= \frac{1}{2}\left(\mathcal{H}^d+\mathcal{H}^i+\mathcal{H}^I+\left[\eta^\dagger-\eta,\mathcal{H}\right]+\eta^\dagger\mathcal{H}\eta-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger\right)
\end{aligned} \tag{1.33}$$

In the last two lines, we expanded the Hamiltonian into the three parts $\mathcal{H}^d, \mathcal{H}^i$ and a third piece $\mathcal{H}^I \equiv c_{q\beta}^\dagger T + T^\dagger c_{q\beta}$.

For reasons that will become apparent, we will split the terms into two groups:

$$\tilde{\mathcal{H}} = \frac{1}{2}\left(\underbrace{\mathcal{H}^d+\mathcal{H}^i+\left[\eta^\dagger-\eta,\mathcal{H}\right]+\eta^\dagger\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger}_{\text{group 1}}+\overbrace{\mathcal{H}^I-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}\eta}^{\text{group 2}}\right) \tag{1.34}$$

Group 2 can be easily shown to be 0. Note that terms that have two η or two $\eta^\dagger$ sandwiching a $\mathcal{H}$ can only be nonzero if the intervening $\mathcal{H}$ has an odd number of creation or destruction operators.

$$\eta\mathcal{H}\eta = \eta c_q^\dagger T\eta \tag{1.35}$$

and

$$\eta^\dagger\mathcal{H}\eta^\dagger = \eta^\dagger T^\dagger c_q\eta^\dagger \tag{1.36}$$

Group 2 becomes

$$\text{group 2} = \mathcal{H}^I - \eta^\dagger T^\dagger c_q\eta^\dagger - \eta c_q^\dagger T\eta = c_q^\dagger T + T^\dagger c_q - \eta^\dagger T^\dagger c_q\eta^\dagger - \eta c_q^\dagger T\eta \tag{1.37}$$

To simplify this, we use the relation

$$\begin{aligned}
\eta c_q^\dagger T\eta &= \frac{1}{\omega_i^0-\mathcal{H}^d}T^\dagger c_q c_q^\dagger T\eta\\
&= T^\dagger c_q\frac{1}{\omega_i^1-\mathcal{H}^d}c_q^\dagger T\eta \quad &&\text{[eq. 1.16]}\\
&= T^\dagger c_q\eta^\dagger\eta &&\text{[eq. 1.15]}\\
&= T^\dagger c_q\hat{n}_q &&\text{[eq. 1.21]}
\end{aligned} \tag{1.38}$$

which gives

$$\eta c_q^\dagger T\eta = T^\dagger c_q \tag{1.39}$$

Taking the Hermitian conjugate of eq. 1.39 gives

$$\eta^\dagger T^\dagger c_q \eta^\dagger = c_q^\dagger T \tag{1.40}$$

Substituting the expressions 1.39 and 1.40 into the expression for group 2, 1.37, shows that is vanishes. This leaves us only with group 1:

$$\tilde{\mathcal{H}} = \frac{1}{2}\left(\mathcal{H}^d + \mathcal{H}^i + \overbrace{\eta^\dagger\mathcal{H}\eta + \eta\mathcal{H}\eta^\dagger}^{\text{group A}} + \underbrace{\left[\eta^\dagger - \eta, \mathcal{H}\right]}_{groupB}\right) \tag{1.41}$$

Group A simplifies in the following way. First note that $\eta^\dagger\mathcal{H}^I\eta = \eta^\dagger\mathcal{H}^I\eta = 0$ must be 0 because it will involve consecutive $c_{q\beta}$ or consecutive $c^\dagger_{q\beta}$. We are therefore left with the diagonal part of $\mathcal{H}$, which is $H_e\hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right)$.

$$\eta^\dagger\left[H_e\hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right)\right]\eta + \eta\left[H_e\hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right)\right]\eta^\dagger = \eta^\dagger H_h\eta + \eta H_e\eta^\dagger \tag{1.42}$$

This can be shown to be equal to the diagonal part:

$$\text{group A} = \eta^\dagger H_h\eta + \eta H_e\eta^\dagger = H_e\hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right) = \mathcal{H}^d + \mathcal{H}^i \tag{1.43}$$

It can also be shown that

$$\text{group B} = \left[\eta^\dagger - \eta, \mathcal{H}\right] = 2\left[c^\dagger_{q\beta}T, \eta\right] \tag{1.44}$$

Putting it all together,

$$\tilde{\mathcal{H}} = \mathcal{H}^d + \mathcal{H}^i + \left[c^\dagger_{q\beta}T, \eta\right] \tag{1.45}$$

The renormalizing in the Hamiltonian is

$$\Delta\mathcal{H} = \tilde{\mathcal{H}} - \mathcal{H}^d - \mathcal{H}^i = \left[c^\dagger_{q\beta}T, \eta\right] \tag{1.46}$$

Because of eq. 1.44, it can also be written as

$$\Delta\mathcal{H} = \frac{1}{2}\left[\eta^\dagger - \eta, \mathcal{H}_X\right] = \frac{1}{2}\left[\eta^\dagger - \eta, \mathcal{H}\right] \tag{1.47}$$

This form will be useful later when we make the connection with one-shot Schrieffer-Wolff transformation and CUT RG.

To check that the renormalised Hamiltonian indeed commutes with $\hat{n}_{q\beta}$,

$$\begin{aligned}
\left[\tilde{\mathcal{H}}, \hat{n}_{q\beta}\right] &= \left[\left[c^\dagger_{q\beta}T, \eta\right], \hat{n}_{q\beta}\right] = \left[c^\dagger_{q\beta}T\eta, \hat{n}_{q\beta}\right] - \left[\eta c^\dagger_{q\beta}T, \hat{n}_{q\beta}\right] \\
&= c^\dagger_{q\beta}T\eta\hat{n}_{q\beta} - \hat{n}_{q\beta}c^\dagger_{q\beta}T\eta && \left[2^{\text{nd}}\ [\ .\ ]\text{ is 0, } \because c^\dagger_{q\beta}\hat{n}_{q\beta} = \hat{n}_{q\beta}\eta = 0\right] \\
&= c^\dagger_{q\beta}T\eta - c^\dagger_{q\beta}T\eta = 0
\end{aligned} \tag{1.48}$$

Fixed point condition

Within the URG, it is a prescription that the fixed point is reached when the denominator of the RG equation vanishes. This is equivalent to either $\omega_i^1 = \mathcal{H}_1^d$ or $\omega_i^0 = \mathcal{H}_0^d$. This shows that at the fixed point, one of the eigenvalues of $\hat{\omega}_i$ matches the corresponding eigenvalue of the diagonal blocks. This also leads to the vanishing of the off-diagonal block, because eqs. 1.12 and 1.13 gives

$$c_{q\beta}^\dagger T \left|0\right\rangle \left|\phi_0^i\right\rangle = \left(\omega_i^1 - \mathcal{H}_1^d\right) \left|1\right\rangle \left|\phi_1^i\right\rangle = 0 \implies c_{q\beta}^\dagger T = 0 \tag{1.49}$$

Multiple off-diagonal terms

There is a subtle assumption in the definitions eq. 1.14. In order for η to be the Hermitian conjugate of $\eta^\dagger$, $\mathcal{H}_d$ cannot have any information that relates to the structure of T. To see why, say the total off-diagonal term is composed of two parts: $T = T_1 + T_2$.

$$\begin{aligned} \eta &= \frac{1}{\omega_0 - \mathcal{H}_d}\left(T_1^\dagger + T_2^\dagger\right) c = \left[\frac{1}{\omega^0 - E_1^0} T_1^\dagger c + \frac{1}{\omega^0 - E_2^0} T_2^\dagger c\right] \\ \eta^\dagger &= \frac{1}{\omega^1 - \mathcal{H}_d} c^\dagger \left(T_1 + T_2\right) = \left[\frac{1}{\omega^1 - E_1^1} c^\dagger T_1 + \frac{1}{\omega^1 - E_2^1} c^\dagger T_2\right] \end{aligned} \tag{1.50}$$

where $\mathcal{H}_d T_i^\dagger c = E_i^0 T_i^\dagger c$ and $\mathcal{H}_d c^\dagger T_i = E_i^1 c^\dagger T_i$. We can now see that in order for $\eta = \left(\eta^\dagger\right)^\dagger$ to hold, two conditions must be met:

$$\omega^0 - E_1^0 = \omega^1 - E_1^1, \quad \omega^0 - E_2^0 = \omega^1 - E_2^1 \tag{1.51}$$

This will not hold generally. The correct solution is to realize that each such off-diagonal term T_i will come with its own quantum fluctuation scale ω_i.

$$\begin{aligned} \eta &= \sum_i \frac{1}{\omega_i^0 - E_i^0} T_i^\dagger c \\ \eta^\dagger &= \sum_i \frac{1}{\omega_i^1 - E_1^1} c^\dagger T_i \end{aligned} \tag{1.52}$$

If we now impose the condition that $\eta = \left(\eta^\dagger\right)^\dagger$, we get the relations

$$\omega_i^0 - \omega_i^1 = E_i^0 - E_1^1 \tag{1.53}$$

and so

$$\eta^\dagger - \eta = \sum_i \frac{1}{\omega_i^0 - E_i^0}\left(c^\dagger T_i - T_i^\dagger c\right) \tag{1.54}$$

The expression for the renormalization will not be just $\left[c^\dagger T, \eta\right]$ in this case. That form will be non-Hermitian. The correct form is obtained from the more general form $\left[\eta^\dagger - \eta, \mathcal{H}_X\right]$:

$$\begin{aligned}\Delta\mathcal{H} &= \frac{1}{2}\left[\eta^\dagger - \eta, c^\dagger T + T^\dagger c\right] = \frac{1}{2}\sum_{ij}\frac{1}{\omega_i^0 - E_i^0}\left[c^\dagger T_i - T_i^\dagger c, c^\dagger T_j + T_j^\dagger c\right] \\ &= \frac{1}{2}\sum_{ij}\frac{1}{\omega_i^0 - E_i^0}\left[\hat{n}\left(T_i T_j^\dagger + T_j T_i^\dagger\right) - (1-\hat{n})\left(T_i^\dagger T_j + T_j^\dagger T_i\right)\right] \\ &= \frac{1}{2}\sum_{ij}\left(\frac{1}{\omega_i^0 - E_i^0} + \frac{1}{\omega_j^0 - E_j^0}\right)\left[\hat{n} T_i T_j^\dagger - (1-\hat{n}) T_i^\dagger T_j\right]\end{aligned} \tag{1.55}$$

Prescription

Given a Hamiltonian

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_0 + c^\dagger T + T^\dagger c \tag{1.56}$$

the goal is to look at the renormalization of the various couplings in the Hamiltonian as we decouple high energy electron states. Typically we have a shell of electrons at some energy D. During the process, we make one simplification. We assume that there is only one electron on that shell at a time, say with quantum numbers q, σ, and calculate the renormalization of the various couplings due to this electron. We then sum the momentum q over the shell and the spin β, and this gives the total renormalization due to decoupling the entire shell.

From eq. 1.45, the first two terms in the rotated Hamiltonian are just the diagonal parts of the bare Hamiltonian; they are unchanged in that part. The renormalization comes from the third term. For one electron $q\beta$ on the shell, the renormalization is

$$\Delta\mathcal{H} = \left[c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right), \eta\right] = c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right)\eta - \eta c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right) \tag{1.57}$$

Since this assumes we have obtained this from U_1, it is fair to tag the η with a suitable label:

$$\Delta\mathcal{H} = c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right)\eta_1 - \eta_1 c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right) \tag{1.58}$$

It is clear that the first term takes into account virtual excitations that start from a filled state ($\hat{n}_{q\beta} = 1$ initially) - such a term is said to be a part of the *particle sector*.

$$\Delta_1\mathcal{H} = c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right)\eta_1 \tag{1.59}$$

The second term, on the other hand, considers excitations that start from an empty state. They constitute the *hole sector*.

$$\Delta_0\mathcal{H} = -\eta_1 c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right) \tag{1.60}$$

To write the total renormalization in a particle-hole symmetric form, we can use the relation $\eta_0 = -\eta_1$, such that both the terms will now come with a positive sign:

$$\Delta\mathcal{H} = c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right)\eta_1 + \eta_0 c^\dagger_{q\beta}\text{Tr}\left(\mathcal{H}c_{q\beta}\right) \tag{1.61}$$

We can make one more manipulation: using eq. 1.16, we get

$$\Delta\mathcal{H} = c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\eta_1 + \mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}\eta^\dagger_0 \tag{1.62}$$

This form of the total renormalization is identical to the one we use in the "Poor Man's scaling"-type of renormalization that was used to get the scaling equations in the Kondo and Anderson models [408, 409]. Writing down the forms of η and $\eta^\dagger$ explicitly, we get

$$\Delta\mathcal{H} = c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\frac{1}{\omega^0_1 - \mathcal{H}^d_0}\mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta} + \mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}\frac{1}{\omega^1_0 - \mathcal{H}^d_1}c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right) \tag{1.63}$$

The renormalization due to the entire shell is obtained by summing over all states on the shell.

$$\Delta\mathcal{H} = \sum_{q\beta}\left[c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\frac{1}{\omega^0_1 - \mathcal{H}^d_0}\mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta} + \mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}\frac{1}{\omega^1_0 - \mathcal{H}^d_1}c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\right] \tag{1.64}$$

These equations will now need to be simplified. For example, in the particle sector, we can set $\hat{n}_{q\beta} = 0$ in the numerator, because there is no such excitation in the initial state. Similarly, in the hole sector, we can set $\hat{n}_{q\beta} = 1$ because that state was occupied in the initial state. Another simplification we typically employ is that $\mathcal{H}^d_{0,1}$ will, in general, have the energies of all the electrons. But we consider only the energy of the on-shell electrons in the denominator. After integrating out these electrons, we can rearrange the remaining operators to determine which term in the Hamiltonian it renormalizes and what is the renormalization.

At first sight, one might think that we must evaluate lots of traces to obtain the terms in $\Delta\mathcal{H}$. A little thought reveals that the terms in the numerator are simply the off-diagonal terms in the Hamiltonian; $\mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}$ is the off-diagonal term that has $c_{q\beta}$ in it, and $c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)$ is the off-diagonal term that has $c^\dagger_{q\beta}$ in it. $\mathcal{H}^D$ is just the diagonal part of the Hamiltonian.

2.2 Example I: URG analysis of the star graph model

The star graph problem has already been analyzed using URG and an extensive study of its entanglement properties has already been carried out, in ref. [410]. Here we focus on just deriving the RG equations. The system consists of N spin-like degrees of freedom (labeled 1 through N) individually talking to a spin at the center (labeled 0). Each spin i ($\in [0, N]$) has an on-site energy ϵ_i. The coupling strength between 0 and i ($\in [1, N]$) is J_i. We choose the on-site energies such that $\epsilon_{i+1} > \epsilon_i, i \in [N-1, 1]$. In this way, ϵ_1 is the infrared limit and ϵ_N is the ultraviolet limit.

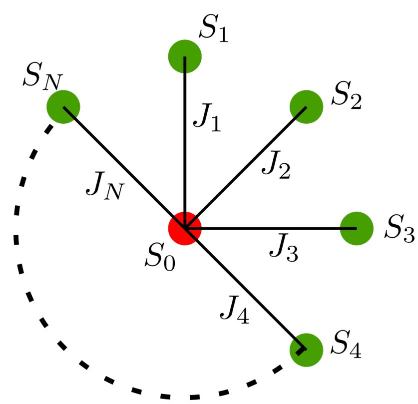

Figure 2.2: *Star Graph model*

$$\mathcal{H} = \epsilon_0 S_0^z + \sum_{i=1}^{N} \left[\epsilon_i S_i^z + J_i \vec{S}_0 \cdot \vec{S}_i\right] \tag{2.1}$$

By converting the last term into S^z and $S^\pm$, we can write the Hamiltonian as

$$\mathcal{H} = \epsilon_0 S_0^z + \sum_{i=1}^{N} \left[\epsilon_i S_i^z + J_i \left(S_0^z S_i^z + \frac{1}{2}\left(S_0^+ S_i^- + S_0^- S_i^+\right)\right)\right] \tag{2.2}$$

Calculation of Renormalization

The RG involves decoupling the nodes N through 1, and looking at the resultant renormalization in ϵ_i and J_i. As a simplification, we will ignore the lower nodes in the denominator and keep only the node currently being decoupled, ie node N. Since node 0 is connected to node N, we will keep node 0 in the denominator as well. Making this simplification gives

$$\mathcal{H}^D = \epsilon_0 S_0^z + \epsilon_N S_N^z + J_N S_0^z S_N^z \tag{2.3}$$

The off-diagonal part in the subspace of the node N is

$$\mathcal{H}_X = S_N^+ T + T^\dagger S_N^- = \frac{1}{2} J_N \left(S_N^+ S_0^- + S_N^- S_0^+\right) \tag{2.4}$$

The renormalization on doing one step of the URG is given by

$$\begin{aligned}\Delta\mathcal{H} &= S_N^+ T \frac{1}{\omega_0^1 - \mathcal{H}^D} T^\dagger S_N^- + T^\dagger S_N^- \frac{1}{\omega_1^0 - \mathcal{H}^D} S_N^+ T \\ &= \frac{J_N^2}{4} S_N^+ S_0^- \frac{1}{\omega_0^1 - \epsilon_0 S_0^z - \epsilon_N S_N^z - J_N S_0^z S_N^z} S_0^+ S_N^- + \frac{J_N^2}{4} S_0^+ S_N^- \frac{1}{\omega_1^0 - \epsilon_0 S_0^z - \epsilon_N S_N^z - J_N S_0^z S_N^z} S_N^+ S_0^-\end{aligned} \tag{2.5}$$

There, N refers to the spin being decoupled. The first Greens function has S_0^+ and S_N^- in front of it, so we substitute $S_0^z = \frac{1}{2}, S_N^z = -\frac{1}{2}$ in that Greens function. For the other Greens function we do the opposite.

$$\Delta\mathcal{H} = \frac{J_N^2}{4} S_N^+ S_0^- \frac{1}{\omega_1^0 - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} S_0^+ S_N^- + \frac{J_N^2}{4} S_0^+ S_N^- \frac{1}{\omega_0^1 + \frac{1}{2}\epsilon_0 - \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} S_N^+ S_0^- \quad (2.6)$$

To relate ω_1^0 and ω_0^1, we use eq. **??**:

$$\omega_1^0 + \omega_0^1 = \mathcal{H}_0^D + \mathcal{H}_1^D = -\frac{1}{2}J_N \implies \omega_1^0 \equiv \omega, \;\; \omega_0^1 \equiv \omega' = -\frac{1}{2}J_N - \omega \quad (2.7)$$

So, the renormalization becomes

$$\Delta\mathcal{H} = \frac{J_N^2}{4} \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \left(S_N^+ S_0^- S_0^+ S_N^- - S_0^+ S_N^- S_N^+ S_0^-\right) = \frac{J_N^2}{4} \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \left(S_N^z - S_0^z\right) \quad (2.8)$$

There we used $S^+S^- = \frac{1}{2} + S^z$ and $S^-S^+ = \frac{1}{2} - S^z$.

We can now read off the renormalizations in ϵ_N and ϵ_0.

$$\begin{aligned} \Delta\epsilon_N &= \frac{1}{4}J_N^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \\ \Delta\epsilon_0 &= -\frac{1}{4}J_N^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon_N + \frac{1}{4}J_N} \end{aligned} \quad (2.9)$$

Nature of flows

We are interested in looking at the renormalization of the central node energy ϵ_0, upon removing the nodes N through 1. We will hence concentrate on the second RG equation. We first make some simplifying assumptions: $J_i = J, \epsilon_i = \epsilon$ for all $i \in \{1, N\}$.

$$\Delta\epsilon_0 = -\frac{1}{4}J^2 \frac{1}{\omega - \frac{1}{2}\epsilon_0 + \frac{1}{2}\epsilon + \frac{1}{4}J} \quad (2.10)$$

Define $\tilde{\omega} = \omega + \frac{1}{2}\epsilon + \frac{1}{4}J$.

$$\Delta\epsilon_0 = -\frac{1}{4}J^2 \frac{1}{\tilde{\omega} - \frac{1}{2}\epsilon_0} \quad (2.11)$$

Our goal here is to look for a fixed-point condition such that the denominator vanishes at some point of the RG. If we start with a bare of ϵ_0 such that $\tilde{\omega} - \frac{1}{2}\epsilon_0 > 0$, the denominator will be positive and the RG equation will be irrelevant. This means that ϵ_0 will keep on decreasing, and the denominator will keep on becoming more and more positive, meaning there cannot be a fixed point in this situation.

If, on other hand, we start with a bare of ϵ_0 such that $\tilde{\omega} - \frac{1}{2}\epsilon_0 < 0$, the denominator will be negative and the RG equation will be relevant. This means that ϵ_0 will keep on increasing, and the

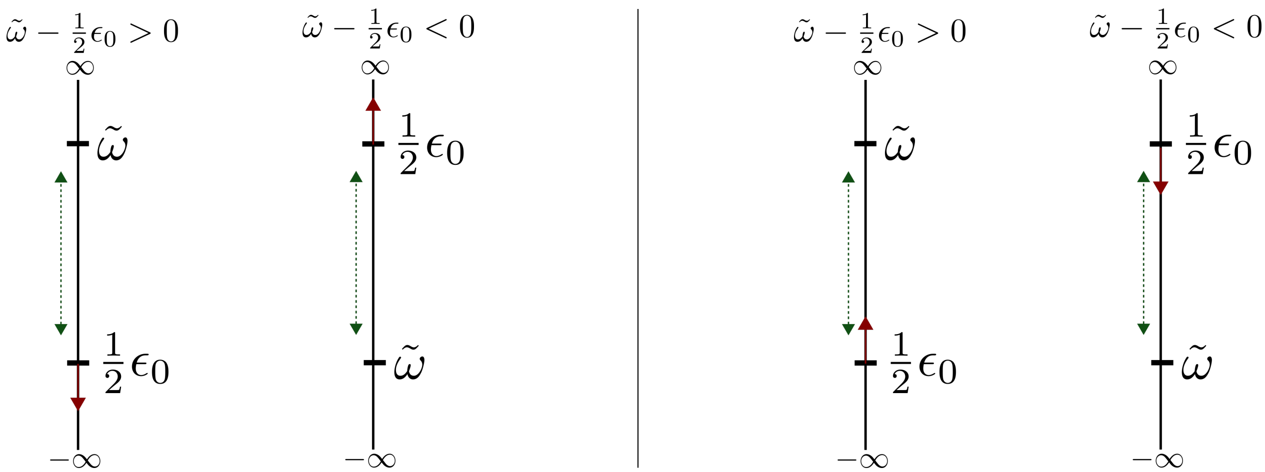

Figure 2.3: *Left: RG flow for the two cases. The green line is the distance between the bare values of the two couplings, and hence also the magnitude of the denominator. The red arrow denotes the direction in which ϵ_0 will flow. Upward flow is increase. In both cases, the flow is such that the distance between the two quantities (and hence the magnitude of the denominator) increases. The RG fixed point occurs when the magnitude of the denominator goes to 0. This happens if the distance vanishes. Since the distance necessarily increases, we cannot get a fixed point in this way. Right: RG flow for the two cases with the new $-\tilde{\omega} = \omega' - \frac{1}{2}\epsilon + \frac{1}{4}J$. Now we can see that in both cases, the flow is such that the distance (green dotted line) between the couplings decreases. A fixed point is reached when this distance vanishes.*

denominator will keep on becoming more and more negative, meaning there cannot be a fixed point in this situation either. These situations are depicted in figure 2.3.

Since we cannot find a fixed point, we will use a different ω. Instead of ω_1^0, we will use ω_1^1. From eq. **??**, we have

$$\omega_1^0 - \omega_1^1 = \mathcal{H}_0^D - \mathcal{H}_1^D = \epsilon_0 - \epsilon_N = \epsilon_0 - \epsilon \tag{2.12}$$

Defining $\omega_1^1 = \omega'$ and substituting this in eq. **2.10** gives

$$\Delta\epsilon_0 = -\frac{1}{4}J^2\frac{1}{\omega' - \frac{1}{2}\epsilon + \frac{1}{2}\epsilon_0 + \frac{1}{4}J} \tag{2.13}$$

We again define $-\tilde{\omega} = \omega' - \frac{1}{2}\epsilon + \frac{1}{4}J$.

$$\Delta\epsilon_0 = \frac{1}{4}J^2\frac{1}{\tilde{\omega} - \frac{1}{2}\epsilon_0} \tag{2.14}$$

We now repeat the exercise of determining the relevance of the flows under various regime. If we start with a bare ϵ_0 such that $\tilde{\omega} + \frac{1}{2}\epsilon_0 > 0$, then the denominator is positive so the renormalization will be irrelevant. ϵ_0 will decrease until we reach $\tilde{\omega}+\frac{1}{2}\epsilon_0 = 0$. This will be a fixed point. However, if we start with a bare ϵ_0 such that $\tilde{\omega}+\frac{1}{2}\epsilon_0 < 0$, then the denominator is negative so the renormalization will be relevant. ϵ_0 will increase until we reach $\tilde{\omega} + \frac{1}{2}\epsilon_0 = 0$. This will again be a fixed point. This new situation is depicted in right panel of figure 2.3.

Effective Hamiltonians

If $\tilde{\omega}$ and ϵ_0 are of the same sign at the bare level, then it is easy to see that since the fixed point is defined by $\tilde{\omega} = \frac{1}{2}\epsilon_0^*$ (* denotes value at fixed point), the effective Hamiltonian at the fixed point will be

$$\mathcal{H}^* = 2\tilde{\omega}S_0^z + \epsilon\sum_i S_i^z + J\sum_i \vec{S}_i\cdot\vec{S}_0, \quad \text{if } \tilde{\omega}\epsilon_0 > 0 \tag{2.15}$$

If, at the bare level, ϵ_0 and $\tilde{\omega}$ are of opposite signs, then ϵ_0 would undergo a change in sign at some point as it flows towards $\tilde{\omega}$. Since we do not expect a coupling to change sign under RG, we will restrict it to 0 in such cases.

$$\mathcal{H}^* = \epsilon\sum_i S_i^z + J\sum_i \vec{S}_i\cdot\vec{S}_0, \quad \text{if } \tilde{\omega}\epsilon_0 < 0 \tag{2.16}$$

Things get much more simpler if we assume the onsite energies of the surrounding nodes are zero.

$$\begin{aligned} \mathcal{H}^* &= 2\tilde{\omega}S_0^z + J\sum_i \vec{S}_i\cdot\vec{S}_0, && \text{if } \tilde{\omega}\epsilon_0 > 0 \\ \mathcal{H}^* &= J\sum_i \vec{S}_i\cdot\vec{S}_0, && \text{if } \tilde{\omega}\epsilon_0 < 0 \end{aligned} \tag{2.17}$$

Fixed points

The fixed points are obtained numerically by solving the RG equation. As mentioned before, there are two types of solutions: The first kind is those in which ϵ_0 and $\tilde{\omega}$ are of the same sign, and the former flows to the latter without crossing the 0 axis. These flows are shown (obtained numerically) in fig. 2.4. The second kind are those where the two couplings have different signs, and so ϵ_0 flows to 0. These are shown in fig. 2.5.

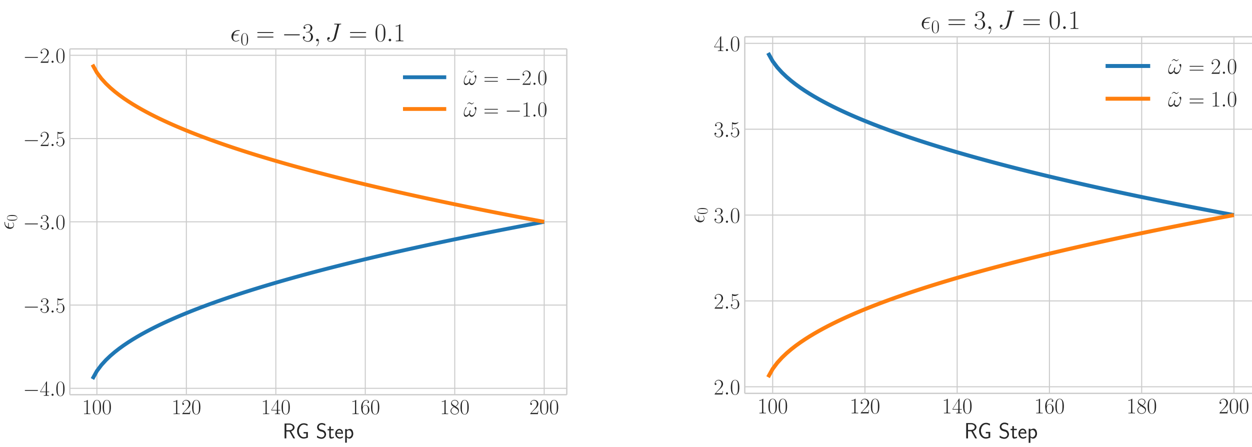

Figure 2.4: *Flows where ϵ_0 and $\tilde{\omega}$ have same sign. The left and right panels show flows starting from negative and positive values respectively. The two plots in each panel correspond to different values of $\tilde{\omega}$, one greater than the bare ϵ_0, the other less than that. The fixed point value is $2\tilde{\omega}$.*

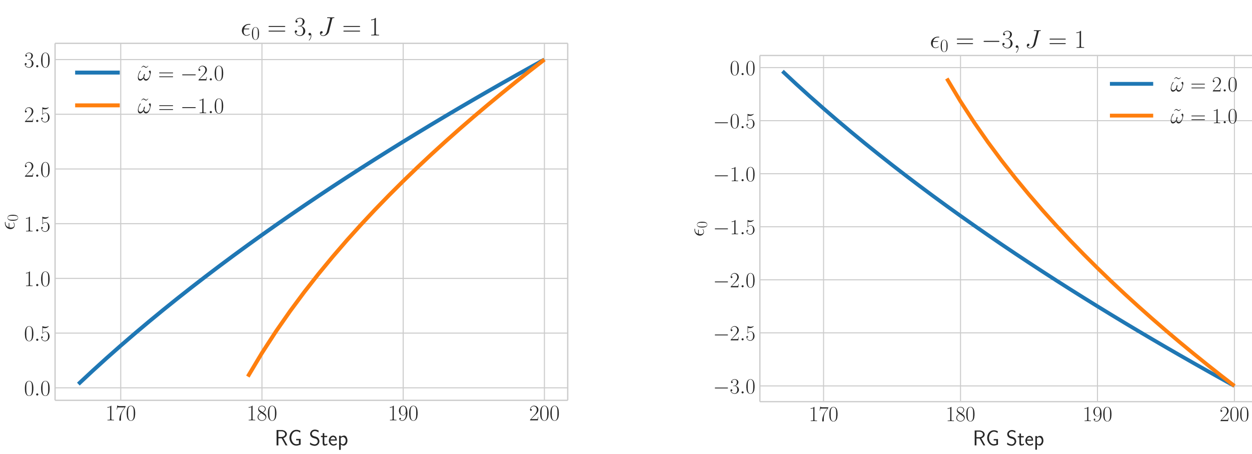

Figure 2.5: *Flows where ϵ_0 and $\tilde{\omega}$ have opposite sign. The left and right panels show flows starting from negative and positive values respectively. The two plots in each panel correspond to different values of $\tilde{\omega}$, one greater than the bare ϵ_0, the other less than that. The fixed point value is 0.*

2.3 Example II: URG analysis of the two-channel Kondo problem

The Hamiltonian for the channel-isotropic MCK model is:

$$H = \sum_l \left[\sum_{\substack{k \\ \alpha=\uparrow,\downarrow}} \epsilon_{k,l} \hat{n}_{k\alpha,l} + \frac{\mathcal{J}}{2} \sum_{\substack{kk' \\ \alpha,\alpha'=\uparrow,\downarrow}} \vec{S}_d \cdot \vec{\sigma}_{\alpha\alpha'} c^\dagger_{k\alpha,l} c_{k'\alpha',l} \right] . \tag{3.1}$$

The URG equation for the single-channel Kondo model [?] shows a stable strong coupling fixed point. Ferromagnetic interactions are irrelevant. Strictly speaking, that RG equation already encodes, in principle, the multi-channel behaviour, through a modified $\hat{\omega}$. To extract this information, we consider the strong coupling fixed-point $J \gg D$ as a fixed point and analyze its stability from the star graph perspective. For the exactly-screened case, the star graph decouples from the conduction bath, leaving behind a local Fermi liquid interaction on the first site. Similarly, in the under-screened regime, the ground state is composed of states where the impurity spin is only partially screened by the conduction channels. If a particular configuration of the bath-impurity system has the total conduction bath spin down, the impurity will have a residual up spin. This induces a ferromagnetic super-exchange coupling that is irrelevant under RG, so this fixed point is stable as well.

We now come to the over-screened case, where there is a residual spin on the conduction channel site. The neighbouring electrons will now hop in with spins opposite to that of the impurity, so an antiferromagnetic interaction will be induced, and such an interaction is relevant under the RG. This shows that the over-screened regime cannot have a stable strong coupling fixed point, and we need to search for an intermediate coupling fixed point. We therefore need the generator of the unitary transformation that incorporates third order scattering scatterings explicitly. We should take account of all possible processes that render the set of states $\left\{ |\hat{n}_{q\beta} = 1\rangle, |\hat{n}_{q\beta} = 0\rangle \right\}$ diagonal. The higher order generator itself has two scattering processes, such that the entire renormalisation term $c^\dagger_{q\beta} T \eta$ has in total three coherent processes. The complete generator up to third order can be written as

$$\eta = \frac{1}{\hat{\omega} - H_D} T^\dagger c \simeq \frac{1}{\omega' - H_D} T^\dagger c + \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X , \tag{3.2}$$

where $H_X = J \sum_{k,k'<\Lambda_j,\alpha,\alpha'} \vec{S}_d \cdot \vec{s}_{\alpha\alpha'} c^\dagger_{k\alpha} c_{k'\alpha'}$ is scattering between the entangled electrons. There are two third order terms in the above equation corresponding to the two possible sequences in which the processes can occur while keeping the total renormalisation $c^\dagger_{q\beta} T \eta$ diagonal in $q\beta$. The second order processes remain unchanged. The total renormalisation takes the form:

$$\begin{aligned} \Delta H_{(j)} = & \underbrace{c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c + \left(c^\dagger \leftrightarrow c\right)}_{\Delta H^{(2)}_{(j)}} \\ & \underbrace{+ c^\dagger T \frac{1}{\omega' - H_D} H_X \frac{1}{\omega' - H_D} T^\dagger c + c^\dagger T \frac{1}{\omega' - H_D} T^\dagger c \frac{1}{\omega' - H_D} H_X + \left(c^\dagger \leftrightarrow c\right)}_{\Delta H^{(3)}_{(j)}} . \end{aligned} \tag{3.3}$$

$\Delta H^{(2)}_{(j)}$ and $\Delta H^{(3)}_{(j)}$ are the renormalisation arising from the second and third order processes respectively.

It is easier to see the RG flow of the couplings if we write the Hamiltonian in terms of the eigenstates of S_d^z. These eigenstates are defined by $S_d^z \left|m_d\right> = m_d \left|m_d\right>, m_d \in [-S, S]$. In terms of these eigenstates, the Hamiltonian becomes

$$\mathcal{H} = \sum_{k\sigma} \epsilon_k \tau_{k\sigma} + \sum_{m_d=-S}^{S} \sum_{\substack{kl,\\ \sigma=\uparrow,\downarrow}} J^\sigma_{m_d} \left|m_d\right>\left<m_d\right| c^\dagger_{k\sigma} c_{l\sigma} + \sum_{kl} \sum_{m_d=-S}^{S-1} J^t_{m_d} \left(\left|m_d+1\right>\left<m_d\right| s^-_{kl} + \text{h.c.}\right) \quad (3.4)$$

where k, l sum over the momentum states, σ sums over the spin indices, $J^\sigma_m = \frac{1}{2}\sigma m J$ in the UV Hamiltonian, and $J^t_m = J\frac{1}{2}\sqrt{S(S+1) - m(m+1)}$ is the coupling that connects $\left|m\right>$ and $\left|m+1\right>$. We first calculate $\Delta H^{(2)}_{(j)}$. There will be two types of processes - those processes that start from an occupied state (particle sector) and those that start from a vacant state (hole sector). Due to particle hole symmetry of the Hamiltonian, they will be equal to each other and we will only calculate the particle sector contribution.

In the particle sector, we have $(\hat{n}_{q\beta} = 1)$, so we will work at a negative energy shell $\epsilon_q = -D$. The renormalisation can schematically represented as $H^I_0 \frac{1}{\omega - H^D_{q\beta}} H^I_1$. Both H^I_0 and H^I_1 have all three operators $S^z_d, S^\pm_d$. We first consider specifically the case of spin-$\frac{1}{2}$ impurity. Those terms that have identical operators on both sides can be ignored because ${S^z_d}^2$ = constant and $S^{\pm 2} = 0$. All the six terms that *will* renormalise the Hamiltonian have a spin flip operator on at least one side of the Greens function. This means that in the denominator of the Greens function, S^z_d and s^z_{qq} have to be anti-parallel in order to produce a non-zero result for that term. This means we can identically replace $S^z_d s^z_{qq} = -\frac{1}{4}$. Also, in the particle sector, the Greens function always has $c_{q\beta}$ in front of it, so $\epsilon_q \tau_{q\beta} = \frac{D}{2}$. The upshot of all this is that the denominator of all scattering processes for the spin-$\frac{1}{2}$ impurity Hamiltonian will be $\omega - \frac{D}{2} + \frac{J}{4}$.

We now come to the general case of spin-S impurity. The various terms that renormalise the Hamiltonian can be described in terms of the bath spin operators that come into them. For example, the term that has s^z on both sides of the intervening Greens function can be represented as $z|z$. There are 7 such terms: $z|z, \pm|\mp, z|\pm, \pm|z$. Each of these terms occurs both in the particle and the hole sectors. We will demonstrate the calculation of two of these terms. The $z|z$ $+|-$ terms evaluate in the following manner.

$$\begin{aligned} z|z : \sum_{kk',m,\sigma} c^\dagger_{q\sigma} c_{k'\sigma} \left|m\right>\left<m\right| \frac{J^{\sigma 2}_m}{\omega - \frac{D}{2} + \frac{J}{2}\sigma S^z_d} \left|m\right>\left<m\right| c^\dagger_{k\sigma} c_{q\sigma} &= -\sum_{kk',m,\sigma} n_{q\sigma} \frac{J^{\sigma 2}_m c^\dagger_{k\sigma} c_{k'\sigma} \left|m\right>\left<m\right|}{\omega_{m,\sigma} - \frac{D}{2} + \frac{J}{2}\sigma m}, \\ +|- : \sum_{kk',m} c^\dagger_{q\uparrow} c_{k'\downarrow} \left|m\right>\left<m+1\right| \frac{{J^t_m}^2}{\omega - \frac{D}{2} + \frac{J}{2} S^z_d} \left|m+1\right>\left<m\right| c^\dagger_{k\downarrow} c_{q\uparrow} &= -n_{q\uparrow} \sum_{kk',m} \frac{{J^t_m}^2 c^\dagger_{k\downarrow} c_{k'\downarrow} \left|m\right>\left<m\right|}{\omega_{m+1,\uparrow} - \frac{D}{2} + \frac{J}{2}(m+1)}. \end{aligned} \quad (3.5)$$

We similarly compute the rest of the terms. We again define $\sum_q \hat{n}_{q\sigma} = n(D)$. To compare with the spin-$\frac{1}{2}$ RG equations, we will transform the general spin-S ω to the spin-$\frac{1}{2}$ ω, using $\omega_{m,\sigma} \to$

$\omega - \frac{J}{2}\left(m\sigma - \frac{1}{2}\right)$. The renormalisation in J_m^σ is

$$\Delta J_m^\sigma = -n(D)\frac{\left(J_m^\sigma\right)^2 + \left(J^t_{m-\frac{1+\sigma}{2}}\right)^2}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.6}$$

Here, we have defined $J_m^t = 0$ for $|m| > S$. Two relations can be obtained from this RG equation, the RG equations for the sum and difference of the couplings: $J_m^\pm = \frac{1}{2}\left(J_m^\uparrow \pm J_m^\downarrow\right)$. The RG equation for the sum of the couplings is

$$\Delta J_m^+ = -n(D)\frac{\sum_\sigma \left(J_m^\sigma\right)^2 + \sum_\sigma \left(J^t_{m-\frac{1+\sigma}{2}}\right)^2}{2\left(\omega - \frac{D}{2} + \frac{J}{4}\right)} = -n(D)\frac{J^2}{4}\frac{S(S+1)}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.7}$$

This is an $m-$independent piece, so it can be summed over to produce an impurity-independent potential scattering term, which we ignore.

The second is the RG equation for the difference of the couplings:

$$\Delta J_m^- = -n(D)\frac{1}{2}\frac{\left(J^t_{m-1}\right)^2 - \left(J_m^t\right)^2}{\omega - \frac{D}{2} + \frac{J}{4}} = -\frac{1}{4}\frac{n(D)mJ^2}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.8}$$

The usual J Kondo coupling is recovered through $J = 2J_m^-/m$. Substituting this gives

$$\Delta_{\text{p sector}} J = -\frac{1}{2}n(D)\frac{J^2}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.9}$$

We can also obtain the RG equation for J from the transverse renormalisation:

$$\Delta J_m^t = -\frac{n(D)J_m^t\left(J_m^\downarrow + J_{m+1}^\uparrow\right)}{\omega - \frac{D}{2} + \frac{J}{4}} = -\frac{1}{2}\frac{n(D)J_m^t J}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.10}$$

Since $J_m^t \propto J$, we have

$$\Delta J_{\text{p sector}} = -\frac{1}{2}\frac{n(D)J^2}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.11}$$

The total renormalisation from both particle and hole sectors, at this order, is

$$\Delta J^{(2)} = -\frac{n(D)J^2}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.12}$$

We now come to the third-order renormalisation. Following eq. (3.3), the next order renormalisation is

$$\Delta H_j^{(3)} = c^\dagger T\frac{1}{\omega' - H_D}H_X\frac{1}{\omega' - H_D}T^\dagger c + c^\dagger T\frac{1}{\omega' - H_D}T^\dagger c\frac{1}{\omega' - H_D}H_X . \tag{3.13}$$

The first term will be of the form

$$\sum_{q,k,l_1} c^\dagger_{q\beta,l_1} c_{k\alpha,l_1} \left|m_1\right\rangle \left\langle m_2\right| \frac{1}{\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J}{4} S_d^z} \left|m_2\right\rangle c^\dagger_{k_1\sigma_1,l_2} c_{k_2\sigma_2,l_2} \left\langle m_3\right| \frac{1}{\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J}{4} S_d^z} \left|m_3\right\rangle \left\langle m_4\right| c^\dagger_{k\alpha,l_1} c_{q\beta,l_1} . \tag{3.14}$$

We have not bothered to write all the summations and the couplings correctly, because we will only simplify the denominator here. Evaluating the inner products gives

$$\sum_{qk,l_1} \frac{\left|m_1\right\rangle \left\langle m_4\right| c^\dagger_{q\beta} c_{k\alpha} c^\dagger_{k_1\sigma_1,l_2}}{\omega_{m_2,\beta} - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J m_2}{4}} \frac{c_{k_2\sigma_2,l_2} c^\dagger_{k\alpha} c_{q\beta}}{\omega_{m_3,\beta} - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{\beta J m_3}{4}} . \tag{3.15}$$

We again use $\omega_{m,\sigma} \to \omega - \frac{J}{2}\left(m\sigma - \frac{1}{2}\right)$.

$$\left|m_1\right\rangle \left\langle m_4\right| c^\dagger_{k_1\sigma_1,l_2} c_{k_2\sigma_2,l_2} \sum_{qk,l_1} \frac{\hat{n}_{q\beta}\left(1 - \hat{n}_{k\alpha}\right)}{\left(\omega - \frac{D}{2} - \frac{\epsilon_k}{2} + \frac{J}{4}\right)^2} . \tag{3.16}$$

We define $\sum_q \hat{n}_{q\beta} = n(D)$. Performing the sums over k and l_1 gives

$$-\frac{1}{2} \left|m_1\right\rangle \left\langle m_4\right| c^\dagger_{k_1\sigma_1,l_2} c_{k_2\sigma_2,l_2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} . \tag{3.17}$$

ρ is the density of states which we have taken to be constant. Reinstating the complete summation and the couplings gives

$$-\frac{1}{2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\substack{m_1,m_4,k_1,k_2,\\ l_2,\sigma_1\sigma_2}} \lambda_1\lambda_2\lambda_3 \left|m_1\right\rangle \left\langle m_4\right| c^\dagger_{k_1\sigma_1,l_2} c_{k_2\sigma_2,l_2} . \tag{3.18}$$

There is no sum over m_2 and m_3 because they are constrained by m_1 and m_4 respectively. λ_i represent the couplings present at the three interaction vertices. $k_{1,2}$ sum over the momenta, $\sigma_{1,2}$ sum over the spin indices and l_2 sums over the channels.

The second term in eq. (3.13) can be evaluated in an almost identical fashion. The integral here will be negative of the first term, because of an exchange in the scattering processes.

$$\frac{1}{2} \frac{\rho n(D) K}{\omega - \frac{D}{2} + \frac{J}{4}} \sum_{\substack{m_1,m_4,k_1,k_2,\\ l_2,\sigma_1\sigma_2}} \lambda_1\lambda_3\lambda_2 \left|m_1\right\rangle \left\langle m_4\right| c^\dagger_{k_1\sigma_1,l_2} c_{k_2\sigma_2,l_2} . \tag{3.19}$$

The first group of terms (those that appear in (3.18)) in the particle sector can be represented as $a|b'|c$, where $a, b, c \in \{z, +, -\}$ and represent the operator for the conduction electrons in the three connected processes. The $'$ on b indicates that it is the state of the electrons *not being decoupled*. The second group of terms (those that appear in (3.19)) are therefore represented as $a|b|c'$, because in this group, the interaction H_X between the electrons that are not being decoupled occur at the very end. We will only calculate the terms in the particle sector, the ones in hole sector will be equal to these because of particle-hole symmetry. The full list of terms is:

$z|z'|z \quad z|z|z' \quad -|z'|+ \quad -|+|z' \quad +|z'|- \quad +|-|z' \quad z|+'|z \quad z|-'|z \quad +|+'|- \quad -|+'|+$
$z|z|+' \quad z|z|-' \quad +|-|+' \quad +|-|-' \quad -|+|+' \quad z|z'|z \quad z|z|z' \quad -|z'|+ \quad -|+|z' \quad +|z'|-$
$+|-|z' \quad z|+'|z \quad z|-'|z \quad +|+'|- \quad -|+'|+ \quad z|z|+' \quad z|z|-' \quad +|-|+' \quad +|-|-' \quad -|+|+'$
.

The total renormalisation in J^σ_m is

$$\begin{aligned}\Delta J^\sigma_m &= \frac{1}{2}\frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}}\left[\left(J^t_{m-1}\right)^2 J^\sigma_m + \left(J^t_m\right)^2 J^\sigma_m - \left(J^t_{m-1}\right)^2 J^\sigma_{m-1} - \left(J^t_m\right)^2 J^\sigma_{m+1}\right] \\ &= \frac{1}{2}\frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 m\sigma \ .\end{aligned} \tag{3.20}$$

Since we had defined $J^\sigma_m \equiv \frac{1}{2}Jm\sigma$, we have $\Delta J = \frac{2}{m\sigma}\Delta J^\sigma_m$, and we get $\Delta_{\text{p sector}} J = \frac{1}{4}\frac{\rho n(D)K}{\omega - \frac{D}{2}+\frac{J}{4}}J^3$. Combining with the hole sector renormalisation, we get

$$\Delta J^{(3)} = \frac{1}{2}\frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 \ . \tag{3.21}$$

The total renormalisation in J after combining all orders is

$$\Delta J = -\frac{n(D)J^2}{\omega - \frac{D}{2} + \frac{J}{4}} + \frac{1}{2}\frac{\rho n(D)K}{\omega - \frac{D}{2} + \frac{J}{4}} J^3 \ . \tag{3.22}$$

2.4 Example III: Particle in a periodic potential

We consider electrons moving in one dimension in the presence of a periodic potential $V(x)$ satisfying $V(x+a) = V(x)$. The full Hamiltonian can be represented as

$$\mathcal{H} = \int_{-\infty}^{\infty} \mathrm{d}x\; c^\dagger(x)\left[\hat{p}^2/2m + V(x)\right] c(x) \ , \tag{4.1}$$

where $c(x)$ destroys a fermion at the position x.

Physically, the model represents a set of N_{well} number of wells (located at the minima of the potential) linked to each other through hopping (brought about by the kinetic energy term). Each well is described a potential $V(x)$, whose spectrum consists of N_{levs} single particle levels. These levels are localised in real space (around each minima of width a), so the corresponding momentum states are widely separated; each of these momentum states eventually give rise to an energy band, we therefore represent them using the band index: $k_n = nG$ ($n = 0, 1, \ldots, N_{\text{levs}} - 1$), where G is the spacing between the momentum states arising from a single potential well, and is equal to the reciprocal lattice vector $2\pi/a$.

These are of course not the only momentum states in the problem; since each potential well is spaced far apart (at distances greater than a), the multiplicity of the wells leads to closely spaced momentum states, N_{well} in number (the number of potential wells). Since these are closely spaced, they lead to the dense set of states within each band: $k_{n,j} = nG + \frac{2\pi j}{N_{\text{well}}}$ $(j = 0, 1, \ldots, N_{\text{well}} - 1)$. The spacing $2\pi/N_{\text{well}}$ between the states within the band is imposed by applying periodic boundary conditions on the system: $\psi(x = 0) = \psi(x = aN_{\text{well}})$. We take the thermodynamic limit by allowing $N_{\text{well}} \to \infty$, keeping all other independent parameters unchanged. Because of the periodicity of the real space potential, it can only scatter states that are separated by a reciprocal lattice vector G; this means that states within a band (that are within a reciprocal lattice vector of each other) do not scatter off each other: $\langle k_{n+m,j+l}|\hat{V}|k_{n,j}\rangle = \delta_{l,0} \langle k_{n+m,j}|\hat{V}|k_{n,j}\rangle$.

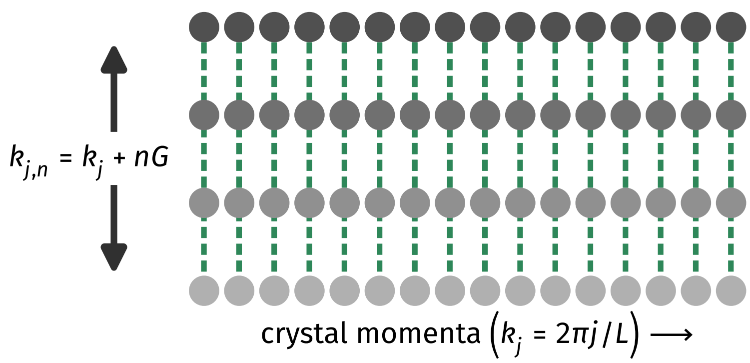

Figure 2.6: The complete set of momentum states in the system. Each row represents a band of states. Each column contains the set of momenta that are connected by a reciprocal lattice vector nG. The states in the lowest band give rise to the crystal momenta. All the momentum states in a column are connected to each other through the potential term (green dotted line). Any two states within a band, on the other hand, are decoupled from each other.

Decomposition into sum of zero-dimensional problems

We begin by writing the Hamiltonian completely in momentum space. A general momentum is represented as $k+nG$, where n is the band index goes from 0 to $N-1$, and k is the crystal momentum that spans through the lowest band. With this notation, we have

$$\begin{aligned}\mathcal{H} =& \frac{\hbar^2}{2m}\int \mathrm{dk} \sum_n c^\dagger(k+nG)\,(k+nG)^2\, c(k+nG) \\ &+ \int \mathrm{dk} \sum_{m,n}\left[\mathcal{V}(nG)c^\dagger(k+nG+mG)c(k+mG) + \text{h.c.}\right] \\ =& \frac{2\pi}{a}\int \mathrm{dk}\,\mathcal{H}(k)\end{aligned} \tag{4.2}$$

where

$$\begin{aligned}\mathcal{H}(k) =&\frac{a}{2\pi}\frac{\hbar^2}{2m}\sum_n c^\dagger(k+nG)\,(k+nG)^2\,c(k+nG)\\&+\frac{a}{2\pi}\sum_{m,n}\left[v(nG)c^\dagger(k+nG+mG)c(k+mG)+\text{h.c.}\right]\\=&\frac{a}{2\pi}\int \mathrm{d}x'\;c^\dagger(x')\left[\frac{1}{2m}\left(-i\hbar\partial_{x'}+\hbar k\right)^2+V(x')\right]c(x')\;.\end{aligned} \tag{4.3}$$

We have used x' to symbolise the position variable conjugate to the new set of momenta $\{nG\}$ =

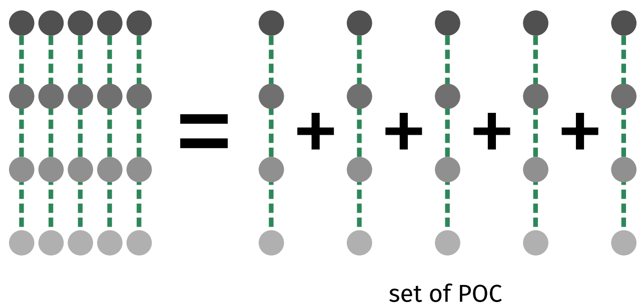

Figure 2.7: *Because of the periodicity of the potential, the problem splits into a sum of particle on a circle (POC) problems, each formed by a specific crystal momentum and other momentum states that are related to it by a reciprocal vector.*

$-i\partial_{x'}$ that span the restricted Hilbert space of $\mathcal{H}(k)$. Also, $\mathcal{V}(k)$ is the Fourier transform of the real space potential $V(x)$.

This decoupling of the full Hamiltonian $\mathcal{H}$ into a block-diagonal form (each block indexed by the crystal momentum k) can be attributed to the discrete translation symmetry present in the problem and could have been anticipated very generally, as we now show. The Hamiltonian satisfies $[\mathcal{H}, T(a)] = 0$, where $T(x) = e^{i\hat{p}x}$ is the translation operator. By choosing a starting location $x = 0$, applying the translation operator cycles through the set of states $|x = na\rangle\,;\; n = 0, 1, \ldots, N_{\text{well}}$; this set is closed under the application of $T(a)$. It is easy enough to guess the eigenstates of $T(a)$. Since the application of $T(a)$ increases the position monotonically, the only way to create an eigenstate is to involve all states. With that in mind, the simplest ansatz is of the form $|\Phi\rangle = \sum_n e^{-i\Phi n}|an\rangle$. Applying the translation operator on this state gives

$$T(a)\,|\Phi\rangle = \sum_{n=0}^{N_{\text{well}}-2} e^{-i\Phi n}\,|a(n+1)\rangle + e^{-i\Phi(N_{\text{well}}-1)}\,|aN_{\text{well}}\rangle = e^{i\Phi}\left[e^{-i\Phi N_{\text{well}}}\,|aN_{\text{well}}\rangle + \sum_{n=1}^{N_{\text{well}}-1} e^{-i\Phi n}\,|an\rangle\right] \tag{4.4}$$

Since we have applied PBC on the system, we used $|aN_{\text{well}}\rangle$ for $|0\rangle$. If the values of Φ are chosen to be some integer multiple of $2\pi/N_{\text{well}}$, we would have $e^{-i\Phi N_{\text{well}}} = 1$, and the entire state can be resummed into the form

$$T(a)\,|\Phi\rangle = e^{i\Phi}\,|\Phi\rangle \tag{4.5}$$

This shows that $|\Phi\rangle$ is an eigenstate of the translation operator. Since the set of states has a dimension of N_{well}, the number of eigenstates and the number of possible values of Φ will also be N_{well}. This fixes the eigenvalues to $\Phi = 2\pi m/N_{\text{well}}, m = 0, 1, \ldots, N_{\text{well}} - 1$. These will also be eigenstates of the Hamiltonian, because they are indexed by different values of Φ and the Hamiltonian cannot change Φ (because it commutes with T). As will be shown later, the eigenvalues Φ are actually related to the crystal momentum, through $\Phi = ka$. The set of allowed values of Φ form the first Brillouin zone.

However, $|\Phi\rangle$ as defined above is not the only eigenstate with eigenvalue $e^{i\Phi}$. One can simply shift the starting location $x = 0$ by any distance smaller than a, and repeat the entire procedure. The eigenstates hence obtained will have the same eigenvalues as above:

$$|\Phi, \delta\rangle = \sum_n e^{-i\Phi n} |an + \delta\rangle \, , \; 0 \leq \delta < a; \tag{4.6}$$

$$T(a) |\Phi, \delta\rangle = \sum_{n=0}^{N_{\text{well}}-2} e^{-i\Phi n} |a(n+1) + \delta\rangle + e^{-i\Phi(N_{\text{well}}-1)+\delta} |aN_{\text{well}} + \delta\rangle = e^{i\Phi} |\Phi, \delta\rangle \tag{4.7}$$

The translation operator is therefore able to at most classify the complete set of states into N_{well} number of blocks, each labelled by a specific value of Φ. Each block has a large number of states (N_{levs}), and the Hamiltonian will general scatter these states into each other. The Hamiltonian for each of these blocks can be represented as $\mathcal{H}(\Phi) = \mathcal{P}_\Phi \mathcal{H} \mathcal{P}_\Phi$ (where $\mathcal{P}_\Phi$ projects onto the block of states with eigenvalue $e^{i\Phi}$), such that the total Hamiltonian can be represented as $\mathcal{H} = \sum_\Phi \mathcal{H}(\Phi)$. Upon rescaling by $2\pi/a$ and taking the limit of $N_{\text{well}} \to \infty$, this is the same Hamiltonian decoupling relation as that obtained in eq. 4.2.

Since the spacing between two adjacent momentum is G, the maximum value of x' has to be $2\pi/G = a$, with the ends identified with each other: $\psi(x' = 0) = \psi(x' = a)$. We can then define a dimensionless variable $\theta = 2\pi x'/a$ that goes from 0 to 2π, and is conjugate to the angular wave number $Q = -i\partial_\theta$. This definition ensures $[\theta, Q] = i$. For reasons that will become clear very soon, we will also define a dimensionless crystal momentum $\Phi = ka$. With these modifications, the Hamiltonian takes the simpler form

$$\mathcal{H}(k) = \int_0^{2\pi} \mathrm{d}\theta \, c^\dagger(\theta) \left[\frac{\hbar^2}{2ma^2} \left(\hat{Q} + \Phi/2\pi\right)^2 + U(\theta)\right] c(\theta) \, . \tag{4.8}$$

By nothing that we have the boundary condition $\psi(\theta = 0) = \psi(\theta = 2\pi)$, we can recognise this Hamiltonian as that describing the dynamics of a particle moving on a circle (POC) with a potential $U(\theta) = \frac{a}{2\pi} V(a\theta/2\pi)$, in the presence of an Aharonov-Bohm flux Φ. Note that $U(\theta)$ has a period of 2π.

Even when reduced to the simpler POC, the problem cannot be simply integrated to obtain the eigenstates. This is because the complete Hamiltonian does not commute with either the position or the momentum operator; it contains quantum fluctuations that need to be resolved in order to obtain the ground state energy. One way of obtaining the energy is to calculate the propagator for transition from $\theta = 0$ to $\theta = 2\pi$, using the method of path integrals [?, ?]. This is shown in Appendix **??**. We, however, will attack the problem using a renormalisation group method, called the unitary renormalisation group in order to track the emergence of the tight-binding metal in the limit when the effective amplitude of the single-particle periodic potential at low-energies has become much larger than the typical kinetic energy of the electron.

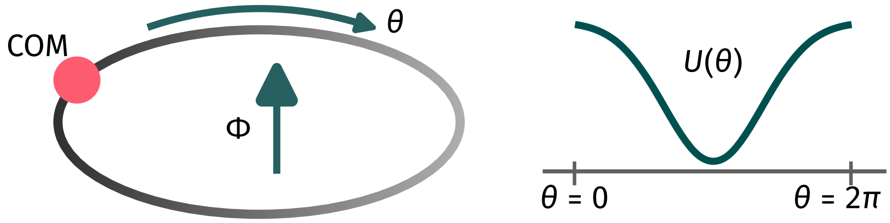

Figure 2.8: *The full problem reduces to that of a particle on a circle, in the presence of potential with period 2π, and a flux Φ.*

URG analysis of the particle on a circle problem

A summary of the unitary RG method is provided in Appendix **??**. For the purposes of studying the model under renormalisation group, we write the particle on a circle Hamiltonian in momentum$(Q-)$space. For that, we introduce the Fourier modes

$$c(\theta) = \frac{1}{\sqrt{2\pi}} \int dQ e^{iQ\theta} c(Q), \text{ and } U(\theta) = \int dQ e^{iQ\theta} \mathcal{U}(Q) , \tag{4.9}$$

so that the Hamiltonian becomes

$$\mathcal{H}(k) = \int \mathrm{dQ}\, \varepsilon\left(Q + \Phi/2\pi\right) \hat{n}(Q) + \int \mathrm{dQ}_1\, \mathrm{dQ}_2\, c^\dagger(Q_1) c(Q_2) \mathcal{U}\left(Q_1 - Q_2\right) . \tag{4.10}$$

where we have defined the dispersion $\varepsilon(Q) = \hbar^2 Q^2/(2ma^2)$. The second term involves scattering between momentum states Q_1 and Q_2. The quantum fluctuations that connect the low-energy states to the high-energy states can be integrated out using the URG method [**?**]. This proceeds through a reduction in the momentum cutoff, ultimately leading to an IR theory. As the large momenta are integrated out, the potential term for scattering between the lower momenta undergo renormalisation. Let, at the l^{th} RG step, the momentum cutoff be Q_l. At this step, fluctuations between all momentum states more energetic than Q_l have been integrated out. The RG equations for the renormalisation of the potential $\mathcal{U}_{ij}^{(l)}$ that scatters any two lower momenta Q_i and Q_j $\left(Q_i, Q_j < Q_l\right)$ is given by [**?**]

$$\Delta \mathcal{U}_{ij}^{(l)}(\omega) = \frac{\mathcal{U}_{il}\mathcal{U}_{lj}}{\omega - \varepsilon(Q_l + \Phi/2\pi)} , \tag{4.11}$$

where ω is an energy scale that tracks the amount of quantum fluctuations present in the system, and is characteristic of the URG method. The dispersion also undergoes renormalisation::

$$\Delta\varepsilon^{(l)}(Q_i, \omega) = \frac{|\mathcal{U}_{il}|^2}{\omega - \varepsilon(Q_l + \Phi/2\pi)} , \tag{4.12}$$

Effective description at low energies

Appearance of bands and band gaps

As the URG proceeds, the quantum fluctuations between the momentum states are resolved, leading to a simpler theory for the IR. We focus on the final two states $|Q_0 + \Phi/2\pi\rangle$, $|Q_1 + \Phi/2\pi\rangle$, for $\Phi = 0$

(Brillouin zone edge). The Hamiltonian for their dynamics, just prior to the decoupling of Q_j, is

$$H^*_{01} = \varepsilon^*(Q_0)\,|Q_0\rangle\langle Q_0| + \varepsilon^*(Q_1)\,|Q_1\rangle\langle Q_1| + \left(\mathcal{U}^*_{01}\,|Q_1\rangle\langle Q_0| + \text{h.c.}\right) \tag{4.13}$$

Since both the states are at the edge of the band, we can assume $\varepsilon^*(Q_0) \simeq \varepsilon^*(Q_1)$, in which case the problem becomes that of a symmetric double well with tunnelling. The eigenstates are $|Q_0\rangle \pm |Q_1\rangle$, with energies $\varepsilon^*(Q_0) \pm |\mathcal{U}^*_{01}|$. We can substitute this estimate of the energy eigenvalues into the RG equations in the form of the quantum fluctuation scale ω. At the last decoupling, the shifts in the kinetic energy are therefore

$$\Delta\varepsilon^*(Q_i, \omega^\pm) \simeq \frac{|\mathcal{U}^*_{01}|^2}{\varepsilon^*(Q_0) \pm |\mathcal{U}^*_{01}| - \varepsilon(Q_0)} = \pm|\mathcal{U}^*_{01}| = \pm|\mathcal{U}^*(G)|\,, \tag{4.14}$$

where we used $\mathcal{U}_{01} = \mathcal{U}(Q_1 - Q_0) = \mathcal{U}(G)$. The difference between the two shifts gives the energy difference between the two lowest states of the POC. In this way, the above RG equation shows the emergence of the single-particle band gap and the formation of the band insulator (if the chemical potential is placed within the gap). If we now reintroduce the flux degree of freedom, the lowest energy eigenstates of all the POCs can be combined to form a band of closely spaced energy levels $\left\{\varepsilon^*(Q_0 + \Phi_j/2\pi)\right\}, \Phi = k_{0,j}a$ (see Fig.2.9). These levels are indexed by $k_{0,j}$ with j ranging through the first Brillouin zone; $k_{0,j}$ are the crystal momenta of the system.

The second band of states is formed similarly by considering the first excited of the POCs: $\left\{\varepsilon^*(Q_1 + \Phi_j/2\pi)\right\}, \Phi$ $k_{0,j}a$. The gap between the first and second band is set by the energy difference between the ground state and first excited state of the POC at $\Phi = 0$. This difference, and hence the gap, is equal to $2|\mathcal{U}^*_{01}|$.

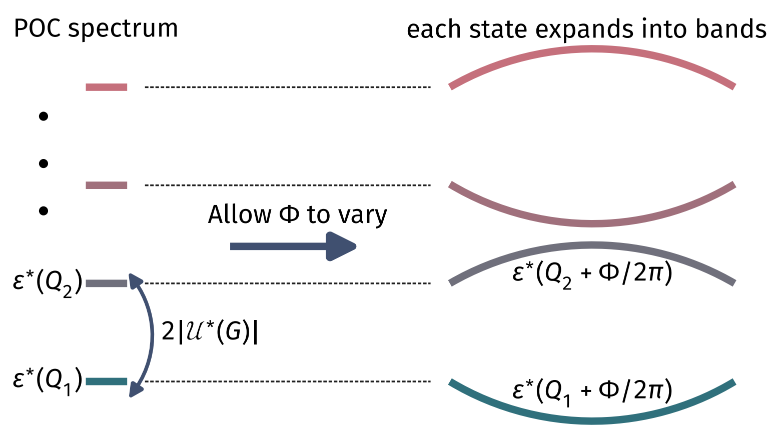

Figure 2.9: *Spectral flow transformations on the POC by tuning the flux Φ generates the full band of delocalised single-particle electronic states whose crystal momenta correpond to $\hbar\Phi/a$.*

Dispersion relation of the lowest band

In order to derive the dispersion for the lowest band, we note that the final fixed point Hamiltonian (obtained upon decoupling all higher momentum states) is described by the single momentum state

$|Q_0\rangle$. In terms of real space hopping, this means that all short wavelength processes have been removed, and the only process that remains is the longest hopping, that from 0 to 2π. The fixed point Hamiltonian can be written in terms of this process:

$$\varepsilon^*(Q_0)\,|Q_0\rangle\,\langle Q_0| = \frac{1}{2}\varepsilon^*(2\pi)\,(|\theta=0\rangle + |\theta=2\pi\rangle)\,(\langle\theta=0| + \langle\theta=2\pi|) = \frac{1}{2}\varepsilon^*(2\pi)\,(\hat{n}(0)+\hat{n}(2\pi)) + \frac{1}{2}\varepsilon^*(2\pi)\left(c^\dagger(0)c(2\pi) + \text{h.c.}\right) \tag{4.15}$$

We now recall that we had mapped the spatial point $x=0$ ($x=a$) to the angular point $\theta=0$ ($\theta=2\pi$) point. We now transform back to spatial coordinates to get

$$\frac{1}{2}\varepsilon^*(2\pi)\,(\hat{n}(0)+\hat{n}(a)) + \frac{1}{2}\varepsilon^*(a)\left(c^\dagger(0)c(a) + \text{h.c.}\right)\ . \tag{4.16}$$

This is computed for a specific value $\Phi = 0$ for the flux (or the crystal momentum). The POC corresponding to a particular flux is built from the states within a particular well. Changing the flux is then equivalent to studying the POC corresponding to the states within a different well. What this means is that tuning the flux by one interval amounts to translating the fixed point Hamiltonian by one period a of the potential. Adding the contributions from all the values of the flux leads to the following tight-binding Hamiltonian:

$$H_{TB} = \varepsilon^*(2\pi)\sum_{j=0}^{N_{\text{well}}-1}\hat{n}(ja) + \frac{1}{2}\varepsilon^*(a)\sum_{j=0}^{N_{\text{well}}-1}\left(c^\dagger(ja)c((j+1)a) + \text{h.c.}\right)\ , \tag{4.17}$$

where the first term corresponds to a constant chemical potential applied to the tight-binding metal in one dimension with a single-particle dispersion $\varepsilon(k) = -\varepsilon^*(a)\cos(ka)$ where k corresponds to the set of crystal momenta $k \in \left[-\frac{\pi}{a}, \frac{\pi}{a}\right]$. The exercise, therefore, shows the emergence of nearest-neighbour hopping in the lowest band by starting from a more microscopic model of a particle in a periodic potential. (An alternate derivation of the tight-binding metal has also been shown from path-integral methods in Appendix **??**).

2.5 Comparison of renormalisation group approaches based on canonical transformations

We have considered three canonical transformations in this section: the Poor Man's scaling (PMS), the Schrieffer-Wolff transformation (SWT) and the continuous unitary transformation renormalization group (CUT-RG). PMS and SWT are more or less identical; they differ in the context in which they are used. PMS is used when there is an entire spectrum of energies in the model, ranging from a low energy limit to a high energy; it is then employed to decouple the highest energy modes in an iterative fashion. SWT is used when the Hamiltonian can be split into one high and one low energy part, and we need to decouple these two modes. It is clear when seen from this perspective that SWT is like a one-shot PMS; it decouples the UV from the IR in one step, compared to the iterative approach of PMS. However, as has been shown in the previous subsections, both PMS and SWT can be formalized in an identical fashion, so that one can be switched for the other in both contexts.

Now that we have established that PMS and SWT are formally identical, we can relate them to URG. URG is similar in philosophy to PMS in the fact that URG also successively decouples high energy modes from the low energy ones. The difference is that URG takes care of the quantum fluctuations, at least in principle, by introducing a new set of energy scales ω. These ω then parameterise the RG flows of URG, compared to the single RG flow of PMS. Since SWT is formally the same as PMS, these is also a comparison between SWT and URG. Both PMS and SWT trivialize the quantum fluctuation scales of URG by replacing it with the bare energy values.

CUT-RG is philosophically different from the other transformations. Its goal is to gradually reduce the contributions of the off-diagonal terms by making them decay along a certain scale l. Off-diagonal terms that connect states with large energy differences decay faster. In this sense, there is no sequential dropping out of off-diagonal terms; all off-diagonal terms disappear at $l = \infty$. In this sense, it is like a continuous version of SWT. While SWT strives to remove an entire off-diagonal term in one-shot, CUT RG does this gradually by introducing a scale l. This separation of scales is absent in SWT. It does exist in URG and PMS, albeit in a different fashion. There, the separation comes in when we decouple single electron states starting from the Brillouin zone boundary Λ_N and come down to the Fermi surface Λ_0. Each Λ provides a natural energy scale for separating the high and low energy physics.

If one integrates the continuum generator η over all the scales, one should recover something analogous to the SWT generator. This generator is however necessarily perturbative in the off-diagonal term, since by definition η will only be of first order in $\mathcal{H}_X$. This is in contrast to the non-perturbative generators in URG and, at least in principle, PMS. This non-perturbative form is encapsulated in the presence of a second completely different set of energy scales ω (or E in PMS). This second energy scale is absent in CUT RG because it takes care of at most the second order term.

Another point to note is that since SWT keeps the entire off-diagonal piece in the generator, new terms will almost certainly by generated at every step, and they have to be truncated. This is not the case with URG or PMS, becasue in those methods, we decouple just a single-electron at each step, and so those electrons become integrals of motion at that step, leading to their removal from the off-diagonal piece, and very often the Hamiltonian retains the same form as the bare model. This makes the philosophy of URG and PMS easier to work with. Tied to this is the fact that CUT RG often takes a certain type of interaction in the generator part, and not the entire off-diagonal piece. Hence, at the limit of the flow parameter going to ∞, the chosen off-diagonal piece goes to zero but the remaining off-diagonal pieces still remain. As a result, the Hamiltonian is atmost block diagonal at this stage. On the other hand, URG progressively decouples single electrons, meaning all scattering terms corresponding to that electron vanish at each step.

One last thing to note is that URG, being unitarily implemented with a well-defined generator, does not accomodate for spontaneous symmetry breaking (SSB). In order to see SSB, the symmetry-breaking term has to be added to the bare model explicitly; if this term grows under the RG, then the ground state will be symmetry-broken. CUT RG, however, does allow for SSB through the idea that the generator is not uniquely defined. If the generator commutes with a particular symmetry, then the family of Hamiltonians will also have the symmetry [411]. However, if the generator is replaced by something that is normal ordered w.r.t a particular expectation value, then the CUT RG flow will usually be towards either the symmetry-preserved state or the symmetry-broken state, depending on whichever is stable [412].

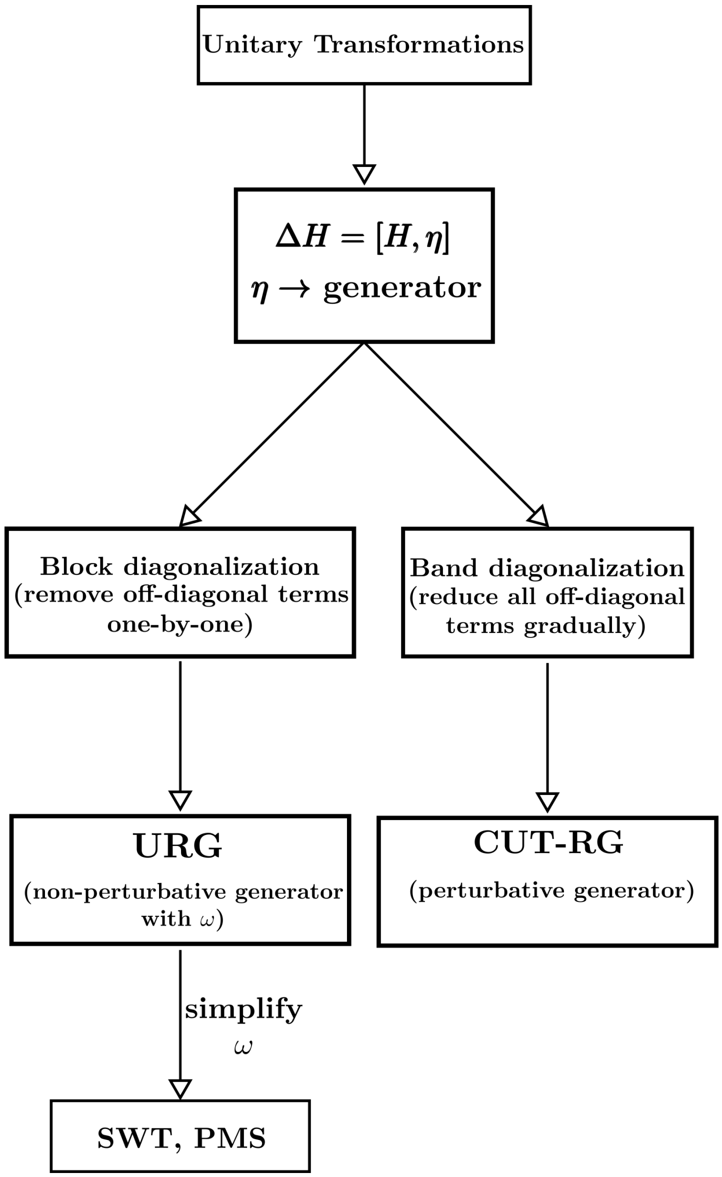

Figure 2.10: *Comparison of the various unitary transformations and their relationships to each other.*

2.6 Iterative Diagonalisation of Quantum Impurity Models

The broad idea is the following. We have a Hamiltonian describing an interacting set of states (in real or momentum space). We can express the total Hamiltonian in an incremental fashion: $H = \sum_i H_i$, where H_{i+1} involves more number of states than H_i. For example, a tight-binding model can be written in that form, with the definition $H_i = c^\dagger_{i+1} c_i$ + h.c.. The iterative diagonalisation method obtains the low-energy spectrum of this problem in the following manner: We first diagonalise the Hamiltonian H for a smaller value of i, small enough such that this can be done exactly. We then truncate the spectrum to a predefined size, and rotate all existing operators to this truncated basis, including the Hamiltonian H. We then consider the "bonding Hamiltonian" ΔH between the existing sites and the new sites, and rotate the same into the truncated basis. Adding the previous rotated Hamiltonian H and the rotated increment Hamiltonian ΔH gives us a truncated but effective Hamiltonian for the increased number of sites. We again diagonalise this, and again retain only a fixed number of states in the spectrum. We keep repeating this until we reach the required number of sites.

Jordan-Wigner Matrix Representation of Fermionic Operators

For a single qubit, the creation/annihilation matrices are

$$c = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, c^\dagger = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} , \tag{6.1}$$

given the convention that $|1\rangle = (0\ \ 1)$ is the occupied state. For a many-body system, these must be replaced with field operators that have the canonical fermionic algebra. For a system of N 1-particle levels, this can be accomplished through a *Jordan-Wigner*-like transformation

$$c_j = \left(\otimes_1^j \sigma_z\right) \otimes c \otimes \left(\otimes_1^{N-j} \mathbb{I}\right) , j \in [0, N-1] , \tag{6.2}$$

where $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and $\mathbb{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$. For example, for spinless electrons on a three-site lattice, we can have three fermionic operators: c_1,c_2, and c_3. Following the above expression, their matrices are

$$c_1 = \begin{bmatrix} 0&0&1&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-1&0 \\ 0&0&0&0&0&0&0&-1 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \end{bmatrix}, c_2 = \begin{bmatrix} 0&0&1&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&-1&0 \\ 0&0&0&0&0&0&0&-1 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \end{bmatrix}, c_3 = \begin{bmatrix} 0&1&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&-1&0&0&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&-1&0&0 \\ 0&0&0&0&0&0&0&0 \\ 0&0&0&0&0&0&0&1 \\ 0&0&0&0&0&0&0&0 \end{bmatrix},$$

For later use, we define the *antisymmetriser matrix* $S(j)$ and identity matrix $\mathbb{I}(j)$,

$$\begin{aligned} S(j) &= \otimes_1^j \sigma_z = \sigma_z \otimes \ldots j \text{ times } \ldots \otimes \sigma_z, \\ \mathbb{I}(j) &= \otimes_1^{N-j} \mathbb{I} = \mathbb{I} \otimes \ldots j \text{ times } \ldots \otimes \mathbb{I} , \end{aligned} \tag{6.3}$$

to express the fermionic representation compactly:

$$c_j = S(j) \otimes c \otimes \mathbb{I}(N-j) . \tag{6.4}$$

Structure of Hamiltonian

We consider a general impurity model, of L number of sites (excluding the impurity site):

$$H_{\text{sys}}(L) = H_{\text{imp}} + H_{\text{imp-bath}} + -t \sum_{\sigma,j=1}^{L-1} \left(c^\dagger_{j,\sigma} c_{j+1,\sigma} + \text{h.c.} \right) , \tag{6.5}$$

where H_{imp} is the Hamiltonian for the decoupled impurity site and $H_{\text{imp-bath}}$ is the impurity-bath coupling. Such a system has $2L + 2$ single-particle levels in the full system (2 spin levels for each of the $L + 1$ sites).

Iteration scheme (the initial number of states is L_0):

$$\begin{aligned} H_0(L_0) &= H_{\text{sys}}(L_0) , \\ H_{r+1}(L_0) &= H_r(L_0) + \Delta H_r(L_0),\ r \geq 0 , \\ \Delta H_r(L_0) &= -t \sum_\sigma \left(c^\dagger_{L_0+r,\sigma} c_{L_0+r+1,\sigma} + \text{h.c.} \right) . \end{aligned} \tag{6.6}$$

H_0 is the initial Hamiltonian, consisting of L_0 lattice sites in the bath, $H_{r+1}(L_0)$ is the Hamiltonian after $r+1$ iterations having started with L_0 sites, and $\Delta H_r(L_0)$ is the increment term that gets added to $H_r(L_0)$ during the $(r + 1)^{\text{th}}$ iteration to give rise to $H_{r+1}(L_0)$.

Iterative Diagonalization

Let the starting Hamiltonian be $H_0(L_0)$, consisting of L_0 single-particle levels (hence a Hilbert space dimension of 2^{L_0}). We start with a single-particle computational basis and construct the L_0 fermionic operators $c_1, c_2, \ldots, c_{L_0}$ in this basis, using eq. 6.4. We also keep track of a large antisymmetriser matrix $S(L_0)$ that will be used to attach new sites when we expand the system. Let M_s be the maximum number of eigenstates we retain in the spectrum at any given step. The value of M_s should be chosen so that a $M_s \times M_s$ matrix can be diagonalised in reasonable time.

S1. Construct the complete Hamiltonian matrix H_0 in our present basis using the field operators $\{c_j\}$. Diagonalise the Hamiltonian (of size $D_0 \times D_0$) and obtain the eigenvalues E_n and eigenstates X_n. Each X_n is a column vector of size D_0.

S2. Retain at most M_s number of eigenstates, preferring the ones with lower energy. The reduced basis for this rotated truncated subspace is constructed by stacking the column vectors X_n horizontally:

$$R = [X_1 X_2 \ldots X_{M_s}]_{D_0 \times M_s} . \tag{6.7}$$

This matrix also acts as the transformation to rotates and truncate all operators from the old basis into the new one.

S3. We rotate our Hamiltonian H_0, our fermionic operators $\{c_j\}$ and the large antisymmetrizer matrix $S(L_0)$ into the new reduced basis, using the transformation $O \to R^\dagger O R$.

S4. We now need to expand our system by adding the increment Hamiltonian ΔH_0. Let the number of *new* 1-particle levels in ΔH_0 be L_Δ. These new levels will be indexed as $L_0+1, \ldots, L_0+L_\Delta$. We need to define antisymmetrized fermionic operators for the new sites:

$$\begin{aligned} c_{L_0+1} &= c \otimes \mathbb{I}(2^{L_\Delta - 1}), \\ c_{L_0+2} &= \sigma_z \otimes c \otimes \mathbb{I}(2^{L_\Delta - 2}), \\ &\ldots \\ c_{L_0+L_\Delta} &= (\otimes_1^{L_\Delta - 1} \sigma) \otimes c \ . \end{aligned} \tag{6.8}$$

These have to be calculated in the local computational basis (of size 2^{L_Δ}) of the new levels.

S5. Combining the new sites with the old sites leads to a combined Hilbert space dimension of $(M_s + 2^{L_\Delta}) \times (M_s + 2^{L_\Delta})$. To allow all operators to act on the enlarged Hilbert space, we expand both sets of operators:

$$\begin{aligned} c_j &= S(L_0) \otimes c_j; \quad j = L_0 + 1, \ldots, L_\Delta \ , \\ c_j &= c_j \otimes \mathbb{I}(2^{L_\Delta}); \quad j = 1, 2, \ldots, L_0 \ , \\ H_0 &\to H_0 \otimes \mathbb{I}(2^{L_\Delta}), \\ S(L_1) &= S(L_0) \otimes \mathbb{I}(2^{L_\Delta}) \ . \end{aligned} \tag{6.9}$$

where $\mathbb{I}(2^{L_\Delta})$ is an identity matrix of dimension $2^{L_\Delta} \times 2^{L_\Delta}$. Note that the operators in the last three equations are the rotated ones (following Step 3).

S6. Using the transformed operators $c_{L_0+1}, \ldots, c_{L_0+L_\Delta}$ for the new sites, construct the difference Hamiltonian matrix ΔH_0 and hence the updated Hamiltonian $H_1 = H_0 + \Delta H_0$ for the next step. Repeat the process starting from step 2 with the new Hamiltonian H_1, the new operators c_j and the new matrix $S(L_1)$ replacing the old counterparts.

Static correlations

We are in general interested in $n-$point correlations of the form $\langle O_1 O_2 \ldots O_n \rangle$, where O_i are operators that act on 1-particle Hilbert spaces. Let the earliest step of the iterative diagonalisation procedure at which all these operators have entered the system be r. At this step r, construct the correlation operator $O_1 O_2 \ldots O_n$ using the fermionic matrices c_j at that step (recall that these matrices will be highly rotated versions of the matrices we started with). Having constructed the operator O, we expand and rotate it after the completion of every future step: $O_{n+1} = (R_n^\dagger O_n R_n) \otimes \mathbb{I}(2^{L_\Delta})$, where R_n is the rotation matrix for the n^{th} step. The last step n^* of the iterative diagonalisation consists of only a diagonalisation and no expansion, resulting in a final set of eigenstates $\{X_i\}$. The form of the correlation operator in this basis is $O_{n^*} = R_{n^*-1}^\dagger O_{n^*-1} R_{n^*}$. The expectation value can now be calculated using the matrix O_{n^*} and the ground state of $\{X_i\}$.

Reduced density matrices

2.7 Optimising Fermi Surface Calculations

Rotated basis for simplified representation

At the renormalisation group fixed point, we have a renormalised theory for the interaction of the impurity spin S_f with the $f-$layer Fermi surface:

$$H^* = \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha,\beta} \sum_{\mathbf{k},\mathbf{q}} J^*_f(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} f_{\mathbf{q},\beta} \ . \tag{7.1}$$

We will now obtain a more minimal representation of this interaction. Each momentum label is summed over all four quadrants Q_1 through Q_4; eq. **??** relates Q_1 with Q_3 and Q_2 with Q_4:

$$\begin{aligned}\sum_{\mathbf{k},\mathbf{q}} J^*_f(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} f_{\mathbf{q},\beta} &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k}\in Q_1,Q_2} J^*_f(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} + \sum_{\mathbf{k}\in Q_3,Q_4} J^*_f(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} \right] f_{\mathbf{q},\beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k}\in Q_1,Q_2} J^*_f(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} + \sum_{\mathbf{k}\in Q_1} J^*_f(\mathbf{k}-\mathbf{Q_1},\mathbf{q}) f^\dagger_{\mathbf{k}-\mathbf{Q_1},\alpha} + \sum_{\mathbf{k}\in Q_2} J^*_f(\mathbf{k}+\mathbf{Q_2},\mathbf{q}) f^\dagger_{\mathbf{k}+\mathbf{Q_2},\alpha} \right] f_{\mathbf{q},\beta} \\ &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k}\in Q_1} J^*_f(\mathbf{k},\mathbf{q}) \left(f^\dagger_{\mathbf{k},\alpha} - f^\dagger_{\mathbf{k}-\mathbf{Q_1},\alpha} \right) + \sum_{\mathbf{k}\in Q_2} J^*_f(\mathbf{k},\mathbf{q}) \left(f^\dagger_{\mathbf{k},\alpha} - f^\dagger_{\mathbf{k}+\mathbf{Q_2},\alpha} \right) \right] f_{\mathbf{q},\beta} \ .\end{aligned} \tag{7.2}$$

For ease of notation, we define new fermionic operators $A_{\mathbf{k},\sigma}$:

$$A_{\mathbf{k},\sigma,\pm} = \begin{cases} \frac{1}{\sqrt{2}} \left(f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}-\mathbf{Q_1},\sigma} \right), \mathbf{k} \in Q_1 \ , \\ \frac{1}{\sqrt{2}} \left(f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}+\mathbf{Q_2},\sigma} \right), \mathbf{k} \in Q_2 \ , \end{cases} \tag{7.3}$$

which satisfy the appropriate algebra: $\left\{ A_{\mathbf{k},\sigma,p}, A_{\mathbf{k}',\sigma',p'} \right\} = \delta_{\mathbf{k},\mathbf{k}'} \delta_{\sigma,\sigma'} \delta p, p'$, with $p = \pm$ denoting the flavours of the A fields. Note that only the $p = -1$ flavour enters the Hamiltonian. Henceforth, we drop the label $\pm$ and it is implied that A refers to the $p = -1$ variant.

Decomposing the sum over $\mathbf{q}$ in a similar fashion, we get

$$\sum_{\mathbf{k},\mathbf{q}} J^*_f(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} f_{\mathbf{q},\beta} = 2 \sum_{\mathbf{k},\mathbf{q}\in Q_1,Q_2} J^*_f(\mathbf{k},\mathbf{q}) A^\dagger_{\mathbf{k},\alpha} A_{\mathbf{q},\beta} \ . \tag{7.4}$$

Correlations in truncated representation

In order to calculate equal-time correlations (such as $\langle S_d^+ f_{\mathbf{k}\downarrow}^\dagger f_{\mathbf{k}'\uparrow}\rangle$), we use eq. 7.3 to express the bare fields $f_{\mathbf{k},\sigma}$ in terms of the new fields $A_{\mathbf{k},\sigma,\pm}$ and $B_{\mathbf{k},\sigma,\pm}$:

$$\begin{aligned} f_{\mathbf{k},\sigma} &= \frac{1}{\sqrt{2}}\left(A_{\mathbf{k},\sigma,+} + A_{\mathbf{k},\sigma,-}\right) \ , \mathbf{k} \in Q_1, Q_2 \\ f_{\mathbf{k}-\mathbf{Q}_1,\sigma} &= \frac{1}{\sqrt{2}}\left(A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}\right) \ , \mathbf{k} \in Q_1 \\ f_{\mathbf{k}+\mathbf{Q}_2,\sigma} &= \frac{1}{\sqrt{2}}\left(A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-}\right) \ , \mathbf{k} \in Q_2 \end{aligned} \tag{7.5}$$

where the four relations act on operators in four quadrants. These relations allow us to express correlations $F_{k_i,k_j} = \langle S_d^+ c_{k_i,\uparrow}^\dagger c_{k_j,\downarrow}\rangle$ for the bare fields f in terms of the rotated fields. Using the fact that the Hamiltonian does not couple the fields B_+ and A_+, we have

$$\begin{aligned} F_{k_1,k_1} = F_{k_3,k_3} = -F_{k_1,k_3} &= \frac{1}{2}\langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-})\rangle \ , \\ F_{k_2,k_2} = F_{k_4,k_4} = -F_{k_2,k_4} &= \frac{1}{2}\langle S_d^+ A_{k_2,\downarrow,-}^\dagger A_{k_2,\uparrow,-})\rangle \ , \\ F_{k_1,k_2} = F_{k_3,k_4} = -F_{k_1,k_4} = -F_{k_2,k_3} &= \frac{1}{2}\langle S_d^+ A_{k_1,\downarrow,-}^\dagger A_{k_1,\uparrow,-})\rangle \ , \end{aligned} \tag{7.6}$$

where the momentum state k_i belongs to the quadrant Q_i.

Effective Star Graph Representation of Kondo Interaction

To make computation less intensive, we want to make the Hamiltonian in eq. 7.4 sparser, by going to a star graph representation where each node of the graph couples only with the impurity spin (central node) and not to the other nodes of the graph. We therefore consider a rotated basis of excitations

$$\gamma_{i,\sigma}^\dagger = \sum_k \Lambda_{i,k} A_{k,\sigma}^\dagger \ , \ \text{where} \ \sum_k \Lambda_{i,k}^* \Lambda_{j,k} = \delta_{i,j} \ . \tag{7.7}$$

These fermions of course satisfy the appropriate algebra:

$$\left\{\gamma_{i,\sigma}, \gamma_{j,\sigma'}^\dagger\right\} = \sum_{k,q} \Lambda_{i,k}^* \Lambda_{j,q} \left\{A_{k,\sigma}, A_{q,\sigma'}^\dagger\right\} = \delta_{i,j}\delta_{\sigma,\sigma'} \ . \tag{7.8}$$

In order to calculate the coefficients $\Lambda_{i,k}$, we impose the star graph simplification condition described above:

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) A_{\mathbf{k},\alpha}^\dagger A_{\mathbf{q},\beta} = \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta} \ , \tag{7.9}$$

where K_i are the eigenvalues of the Kondo coupling matrix $J_f^*(\mathbf{k},\mathbf{q})$. Substituting the definition of the γ fields (eq. 7.7) leads to the equation

$$J_f^*(\mathbf{k},\mathbf{q}) = \sum_i \Lambda_{k,i} K_i \Lambda_{i,q}^\dagger \implies \mathcal{J} = \Lambda K \Lambda^\dagger \ , \tag{7.10}$$

where $\mathcal{J}$ is the Kondo coupling matrix with matrix elements $J_f^*(\mathbf{k}, \mathbf{q})$, K is a diagonal matrix with the eigenvalues (K_i) of $\mathcal{J}$ along its diagonal, and Λ is the matrix $\Lambda_{i,k}$ (the rotated indices i along the row, and the earlier momentum indices along the column). From eq. 7.10, it is clear that since $\mathcal{J}$ is Hermitian, the matrix Λ is simply the unitary matrix which diagonalises the Kondo matrix $\mathcal{J}$:

$$\Lambda^\dagger \mathcal{J} \Lambda = K \ . \tag{7.11}$$

The prescription is therefore as follows. We compute the unitary transformation Λ that diagonalises $\mathcal{J}$ and obtain the eigenvalues $\{K_i\}$. Using the eigenvalues, we work with the sparser Hamiltonian

$$\vec{S}_f \cdot \sum_{\alpha,\beta} \vec{\sigma}_{\alpha,\beta} \sum_i K_i \gamma_{i,\alpha}^\dagger \gamma_{i,\beta} \ . \tag{7.12}$$

If we have to calculate a certain correlation $C_{k,q}$, we instead calculate the correlation matrix $\tilde{C}_{i,j}$, and then transform back using the unitary transformation:

$$C = \Lambda \tilde{C} \Lambda^\dagger \ . \tag{7.13}$$

2.8 Measures of entanglement and their computation

The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [201–206]. By tracking quantum correlations among particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [207, 208, 211, 215]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [203, 228, 229].

2.8.1 Reduced density matrix elements as equal-time correlation functions

A density matrix plays the role of a probability density function (PDF) for a quantum system. A classical variable X that can take values $x \in [-X^*, X^*]$ has a PDF $F(X = x)$, which is used, for example, for calculating the average value (expectation value) of the variable:

$$\langle X \rangle = \int_{-X^*}^{X^*} x F(x) dx \ . \tag{8.1}$$

In quantum mechanics, the variables we are interested in are operators O, and a quantum PDF ρ will then give the expectation value of the operator, in a fashion similar to its classical counterpart:

$$\langle O \rangle = \sum_\phi \langle \phi | \rho O | \phi \rangle \ ; \tag{8.2}$$

here, the integral over the classical values x is replaced by a sum over the all possible quantum states $|\phi\rangle$ (complete orthonormal basis) of the system. However, we already know that the expectation value of a quantum operator is given by

$$\langle O\rangle = \langle\Psi|O|\Psi\rangle \ . \tag{8.3}$$

To bring this form closer to the one in terms of ρ (eq. 2), we insert an identity resolution: $1 = \sum_\phi |\phi\rangle\langle\phi|$:

$$\langle O\rangle = \sum_\phi \langle\Psi|O|\phi\rangle\langle\phi|\Psi\rangle = \sum_\phi \langle\phi|\Psi\rangle\langle\Psi|O|\phi\rangle \ . \tag{8.4}$$

A comparison of this equation to eq. (2) gives

$$\rho = |\Psi\rangle\langle\Psi| \ ; \tag{8.5}$$

a density matrix (also called density operator) is therefore formally defined as the outer product of the wavefunction with its conjugate. It encodes all information about the state; for example, the probability of measuring the system to be at a position x (hence in a state $|x\rangle$) is given by $P(x) = \langle x|\rho|x\rangle$. From its definition in eq. (5), the DM has the very important property that it has unity trace:

$$\mathrm{Tr}[\rho] = 1 \ , \tag{8.6}$$

where $\mathrm{Tr}[\cdot]$ is the trace of an operator. This is seen very easily be taking the trace in an orthonormal basis in which the state $|\Psi\rangle$ is one of the basis states. For example, consider the DM

$$\rho = \frac{1}{\sqrt{2}}\left[|\Psi_1\rangle\langle\Psi_1| + |\Psi_2\rangle\langle\Psi_2|\right] \ , \tag{8.7}$$

where $|\Psi_{1,2}\rangle$ are orthogonal normalised states. The trace of ρ^2 for such a state is only $1/2$; such states satifying $\mathrm{Tr}[\rho^2] < 1$ are termed *mixed states*; the DM describing such states cannot be written in the form $\rho = |\Psi\rangle\langle\Psi|$ for any wavefunction $|\Psi\rangle$. On the other hand, the DMs that *can* be expressed like that are termed *pure states*, and have $\mathrm{Tr}[\rho^2] = 1$.

One way in which a pure state can be realistically converted into a mixed state is through a measurement process. Consider the zero total spin state formed by two spins:

$$|\Psi\rangle = \frac{1}{2}\left[|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle\right] \ , \rho = |\Psi\rangle\langle\Psi| \ . \tag{8.8}$$

The state $|\uparrow,\downarrow\rangle$ has the first spin pointing up and the second one down, and this is in a superposition with the its flipped counterpart $|\downarrow,\uparrow\rangle$. This is clearly a pure state.

If we now measure the state of the second spin, it will collapse into one of it's two states; the other spin will collapse into the opposite state (given the antiparallel configurations of the spins). What this means is that until one checks what the result of the measurement is, the first spin remains in a statistical superposition of both outcomes (up and down). This is formally expressed through a partial trace operation: the state of the first spin after a measurement of the second spin is obtained by taking the full density matrix ρ and computing its trace over the states $|\sigma_B\rangle$ of the second spin, leaving out the states of the first spin from the trace:

$$\rho_1 = \text{Tr}_2[\rho] = \sum_{\sigma_2} \langle\sigma_2|\rho|\sigma_2\rangle \ ; \tag{8.9}$$

where $|\sigma_2\rangle$ sums over the states $|\uparrow\rangle$ and $|downarrow\rangle$ of the second spin. this partial tracing operation Tr_2 codifies the measurement process, and the partially traced density matrix ρ_1 is termed a reduced density matrix (RDM) of the first spin, because it only captures information about the first spin (the second spin has been traced out).

To see the connection with mixedness, let us compute the RDM for the first spin explicitly:

$$\rho_1 = \langle\uparrow_2|\rho|\uparrow_2\rangle + \langle\downarrow_2|\rho|\downarrow_2\rangle = \frac{1}{2}\left[|\uparrow_1\rangle\langle\uparrow_1| + |\downarrow_1\rangle\langle\downarrow_1|\right] \ , \tag{8.10}$$

where the numbered subscripts on the kets and bras indicate that they only act on the respective Hilbert spaces. To see that ρ_1 describes a mixed state, we adopt a specific representation:

$$\begin{aligned} |\uparrow_1\rangle &= \begin{pmatrix}1\\0\end{pmatrix}, |\downarrow_1\rangle = \begin{pmatrix}0\\1\end{pmatrix} \ , \\ \implies \rho_1 &= \frac{1}{2}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}, \ \rho_1^2 = \frac{1}{4}\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix} \ . \end{aligned} \tag{8.11}$$

The trace of ρ_1^2 is half, hence it is a mixed state.

The deviation of trace of ρ^2 from its maximum value of unity can be thought of as a measure of the lost information. In the example above, when we traced out the states of the second spin, we lost some information about the state of the first spin as well. This is because the two spins were *entangled*; they had quantum correlations between them. The entanglement arises from the fact that the states of the two spins are anti-correlated.

If instead we had chosen a non-entangled parent state to start from, the reduced density matrix would not have been mixed. For example, consider the state

$$|\Psi\rangle = \frac{1}{2}\left(|\uparrow_1\rangle + |\downarrow_1\rangle\right) \otimes \left(|\uparrow_2\rangle + |\downarrow_2\rangle\right) \ . \tag{8.12}$$

It is easy to see that this state does not contain any correlations between the two spins - no matter the configuration of the second spin, the first spin is always in the state $|\uparrow_1\rangle + |\downarrow_1\rangle$. The RDM for the first spin comes out to be

$$\rho_1 = \frac{1}{2}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix}, \ \rho_1^2 = \frac{1}{4}\begin{pmatrix}1 & 1\\1 & 1\end{pmatrix} \ , \tag{8.13}$$

revealing that ρ_1 is not mixed.

The above method of calculating an RDM works for simple cases but can fail if (i) we have fermion signs to account for, or (ii) the full basis cannot be expressed as a direct product of local basis sets. The former arises if we are dealing with fermions, and the latter arises if we are working in a rotated basis. We now derive an alternative method for calculating the RDM, one that automatically takes care of the fermion signs and works even if the basis is rotated and complicated to write down. We start by defining the RDM more precisely. Given a state $|\Psi\rangle$ and a bipartition into subsystems S_1 and its complement S_2, the density matrix for the state is $\rho = |\Psi\rangle\langle\Psi|$. A basis state of the full system can be written as $|A,\alpha\rangle = |A\rangle \otimes |\alpha\rangle$, where $\{A\}$ and $\{\alpha\}$ are complete sets of basis states describing S_1 and S_2 respectively. The RDM for S_2 is then defined as

$$\rho_1 = \mathrm{Tr}_\alpha \rho = \sum_\alpha \langle\alpha|\rho|\alpha\rangle \ . \tag{8.14}$$

The matrix elements of ρ_1 in the basis $\{A\}$ can be expressed as

$$\langle A|\rho_1|B\rangle = \sum_\alpha \langle A,\alpha|\, \rho \,|B,\alpha\rangle \ . \tag{8.15}$$

We next define an operator $O_{A,B}$ that represents scattering processes between the various states of subsystem S_1: $O_{A,B} = |A\rangle\langle B| \otimes I_2$, but acts as identity on S_2. The expectation value of this operator is

$$\langle O_{A,B}\rangle = \mathrm{Tr}(\rho O_{A,B}) = \sum_{C,\beta} \langle C,\beta|\rho|A\rangle\langle B|C,\beta\rangle = \sum_\beta \langle B,\beta|\rho|A,\beta\rangle \ . \tag{8.16}$$

Comparing with eq. 8.15, we obtain the relation

$$\langle A|\rho_1|B\rangle = \langle O_{B,A}\rangle \ . \tag{8.17}$$

This result is significant because it guarantees that in order to calculate the RDM for a subsystem, it is enough to calculate the expectation values of all inter-state off-diagonal transition operators. For example, we take subsystem S_1 to be a two fermion system represented by the operators c_1 and c_2. Assuming a basis $\{|0,0\rangle, |0,1\rangle, |1,0\rangle, |1,1\rangle\}$, where the two entries represent the occupancies of the two fermions, the RDM for this subsystem is determined by the following matrix:

$$\rho_1 = \begin{pmatrix} \langle h_1 h_2\rangle & \langle h_1 c_2^\dagger\rangle & \langle c_1^\dagger h_2\rangle & \langle c_1^\dagger c_2^\dagger\rangle \\ & \langle h_1 n_2\rangle & \langle c_1^\dagger c_2\rangle & \langle c_1^\dagger n_2\rangle \\ & & \langle n_1 h_2\rangle & \langle n_1 c_2^\dagger\rangle \\ & & & \langle n_1 n_2\rangle \end{pmatrix} , \tag{8.18}$$

where the blank entries can be determined from hermiticity. To see how the operators are determined, take the example of the top right element. The two basis states for that matrix element are $|0,0\rangle$ and $|1,1\rangle$.

Quantum Fisher information: a witness of multipartite entanglement

The quantum Fisher information (QFI) F_Q [413,414] quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle\langle\psi|$, $F_Q(\psi,\hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4\left(\langle\psi|\hat{O}^2|\psi\rangle - \langle\psi|\hat{O}|\psi\rangle^2\right) , \tag{8.19}$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}\,(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [413,414]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. If $\varepsilon_m(\psi) \equiv 1 - |\langle m|\psi\rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4\left[\frac{1}{2} - \sum_m m^2\varepsilon_m - \left(\sum_m m\varepsilon_m\right)^2\right] . \tag{8.20}$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (8.20) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0, \varepsilon_{m\neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and a reduced susceptibility towards symmetry-breaking.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4\left(\langle\hat{O}_2\hat{O}_1^\dagger\rangle\Big|_{\hat{O}_1^\dagger=\hat{O}_2} - \langle\hat{O}_2\hat{O}_1^\dagger\rangle^2\Big|_{\hat{O}_1=1}\right) = 4\left[G^<_{O_2,O_2}(t\to\infty) - \left(G^<_{1,O_2}(t\to\infty)\right)^2\right] \tag{8.21}$$

Further, the retarded Greens function defined above in eq. (??) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2}G^R_{O_2,O_2}(t\to\infty) = \langle(O_2)^2\rangle$ and $\frac{i}{2}G^R_{1,O_2}(t\to\infty) = \langle O_2\rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2iG^R_{O_2,O_2}(t \to \infty) + \left(G^R_{1,O_2}(t \to \infty)\right)^2 . \tag{8.22}$$

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [414]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n \left|n\right\rangle \left\langle n\right|$ where $(E_n, \left|n\right\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n}/\sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [414]

$$F_Q(\rho, \hat{O}) = 2\sum_{m,n} \frac{(f_m - f_n)^2}{f_m + f_n} |\left\langle m\right|\hat{O}\left|n\right\rangle|^2 . \tag{8.23}$$

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2\int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n)\, \delta(\omega + E_m - E_n) |\left\langle m\right|\hat{O}\left|n\right\rangle|^2 . \tag{8.24}$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^<$ and the greater Greens function $G^>$ that we will now define admit spectral representations in terms of the eigenstates $\left|n\right\rangle$ of the full problem:

$$G^<_{O^\dagger,O}(t) \equiv i\left\langle \hat{O}^\dagger \hat{O}(t)\right\rangle \implies G^<_{O^\dagger,O}(\omega) = 2\pi i \textstyle\sum_{m,n} f_n \delta(\omega + E_m - E_n) |\left\langle m\right|\hat{O}^\dagger\left|n\right\rangle|^2 , \tag{8.25}$$

$$G^>_{O^\dagger,O}(t) \equiv -i\left\langle \hat{O}(t)\hat{O}^\dagger\right\rangle \implies G^>_{O^\dagger,O}(\omega) = -2\pi i \textstyle\sum_{m,n} f_m \delta(\omega + E_m - E_n) |\left\langle m\right|\hat{O}\left|n\right\rangle|^2 . \tag{8.26}$$

If $\hat{O}$ is a Hermitian operator, eqs. (8.25) can be used to rewrite the QFI in terms of $G^{\lessgtr}_{O^\dagger,O}(\omega)$. Indeed, by combining eqs. (8.25) and eq. (8.24) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i}\int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G^>_{O,O}(\omega) + G^<_{O,O}(\omega)\right) . \tag{8.27}$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t')\left\langle\left[\hat{O}(t), \hat{O}(t')\right]\right\rangle$. By going to the frequency domain, the imaginary part of such a function can be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned}
C_{\hat{O}}(\omega) &= i\int_0^{\infty} d(t - t')e^{i\omega(t-t')}\left(\left\langle\hat{O}(t)\hat{O}(t')\right\rangle - \left\langle\hat{O}(t')\hat{O}(t)\right\rangle\right) \\
\implies C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i\int_0^{\infty} d(t - t')\left(e^{i\omega(t-t')} - e^{-i\omega(t-t')}\right)\left(\left\langle\hat{O}(t)\hat{O}(t')\right\rangle - \left\langle\hat{O}(t')\hat{O}(t)\right\rangle\right) \\
&= i\int_{-\infty}^{\infty} d(t - t')e^{i\omega(t-t')}\left(\left\langle\hat{O}(t)\hat{O}(t')\right\rangle - \left\langle\hat{O}(t')\hat{O}(t)\right\rangle\right) \\
&= -\left(G^>_{O,O}(\omega) + G^<_{O,O}(\omega)\right) .
\end{aligned}$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [414]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh\left(\beta\omega/2\right) \left(C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^*\right) . \tag{8.28}$$

2.8.2 Decomposition of entanglement entropy in terms of multipartite information I_n

We first define a partitioning of the entire system into N disjoint regions, $\mathcal{S}_N = \{\nu_1, \nu_2, \ldots \nu_N\}$. Going beyond bipartite mutual information, one can define the $n-$partite information I_n between the members of $\mathcal{S}_n = \{\nu_1, \nu_2, \ldots, \nu_n\}$, which is itself a subset of the exhaustive set of regions $\mathcal{S}_N$:

$$I_n(\nu_1, \ldots, \nu_n) = \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{\text{EE}}(\sigma) , \tag{8.29}$$

where σ is a subset drawn from the power set of $\mathcal{S}_n$, and $|\sigma|$ is the number of members in the subset σ. The 4-partite information can be written as

$$\begin{aligned} &I_4(\nu_1, \nu_2, \nu_3, \nu_4) \\ &= S_{\text{EE}}(\nu_1) + S_{\text{EE}}(\nu_2) + S_{\text{EE}}(\nu_3) + S_{\text{EE}}(\nu_4) - S_{\text{EE}}(\nu_1 \cup \nu_2) - S_{\text{EE}}(\nu_1 \cup \nu_3) \\ &- S_{\text{EE}}(\nu_1 \cup \nu_4) - S_{\text{EE}}(\nu_2 \cup \nu_3) - S_{\text{EE}}(\nu_2 \cup \nu_4) - S_{\text{EE}}(\nu_3 \cup \nu_4) + S_{\text{EE}}(\nu_1 \cup \nu_2 \cup \nu_3) \\ &+ S_{\text{EE}}(\nu_1 \cup \nu_2 \cup \nu_4) + S_{\text{EE}}(\nu_1 \cup \nu_3 \cup \nu_4) + S_{\text{EE}}(\nu_2 \cup \nu_3 \cup \nu_4) + S_{\text{EE}}(\nu_1 \cup \nu_2 \cup \nu_3 \cup \nu_4) . \end{aligned} \tag{8.30}$$

One can also define the average $n-$partite information $\bar{I}_n$ by considering all sets of n regions:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{\sigma \in \mathcal{P}(S_n)} (-1)^{1+|\sigma|} S_{\text{EE}}(\sigma) , \tag{8.31}$$

where $C_n(\mathcal{S}_N)$ represents all ways of choosing n distinct elements out of the full set $\mathcal{S}_N$ of subsystems.

The expression for $\bar{I}_n$ can be simplified by splitting the summation over the elements of the power set $\mathcal{P}$ into sets of definite number:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{m=1}^{n} \sum_{\sigma_m \in \mathcal{P}_m(S_n)} (-1)^{1+|\sigma|} S_{\text{EE}}(\sigma_m) , \tag{8.32}$$

where $\mathcal{P}_m$ represents all subsets of the power set that have m elements. With some combinatorial magic, it can be shown that

$$\sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{\sigma_m \in \mathcal{P}_m(S_n)} = \binom{N-m}{n-m} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} . \tag{8.33}$$

To see how this comes about, note that what this equation says is that summing over combinations of subsets of length n of the super set S_N and further summing over all subsets of length $m \leq n$ for each set S_n is equivalent to summing just once over subsets of length m of the super set S_N but now each subset of length m appears with a frequency $f_m = \binom{N-m}{n-m}$. The value for the frequency f_m is simply the number of subsets S_n in which the smaller subset σ_m appears. This can be calculated by allowing m of the n slots in S_n to be filled by the members of σ_m, and the remaining $n-m$ slots then have to be filled by choosing from the remaining $N-m$ members of the super set S_N; there's obviously $\binom{N-m}{n-m}$ ways of carrying this out.

Implementing this into the expression for $\bar{I}_n$ gives

$$\bar{I}_n = \sum_{m=1}^{n} (-1)^{1+m} \frac{\binom{n}{m}}{\binom{N}{m}} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} S_{\text{EE}}(\sigma_m) = \sum_{m=1}^{n} (-1)^{1+m} \binom{n}{m} S_{\text{EE}}^{(m)} , \tag{8.34}$$

where we have define the average entanglement entropy $S_{\text{EE}}^{(m)}$ for all $m-$member sets of regions:

$$S_{\text{EE}}^{(m)} = \frac{1}{\binom{N}{m}} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} S_{\text{EE}}(\sigma_m) . \tag{8.35}$$

With most of the machinery in place, we would now like to express the entanglement entropy S_{EE} purely in terms multipartite measures I_n, for the reasons stated above. For a system that is invariant under real-space translation (as our lattice model is), the entanglement entropy of any real-space interval should only depend on the extent of the interval. It therefore suffices to calculate the entanglement entropy averaged over all such intervals, and the corresponding measure in our formulation is $S_{\text{EE}}^{(1)}$.

We will now express $S_{\text{EE}}^{(1)}$ in terms of $\bar{I}_n$, by eliminating $S_{\text{EE}}^{(m)} (m > 1)$ from the equations. We adopt the ansatz

$$S_{\text{EE}}^{(1)} = \sum_{n=2}^{N} \lambda_n \bar{I}_n = \sum_{n=2}^{N} \lambda_n \binom{n}{m} \sum_{m=1}^{n} (-1)^{1+m} S_{\text{EE}}^{(m)} . \tag{8.36}$$

In order to separate $S_{\text{EE}}^{(1)}$ from the rest of the $S_{\text{EE}}^{(m)}$, we exchange the order of summation:

$$S_{\text{EE}}^{(1)} = \sum_{m=1}^{N} (-1)^{1+m} S_{\text{EE}}^{(m)} \sum_{n=m}^{N} \lambda_n \binom{n}{m} - S_{\text{EE}}^{(1)} \lambda_1 = S_{\text{EE}}^{(1)} \sum_{n=2}^{N} n\lambda_n - \sum_{m=2}^{N-1} S_{\text{EE}}^{(m)} (-1)^m \sum_{n=m}^{N} \binom{n}{m} \lambda_n , \tag{8.37}$$

where the constraint of $m \leq n$ in the previous form is now enforced by letting n run from m to N. The $N-1$ coefficients $\{\lambda_n\}$ will be determined by requiring that the $S_{\text{EE}}^{(m)} (m > 1)$ vanish on the right-hand side, as well as by matching the coefficients of $S_{\text{EE}}^{(1)}$ on both sides. Note that m runs only upto $N-1$ in the second term; this is because, $S_{\text{EE}}^{(N)}$ is the entanglement entropy of the full system and hence automatically zero. The equations are

$$\sum_{n=2}^{N} n\lambda_n = 1, \tag{8.38}$$

$$\sum_{n=m}^{N} \binom{n}{m} \lambda_n = 0; \; m = 2, \ldots, N-1 . \tag{8.39}$$

We focus on the second equation first, since it is very short for m close to N. For $m = N-1$ and $m = N-2$, we get

$$\begin{aligned} m &= N-1 : \lambda_{N-1} + N\lambda_N = 0 \implies \lambda_{N-1} = -N\lambda_N \\ m &= N-2 : \lambda_{N-2} + (N-1)\lambda_{N-1} + \frac{N(N-1)}{2}\lambda_N = 0 \implies \lambda_{N-2} = \frac{N(N-1)}{2}\lambda_N \ . \end{aligned} \tag{8.40}$$

We extrapolate this to the general solution,

$$\lambda_m = (-1)^m \binom{N}{m}\lambda_N, \ m < N \ , \tag{8.41}$$

and then prove that if this is true for λ_{N-m}, it also solves the equation for λ_{N-m-1}, thus amounting to a proof by mathematical induction. The general equation for λ_{N-m-1} is

$$\sum_{n=N-m-1}^{N} \binom{n}{N-m-1}\lambda_n = 0 \implies \lambda_{N-m-1} + \sum_{m'=0}^{m}\binom{N-m'}{N-m-1}\lambda_{N-m'} = 0 \ . \tag{8.42}$$

Substituting the extrapolated solution gives

$$\begin{aligned} \lambda_{N-m-1} = -\sum_{m'=0}^{m}(-1)^{N-m'}\binom{N-m'}{N-m-1}\binom{N}{m'}\lambda_N &= \binom{N}{m+1}\lambda_N\sum_{m'=0}^{m}(-1)^{N-m'-1}\binom{m+1}{m'} \\ &= (-1)^{N-m-1}\binom{N}{m+1}\lambda_N \ , \end{aligned} \tag{8.43}$$

which is consistent with the general solution eq. 8.41. This determines all coefficients λ_n for $n < N$ in terms of λ_N. To determine λ_N itself, we use eq. 8.38. Substituting eq. 8.41 into it gives

$$\lambda_N \sum_{n=2}^{N} n(-1)^n \binom{N}{n} = 1 \implies \lambda_N = \frac{(-1)^N}{N} \ . \tag{8.44}$$

The decomposition of entanglement entropy into multipartite measures is now complete:

$$\begin{aligned} S^{(1)}_{\text{EE}} &= \frac{1}{N}\sum_{n=2}^{N}(-1)^n\binom{N}{n}\bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!}\sum_{S_n\in C_n(\mathcal{S}_N)} I_n(S_n), \end{aligned} \tag{8.45}$$

where we have assumed N is even in order to drop the global phase factor $(-1)^N$.

2.9 Wavefunction Renormalisation Group

2.9.1 General idea

A reverse URG analysis involves studying the evolution of wavefunctions from the IR to the UV, by implementing inverse renormalisation group transformations. This is made possible by the fact that the IR fixed point Hamiltonians are often tractable and allow the computation of wavefunctions. Such an analysis gives insight into how states change as high-energy quantum fluctuations are resolved during the Hamiltonian renormalisation from the UV to the IR. It also allows studying the evolution of entanglement measures and correlation functions.

We will start with a very simple IR ground state wavefunction, and then go back towards the UV ground state by applying the inverse unitary operator $U^\dagger$:

$$U : \underbrace{|1, 2, ..., N\rangle}_{\text{UV ground state}} \to |1, 2, ..., N-1\rangle |N\rangle \to ... \to \underbrace{|1, 2, ..., N^*\rangle |N^*+1\rangle ... |N\rangle}_{\text{IR ground state}}$$

$$U^\dagger : \underbrace{|1, 2, ..., N^*\rangle |N^*+1\rangle ... |N\rangle}_{\text{IR ground state}} \to |1, 2, ..., N^*+1\rangle |N^*+2\rangle ... |N\rangle \to ... \to \underbrace{|1, , 2, ..., N\rangle}_{\text{UV ground state}}$$

The first process is the forward RG which we used to obtain the scaling equations. The second process is the reverse RG which we will undertake now. The method has already been applied to study the renormalisation of entanglement in various models of strong correlation. In general, we start with an IR wavefunction that consists a certain number of momentum states n_1 still entangled with the impurity, and the remaining momentum states n_2 disentangled from the impurity. The former are said to be part of the emergent Kondo cloud, while the latter are said to be part of the integrals of motion (IOMs) and appear in direct product with the cloud+impurity system in the ground state wavefunction. Each momentum state is tagged with two conduction bath levels, one above the Fermi surface and one below. These will be represented as $k, \pm$. If we represent the emergent cloud momentum states as $\{k_i\}$ and the IOM states as $\{q_i\}$, the ground state wavefunction can be written as

$$|\Psi\rangle_0 = \left(\otimes_{i=1}^{n_2} |q_i, -, \uparrow\rangle |q_i, -, \downarrow\rangle\right) |\Phi\rangle_{\text{cloud}} \left(\otimes_{i=1}^{n_2} |q_i, +, \uparrow\rangle |q_i, +, \downarrow\rangle\right) \tag{9.1}$$

Throughout, we have assumed that the IOMs below the Fermi surface are occupied while those above are unoccupied. This means:

$$|\Psi\rangle_0 = \left(\otimes_{i=1}^{n_2} |\hat{n}_{q_i,-,\uparrow} = 1\rangle |\hat{n}_{q_i,-,\downarrow} = 1\rangle\right) |\Phi\rangle_{\text{cloud}} \left(\otimes_{i=1}^{n_2} |\hat{n}_{q_i,+,\uparrow} = 0\rangle |\hat{n}_{q_i,+,\downarrow} = 0\rangle\right) \tag{9.2}$$

$|\Phi\rangle_{\text{cloud}}$ will be obtained by diagonalising a small system of n_1 conduction bath k states. This completes the construction of the IR ground state $|\Psi\rangle_0$. The algorithm of reverse RG then involves applying unitary transformations on this IR wavefunction; the unitary transformations will be the inverse of those that were applied during the forward RG algorithm. Since the forwards RG transformations decoupled k states and transformed them into the IOMS, the inverse transformations will take the momentum states in the IOMs and re-entangle them with the impurity+cloud system, one at a time. This is shown in fig. 2.11.

The next step is to write down the unitaries that will take us from the IR ground state to the UV ground state. In the forward RG, we used the following unitary for decoupling the shell ϵ_q and spin β is

$$U_{q\beta} = \frac{1}{\sqrt{2}} \left[1 + \eta_{0\beta} + \eta^\dagger_{1\beta}\right] . \tag{9.3}$$

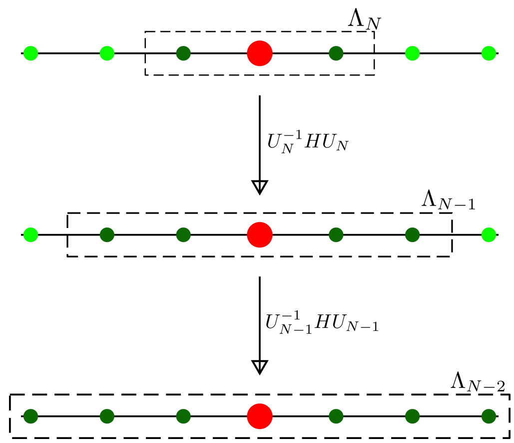

Figure 2.11: *We start with a Hamiltonian with an impurity site (red) coupled with two conduction electrons (dark green), with four other decoupled electrons (bright green). The dotted rectangle represents the emergent window $\left(-\Lambda_j, \Lambda_j\right)$ at each step; the electrons inside that rectangle are still entangled with the impurity, while the ones inside have been decoupled. The next step of reverse RG involves applying the inverse transformation on the Hamiltonian, which will couple two more electrons from the IOMS (hence four dark green circles in the second step), leading to an enlargement of the emergent window. The unitary varies for each step, hence the notation U_j.*

Here, the subscripts 0 and 1 indicate it decouples an electron above and below the Fermi surface respectively. The inverse transformation for re-entangling $q\beta$ is

$$U_{q\beta}^{\dagger} = \frac{1}{\sqrt{2}}\left[1 + \eta_{0\beta}^{\dagger} + \eta_{1\beta}\right] . \tag{9.4}$$

The wavefunction after reversing one step of the RG will thus be

$$|\Psi\rangle_1 = U_{q\uparrow}^{\dagger} U_{q\downarrow}^{\dagger} |\Psi\rangle_0 \tag{9.5}$$

Here, q is the first momentum state immediately outside the emergent cloud. Re-entangling further momentum states using their respective unitaries lead to the sequence of wavefunctions $|\Psi\rangle_2 , |\Psi\rangle_3$ and so on. A straightforward way of calculating entanglement and correlations across the RG flow is to create a second-quantised prototypical form of the IR state $|\Psi_0\rangle$, and then apply the inverse unitary operators $\{U_i\}$ to this wavefunction in order to generate the family of states $|\Psi_i\rangle = U_i |\Psi_{i-1}\rangle , i > 0$. Quantities like entanglement entropy and correlations can then be calculated by using this family of states. If the unitary is of a specific form, there is instead a more efficient method of doing this, which is what we will describe in the next subsection.

2.9.2 Evolution equation for state tensor coefficients

In the following, we restrict ourselves to the reverse RG analysis of an impurity problem hybridising with the local degree of freedom of a conduction bath. At every step, we allow two states from the IOMS to interact with the already-entangled states (one above the Fermi energy and one below, in order to maintain particle-hole symmetry). After the j^{th} application of the reverse RG transformation, $2j$ number of states are entangled with the impurity site, and the state can be represented as

$$|\Psi_j\rangle = \sum_{\nu_d,\nu_1,\ldots,\nu_{2j}} C^{(j)}_{\nu_d,\nu_1,\ldots,\nu_{2j},I_{2j+1}} |\nu_d\rangle |\nu_1\rangle |\nu_2\rangle \ldots |\nu_{2j}\rangle \otimes |I_{2j+1}\rangle \ , \tag{9.6}$$

where $|\nu_d\rangle$ and $|\nu_i\rangle$ ($i \in [1, 2j]$) sum over the configurations of the impurity site and the effective zeroth site configuration at the j^{th} step. $|I_{2j+1}\rangle$ is the configuration of the integrals of motion from the $(2j+1)^{\text{th}}$ site and beyond. We define the product state $|\psi_{dz}(j)\rangle = |\nu_d\rangle |\nu_1\rangle \ldots |\nu_{2j}\rangle$ for future reference.

We consider a restricted form of RG evolution, where the unitary transformation from IR towards UV can be expressed at any step as

$$U_j^\dagger = 1 + \sum_{i\in[1,2j]} \sum_{\nu_d,\mu_d,\nu_i,\mu_i} \mathcal{T}^{\mu_d}_{\nu_d} \left(J^{\mu_d,\mu_i}_{\nu_d,\nu_i} c_{2j+1} \mathcal{T}^{\mu_i}_{\nu_i} + K^{\mu_d,\mu_i}_{\nu_d,\nu_i} c^\dagger_{2j+2} \mathcal{T}^{\mu_i}_{\nu_i}\right) , \tag{9.7}$$

where $\mathcal{T}^{\nu}_{\alpha} = |\nu\rangle\langle\alpha|$ is a transition operator and sums over all transition operators of the specific subspace (impurity or bath sites). We first obtain the action of the transformation $U_j^\dagger$ on the basis states. We have

$$\begin{aligned}
&\mathcal{T}^{\nu_d}_{\mu_d} c^\dagger_{2j+2} \mathcal{T}^{\nu_i}_{\mu_i} |\mu_d\rangle \ldots |\mu_i\rangle \ldots |2_{2j+1}\rangle |0_{2j+2}\rangle \\
&= \mathcal{T}^{\nu_d}_{\mu_d} c^\dagger_{2j+2} (-1)^{\mu_d + \sum_{k<i}\nu_i} |\mu_d\rangle \ldots |\nu_i\rangle \ldots |2_{2j+1}\rangle |0_{2j+2}\rangle \\
&= \mathcal{T}^{\nu_d}_{\mu_d} (-1)^{\mu_d + \sum_{k<i}\nu_i} (-1)^{\mu_d + \sum_{k}\nu_i} |\mu_d\rangle \ldots |\nu_i\rangle \ldots |2_{2j+1}\rangle |1_{2j+2}\rangle \\
&= (-1)^{\sum_{k\geq i}\nu_i} |\nu_d\rangle \ldots |\nu_i\rangle \ldots |2_{2j+1}\rangle |1_{2j+2}\rangle \ ,
\end{aligned} \tag{9.8}$$

and

$$\mathcal{T}^{\nu_d}_{\mu_d} c_{2j+1} \mathcal{T}^{\nu_i}_{\mu_i} |\mu_d\rangle \ldots |\mu_i\rangle \ldots |2_{2j+1}\rangle |0_{2j+2}\rangle = (-1)^{\sum_{k\geq i}\nu_i} |\nu_d\rangle \ldots |\nu_i\rangle \ldots |1_{2j+1}\rangle |0_{2j+2}\rangle \ . \tag{9.9}$$

Using these expressions, we construct the state at the $(j+1)^{\text{th}}$ step:

$$\begin{aligned}
|\Psi_{j+1}\rangle = &\sum_{\nu_d,\nu_1,\ldots,\nu_j} C^{(j)}_{\nu_d,\nu_1,\ldots,} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\
&+ \sum_i \sum_{\mu_d,\mu_i,\nu_d,\nu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,} J^{\nu_d,\nu_i}_{\mu_d,\mu_i} (-1)^{\sum_{k\geq i}\nu_k} |\psi_{dz}(j)\rangle |1_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\
&+ \sum_i \sum_{\mu_d,\mu_i,\nu_d,\nu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots} K^{\nu_d,\nu_i}_{\mu_d,\mu_i} (-1)^{\sum_{k\geq i}\nu_k} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |1_{2j+2}\rangle |I_{2j+3}\rangle
\end{aligned} \tag{9.10}$$

Gathering similar terms and normalising the state, we can write it as

$$\begin{aligned}
|\Psi_{j+1}\rangle = &\frac{1}{L_j} \sum_{\nu_d,\{\nu_i\}} |\psi_{dz}(j)\rangle \Big[C^{(j)}_{\nu_d,\{\nu_i\},1_{2j+1},0_{2j+2},I_{j+1}} |2_{2j+1}\rangle |0_{2j+2}\rangle + C^{(j)}_{+,\nu_d,\{\nu_i\}} |2_{2j+1}\rangle |1_{2j+2}\rangle \\
&+ C^{(j)}_{-,\nu_d,\{\nu_i\}} |1_{2j+1}\rangle |0_{2j+2}\rangle \Big] |I_{2j+3}\rangle \ ,
\end{aligned} \tag{9.11}$$

where $L_j^2 = \sum_{\nu_d,\{\nu_i\}} \left({C^{(j)}_{\nu_d,\{\nu_i\}}}^2 + {C^{(j)}_{+,\nu_d,\{\nu_i\}}}^2 + {C^{(j)}_{-,\nu_d,\{\nu_i\}}}^2\right)$ defines the normalisation factor, and

$$\begin{aligned} C_{+,\nu_d,\{\nu_i\}} &= \sum_{i,\mu_d,\mu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} K^{\nu_d,\nu_i}_{\mu_d,\mu_i} (-1)^{\sum_{k\geq i}\nu_k} \\ C_{-,\nu_d,\{\nu_i\}} &= \sum_{i,\mu_d,\mu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} J^{\nu_d,\nu_i}_{\mu_d,\mu_i} (-1)^{\sum_{k\geq i}\nu_k} \ . \end{aligned} \tag{9.12}$$

We can now read off the renormalisation of the state tensor coefficients:

$$\begin{aligned} C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},0_{2j+2},I_{2j+3}} &= \frac{1}{L_j} C^{(j)}_{\nu_d,\nu_1,\ldots,\nu_{2j},2_{2j+1},0_{2j+2},I_{2j+3}} \ , \\ C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},2_{2j+1},1_{2j+2},I_{2j+3}} &= \frac{1}{L_j} \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\mu_d,\mu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} K^{\nu_d,\nu_i}_{\mu_d,\mu_i} \ , \\ C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},0_{2j+2},I_{2j+3}} &= \frac{1}{L_j} \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\mu_d,\mu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} J^{\nu_d,\nu_i}_{\mu_d,\mu_i} \ , \end{aligned} \tag{9.13}$$

We now make an important assumption: the coefficients $K^{\nu_d,\nu_i}_{\mu_d,\mu_i}$ and $J^{\nu_d,\nu_i}_{\mu_d,\mu_i}$ are uniform for the set of states $S(\nu_i)$ and zero for other states. That gives

$$\begin{aligned} C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},1_{2j+2},I_{2j+3}} &= \frac{1}{L_j} K \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\substack{\mu_d\in S(\nu_d),\\ \mu_i\in S(\nu_i)}} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} \ , \\ C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},0_{2j+2},I_{2j+3}} &= \frac{1}{L_j} J \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\substack{\mu_d\in S(\nu_d),\\ \mu_i\in S(\nu_i)}} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} \ , \end{aligned} \tag{9.14}$$

One can also look at renormalisation of expectation values of operators; this is useful for tracking the evolution of correlations and entanglement measures. Consider an operator $O^{\nu_k}_{\mu_k}$ that acts on the k^{th} bath site. The expectation value of this operator at any step can be written as

$$\begin{aligned} \langle O^{\mu_k}_{\nu_k} \rangle_j &= \sum_{\nu_d,\nu_1,\ldots,\nu_{k-1},\nu_{k+1},\ldots,\nu_{2j}} C^{(j)}_{\nu_d,\ldots,\mu_k,\ldots} C^{(j)}_{\nu_d,\ldots,\nu_k,\ldots} \ , \\ \langle O^{\mu_k}_{\nu_k} \rangle_{j+1} &= \sum_{\nu_d,\nu_1,\ldots,\nu_{k-1},\nu_{k+1},\ldots,\nu_{2j+2}} C^{(j+1)}_{\nu_d,\ldots,\mu_k,\ldots} C^{(j+1)}_{\nu_d,\ldots,\nu_k,\ldots} \end{aligned} \tag{9.15}$$

2.10 Kondo model spectral function

For the Kondo model defined by the Hamiltonian

$$H_K = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \mathbf{S}_d \cdot \sigma_{\alpha,\beta} c^\dagger_{k_1,\alpha} c_{k_2,\beta} \ , \tag{10.1}$$

the impurity spin S_d does not have any fermionic operator. In order to define a spectral function for the impurity, we therefore look at the scattering of conduction electrons off it. This is captured by the conduction bath Greens function $G_{k,k'}$, which can be expressed in terms of a $T-$matrix when there are two-particle scattering processes present in the system:

$$G_{k,k'}(\omega) = G^{(0)}_k(\omega) + G^{(0)}_k(\omega)T_{k,k'}(\omega)G^{(0)}_{k'}(\omega) \ , \tag{10.2}$$

where $T_{k,k'}$ is non-zero because of the presence of the impurity spin which scatters momentum states into each other. Because of the local nature of the impurity scattering, the $T-$matrix is also local: $T_{k,k'}(\omega) = T(\omega)$. We therefore use this to define a spectral function:

$$A_{\text{imp}} = -\frac{1}{\pi}\text{Im}\, T(\omega) \ . \tag{10.3}$$

In order to calculate $T(\omega)$, we evaluate $G_{k,k'}$ using the equation of motion method. It is define as

$$G_{k,k'}(t) = -i\theta(t)\langle\langle c_{k,\sigma}(t), c^\dagger_{k',\sigma}\rangle\rangle \ . \tag{10.4}$$

Differentiating against time gives

$$\partial_t G_{k,k'}(t) = -i\delta(t)\langle\langle c_{k,\sigma}, c^\dagger_{k',\sigma}\rangle\rangle - \theta(t)\langle\langle [c_{k,\sigma}(t), H_K], c^\dagger_{k',\sigma}\rangle\rangle \ . \tag{10.5}$$

Evaluating the commutator gives

$$[c_{k,\sigma}(t), H_K] = \epsilon_k c_{k,\sigma} + J\sum_{q,\beta}\mathbf{S}_d\cdot\sigma_{\sigma,\beta}c_{q\beta} \ . \tag{10.6}$$

For convenience we define the local bath operator $c_{0,\beta} = \sum_q c_{q,\beta}$. Substituting the commutator back into the expression gives

$$\partial_t G_{k,k'}(t) = -i\delta(t)\delta_{k,k'} - \theta(t)\epsilon_k\langle\langle c_{k,\sigma}(t), c^\dagger_{k',\sigma}\rangle\rangle - \theta(t)J\sum_\beta\langle\langle \mathbf{S}_d\cdot\sigma_{\sigma,\beta}c_{0,\beta}, c^\dagger_{k',\sigma}\rangle\rangle(t) \ . \tag{10.7}$$

Defining a new Greens function $\tilde{G}$,

$$\tilde{G}_{k'}(t) = -i\theta(t)\sum_\beta\langle\langle \mathbf{S}_d\cdot\sigma_{\sigma,\beta}c_{0,\beta}, c^\dagger_{k',\sigma}\rangle\rangle \ , \tag{10.8}$$

we have the compact form

$$\partial_t G_{k,k'}(t) = -i\delta(t)\delta_{k,k'} - \theta(t)\epsilon_k\langle\langle c_{k,\sigma}(t), c^\dagger_{k',\sigma}\rangle\rangle - i\tilde{G}_{k'}(t) \ . \tag{10.9}$$

We now transform to frequency space, using the definition $f(t) = \int d\omega e^{-i\omega t}g(\omega)$:

$$\omega G_{k,k'}(\omega) = \delta_{k,k'} + \epsilon_k G_{k,k'}(\omega) + J\tilde{G}_{k'}(\omega) \ . \tag{10.10}$$

We setup the equation of motion for $\tilde{G}$ in a similar fashion. We use the Hermitian conjugate $-i\theta(t)\sum_\beta\langle\langle c_{k',\sigma}(t), \mathbf{S}_d\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle$, because it is easier to work with:

$$\begin{aligned}
\partial_t\tilde{G}_{k'}(t) &= -i\delta(t)\sum_\beta\delta_{\sigma,\beta}\langle\mathbf{S}_d\cdot\sigma_{\sigma,\beta}\rangle - \theta(t)\langle\langle [c_{k',\sigma}(t), H_K], \sum_\beta\mathbf{S}_d\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle \\
&= -i\delta(t)\langle\mathbf{S}^z_d\rangle - \theta(t)\langle\langle(\epsilon_{k'}c_{k',\sigma}(t) + J\sum_\alpha\mathbf{S}_d\cdot\sigma_{\sigma,\alpha}c_{0,\alpha}), \sum_\beta\mathbf{S}_d\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle \\
&= -i\delta(t)\langle\mathbf{S}^z_d\rangle - i\epsilon_{k'}G_{k'}(t) - J\theta(t)\langle\langle\sum_\alpha\mathbf{S}_d\cdot\sigma_{\sigma,\alpha}c_{0,\alpha}, \sum_\beta\mathbf{S}_d\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle \ .
\end{aligned} \tag{10.11}$$

At this point, we define a Kondo excitation operator $O_\sigma = J \sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha} c_{0,\alpha}$. Transforming to frequency space gives

$$\omega \tilde{G}_{k'}(\omega) = \langle S_d^z \rangle + \epsilon_{k'} G_{k'}(\omega) + \frac{1}{J} \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \ , \tag{10.12}$$

which gives an expression for $\tilde{G}_{k'}(\omega)$:

$$(\omega - \epsilon_{k'}) \tilde{G}_{k'}(\omega) = \langle S_d^z \rangle + \frac{1}{J} \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \ . \tag{10.13}$$

Substituting this into eq. 10.10 and noting that the non-interacting Greens function of the conduction bath is $G_k^{(0)} = (\omega - \epsilon_k)^{-1}$, we get

$$(G_k^{(0)})^{-1} G_{k,k'}(\omega) = \delta_{k,k'} + J \left[\langle S_d^z \rangle + \frac{1}{J} \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \right] G_{k'}^{(0)} \ , \tag{10.14}$$

which can be finally compared with the general expression in eq. 10.2:

$$G_{k,k'}(\omega) = G_k^{(0)} + G_k^{(0)} \left[J \langle S_d^z \rangle + \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \right] G_{k'}^{(0)} \ . \tag{10.15}$$

The $T-$matrix for scattering of conduction electrons off the impurity can be read off:

$$T_\sigma(\omega) = J \langle S_d^z \rangle + \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \ , \tag{10.16}$$

2.11 Zero temperature Greens function in frequency domain

2.11.1 Spectral representation of Greens function

The impurity retarded Green's function is defined as

$$G_{d\sigma}(t) = -i\theta(t) \left\langle \left\{ O_\sigma(t), O_\sigma^\dagger \right\} \right\rangle \tag{11.1}$$

where the average $\langle\rangle$ is over a canonical ensemble at temperature T, and O_σ is the excitation whose spectral function we are interested in. What follows is a standard calculation where we write the Green's function in the spectral representation. The ensemble average for the operator O_σ can be written in terms of the exact eigenstates of the Hamiltonian:

$$H \left| n \right\rangle = E_n \left| n \right\rangle , \quad \langle O_\sigma \rangle \equiv \frac{1}{Z} \sum_n \langle n | O_\sigma | n \rangle \, e^{-\beta E_n} \tag{11.2}$$

where $Z = \sum_n e^{-\beta E_n}$ is the partition function and $\{|n\rangle\}$ is the set of eigenfunctions of the Hamiltonian. We can therefore write

$$
\begin{aligned}
&\left\langle\left\{O_\sigma(t), O_\sigma^\dagger\right\}\right\rangle \\
&= \frac{1}{Z}\sum_m e^{-\beta E_m} \langle m| \left\{O_\sigma(t), O_\sigma^\dagger\right\} |m\rangle \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_m} \langle m| \left(O_\sigma(t)\,|n\rangle \langle n|\, O_\sigma^\dagger + O_\sigma^\dagger\,|n\rangle \langle n|\, O_\sigma(t)\right) |m\rangle && \left[\sum_n |n\rangle \langle n| = 1\right] \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_m} \langle m| \left(e^{iH^* t} O_\sigma e^{-iH^* t}\,|n\rangle \langle n|\, O_\sigma^\dagger + O_\sigma^\dagger\,|n\rangle \langle n|\, e^{iH^* t} O_\sigma e^{-iH^* t}\right) |m\rangle \\
&= \frac{1}{Z}\sum_{m,n} e^{-\beta E_m} \left(e^{i(E_m - E_n)t} \langle m| O_\sigma |n\rangle \langle n| O_\sigma^\dagger |m\rangle + e^{i(E_n - E_m)t} \langle m| O_\sigma^\dagger |n\rangle \langle n| O_\sigma |m\rangle\right) \\
&= \frac{1}{Z}\sum_{m,n} e^{i(E_m - E_n)t} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right)
\end{aligned}
\tag{11.3}
$$

The time-domain impurity Green's function can thus be written as (this is the so-called Lehmann-Kallen representation)

$$
G_{d\sigma} = -i\theta(t)\frac{1}{Z}\sum_{m,n} e^{i(E_m - E_n)t} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \tag{11.4}
$$

We are interested in the frequency domain form.

$$
\begin{aligned}
G_{dd}^\sigma(\omega) &= \int_{-\infty}^{\infty} dt e^{i(\omega)t} G_{dd}^\sigma(t) \\
&= \frac{1}{Z}\sum_{m,n} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + E_m - E_n)t}
\end{aligned}
\tag{11.5}
$$

To evaluate the time-integral, we will use the integral representation of the Heaviside function:

$$
\theta(t) = -\frac{1}{2\pi i} \lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \frac{1}{x + i\eta} e^{-ixt} dx \tag{11.6}
$$

With this definition, the integral in $G_{d\sigma}(\omega)$ becomes

$$
\begin{aligned}
(-i)\int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + i0^+ + E_m - E_n)t} &= (-i)\frac{-1}{2\pi i} \lim_{\eta\to 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} \int_{-\infty}^{\infty} dt e^{i(\omega + i0^+ + E_m - E_n - x)t} \\
&= \frac{1}{2\pi} \lim_{\eta\to 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} 2\pi\delta\left(\omega + i0^+ + E_m - E_n - x\right) \\
&= \frac{1}{\omega + E_m - E_n + i0^+}\ .
\end{aligned}
\tag{11.7}
$$

The frequency-domain Green's function is thus

$$G^{\sigma}_{dd}(\omega) = \frac{1}{Z}\sum_{m,n} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \frac{1}{\omega + E_m - E_n + i0^+} \tag{11.8}$$

The infinitesimally positive imaginary part in the denominator shifts the poles onto the lower half of the complex plane, leaving it analytic in the upper half: this is necessary to make the retarded Greens function causal.

2.11.2 Real and imaginary parts - The Sokhotski-Plemelj theorem

In order to write down the real and imaginary parts of the Greens function, we first prove the *Sokhotski-Plemelj* formula:

$$\lim_{\eta\to 0^+} \frac{1}{x + i\eta} = P\left(\frac{1}{x}\right) - i\pi\delta(x) \ . \tag{11.9}$$

where $P(f(x))$ is the Cauchy principal value, defined as

$$P\left[\int_{-\infty}^{\infty} \frac{dx}{x}\right] = \int_{-\infty}^{0^-} \frac{dx}{x} + \int_{0^+}^{\infty} \frac{dx}{x} \ . \tag{11.10}$$

To prove the above identity, we integrate the left-hand side using a test function $f(x)$:

$$\begin{aligned}\lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\, \frac{f(x)}{x+i\eta} &= \lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\, \frac{f(x)\,(x-i\eta)}{x^2+\eta^2} \\ &= \lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\, f(x)\left[\frac{x}{x^2+\eta^2} - i\eta\frac{1}{x^2+\eta^2}\right]\end{aligned} \tag{11.11}$$

The first integral can be split into three parts:

$$\lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\, f(x)\frac{x}{x^2+\eta^2} = \lim_{\eta\to 0^+}\lim_{\varepsilon\to 0^+}\left[\int_{-\infty}^{-\varepsilon} \mathrm{dx}\, \frac{f(x)x}{x^2+\eta^2} + \int_{\varepsilon}^{\infty} \mathrm{dx}\, \frac{f(x)x}{x^2+\eta^2} + \int_{-\varepsilon}^{\varepsilon} \mathrm{dx}\, \frac{f(x)x}{x^2+\eta^2}\right] \tag{11.12}$$

For the first two parts, it is safe to take the limit of η, because x is always non-vanishing there. For the third part, we can approximate $f(x)$ as $f(0)$ in the neighbourhood $x \in [-\varepsilon, \varepsilon], \varepsilon \to 0$. This leaves an odd function $x/(x^2+\eta^2)$ as the integrand, being integrated over a symmetric range. The third term therefore vanishes. In total, we get

$$\lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\, f(x)\frac{x}{x^2+\eta^2} = \int_{-\infty}^{-\varepsilon} \mathrm{dx}\, \frac{f(x)}{x} + \int_{\varepsilon}^{\infty} \mathrm{dx}\, \frac{f(x)}{x} = P\left(\int_{-\infty}^{\infty} \mathrm{dx}\, \frac{f(x)}{x}\right) \ . \tag{11.13}$$

To evaluate the second integral of eq. 11.11, we note that the only region in which the integrand is non-zero is when $|x| \simeq 0$; there, we again approximate $f(x)$ as $f(0)$. The remaining integral can then be evaluated easily:

$$-i\lim_{\eta\to 0^+}\eta\int_{-\infty}^{\infty} \mathrm{dx}\, \frac{f(x)}{x^2+\eta^2} = -i\lim_{\eta\to 0^+}\eta f(0)\int_{-\infty}^{\infty} \frac{\mathrm{dx}}{x^2+\eta^2} = -i\pi f(0), \tag{11.14}$$

where we used $\int \frac{dx}{x^2+\eta^2} = \frac{1}{\eta}\arctan\left(x/\eta\right)$. Combining eqs. 11.11,11.13 and 11.14, we get

$$\lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x+i\eta} = P\left(\int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x}\right) - i\pi\int_{-\infty}^{\infty} \mathrm{dx}\ f(x)\delta(x)\ . \tag{11.15}$$

This proves eq. 11.9.

The Sokhotski-Plemelj formula allows us to split the spectral representation into a real and an imaginary part:

$$\begin{aligned} G'_{d\sigma} &= \frac{1}{Z}\sum_{m,n} \left|\left| \langle m| O_\sigma |n\rangle \right|\right|^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) P\left(\frac{1}{\omega + E_m - E_n}\right) \\ G''_{d\sigma} &= -\pi\frac{1}{Z}\sum_{m,n} \left|\left| \langle m| O_\sigma |n\rangle \right|\right|^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \delta\left(\omega + E_m - E_n\right)\ . \end{aligned} \tag{11.16}$$

The spectral function, specifically, is defined as follows:

$$A_{d\sigma}(\omega) = -\frac{1}{\pi}G'_{d\sigma} = \frac{1}{Z}\sum_{m,n} \left|\left| \langle m| O_\sigma |n\rangle \right|\right|^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \delta\left(\omega + E_m - E_n\right) \tag{11.17}$$

2.11.3 Zero temperature spectral function

We specialise to zero temperature by taking the limit of $\beta \to \infty$. In both the partition function as well as inside the summation, the only term that will survive is the exponential of the ground state energy E_{GS}.

$$Z \equiv \sum_m e^{-\beta E_m} \implies \lim_{\beta\to\infty} Z = d_{\mathrm{GS}} e^{-\beta E_{\mathrm{GS}}}, \quad E_{\mathrm{GS}} \equiv \min\left\{E_n\right\}$$

where d_{GS} is the degeneracy of the ground state. The spectral function then simplifies to

$$\begin{aligned} A_{d\sigma} &= \frac{1}{d_{\mathrm{GS}} e^{-\beta E_{\mathrm{GS}}}}\sum_{m,n} \left|\left| \langle m| O_\sigma |n\rangle \right|\right|^2 \left[e^{-\beta E_m}\delta_{E_m,E_{\mathrm{GS}}} + e^{-\beta E_n}\delta_{E_n,E_{\mathrm{GS}}}\right] \delta\left(\omega + E_m - E_n\right) \\ &= \frac{1}{d_{\mathrm{GS}}}\sum_{n,n_{\mathrm{GS}}} \left[\left|\left| \langle n_{\mathrm{GS}}| O_\sigma |n\rangle \right|\right|^2 \delta\left(\omega + E_{\mathrm{GS}} - E_n\right) + \left|\left| \langle n| O_\sigma |n_{\mathrm{GS}}\rangle \right|\right|^2 \delta\left(\omega - E_{\mathrm{GS}} + E_n\right)\right] \end{aligned} \tag{11.18}$$

The label n_{GS} sums over all states $|n_{\mathrm{GS}}\rangle$ with energy E_{GS}. In practise, we evaluate this expression by replacing the formal Dirac delta functions with "nascent" ones, such as a Lorentzian function with a sufficiently sharp peak.

2.11.4 Reconstructing full Green function from spectral function: Kramers-Kronig relations

As mentioned earlier, the retarded Greens function $G_R(\omega + i0^+)$ has a complex pole in the lower half of the complex plane. In the following, the frequency ω is assumed to be complex. Consider the integral

$$I(\omega) = \oint_C d\omega' \frac{G_R(\omega')}{\omega' - \omega},\ \omega \in \mathbb{R}\ . \tag{11.19}$$

The contour C encloses the entirety of the upper half of the complex plane as well as almost all points of the real line but avoids the pole at $\omega' = \omega$ by forming a semicircle about that point that extends into the upper half of the plane. Since the integrand is completely analytic in the region enclosed by the contour (recall again that G_R is analytic in the upper half), the integral evaluates to zero (by Cauchy's theorem):

$$I(\omega) = 0 \,. \tag{11.20}$$

One can also evaluate the integral along each part of the contour. The semicircular part contributes zero: this is because while the length of the semicircular arc goes as $|\omega'|$, the integrand vanishes faster than $1/|\omega'|$ as $|\omega'| \to \infty$ (this assumes that the Green function vanishes in the limit of $\omega \to \infty$). The part of the contour along the real axis can now be evaluated. The two parts of the contour on either side of the pole contribute

$$\int_{-\infty}^{\omega-0^+} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} + \int_{\omega+0^+}^{\infty} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} = P \int_{-\infty}^{\infty} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} \tag{11.21}$$

To evaluate the part of the integral along the small semicircular around $\omega' = \omega$ (that extends into the upper half), we argue that this contribution should be equal to that obtained by taking the semicircle that goes instead into the lower half but circulates in the opposite direction. Each of these contributions should, in turn, be equal to that obtained from joining these two contours. But joining these contours leads to a closed integral about the pole at $\omega' = \omega$, which is equal to $2\pi G_R(\omega)$.

$$\int_{\text{upper}} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} = \int_{\text{lower}} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} = \frac{1}{2} \oint_{C(\omega)} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} = i\pi G_R(\omega) \,. \tag{11.22}$$

Combining the last two equations, we get

$$I(\omega) = P \int_{-\infty}^{\infty} d\omega' \, \frac{G_R(\omega')}{\omega' - \omega} + i\pi G_R(\omega) \,. \tag{11.23}$$

Comparing with eq. 11.20 and separating into real part G_R' and imaginary part G_R'' gives

$$P \int_{-\infty}^{\infty} d\omega' \, \frac{\left[G_R'(\omega') + iG_R''(\omega')\right]}{\omega' - \omega} + \pi \left[-G_R''(\omega) + iG_R'(\omega)\right] = 0 \,. \tag{11.24}$$

Comparing the imaginary parts gives

$$G_R'(\omega) = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \, \frac{G_R''(\omega')}{\omega' - \omega} = -\mathcal{H}\left[G_R''(\omega)\right] \,, \tag{11.25}$$

where $\mathcal{H}\left[g(\omega)\right] = \frac{1}{\pi} P \int_{-\infty}^{\infty} d\omega' \, \frac{g(\omega')}{\omega - \omega'}$ is the Hilbert transform of the function $g(\omega)$. The spectral function was defined above as $-1/\pi$ times the imaginary part of the Greens function. Exchanging G_R'' for A gives

$$G_R'(\omega) = \pi \mathcal{H}\left[A(\omega)\right] \,. \tag{11.26}$$

This provides a method for calculating the real part of a Greens function if the spectral function is already known.

Bibliography

[1] John Hubbard. Electron correlations in narrow energy bands. In *Proceedings of the royal society of london a: mathematical, physical and engineering sciences*, volume 276, pages 238–257. The Royal Society, 1963.

[2] N. F. Mott. Metal-insulator transition. *Rev. Mod. Phys.*, 40:677–683, Oct 1968.

[3] R F Milligan and G A Thomas. The metal-insulator transition. *Annual Review of Physical Chemistry*, 36(1):139–158, 1985.

[4] Masatoshi Imada, Atsushi Fujimori, and Yoshinori Tokura. Metal-insulator transitions. *Reviews of modern physics*, 70(4):1039, 1998.

[5] Patrick A. Lee, Naoto Nagaosa, and Xiao-Gang Wen. Doping a mott insulator: Physics of high-temperature superconductivity. *Rev. Mod. Phys.*, 78:17–85, Jan 2006.

[6] A. Pustogow, M. Bories, A. Löhle, R. Rösslhuber, E. Zhukova, B. Gorshunov, S. Tomić, J. A. Schlueter, R. Hübner, T. Hiramatsu, Y. Yoshida, G. Saito, R. Kato, T.-H. Lee, V. Dobrosavljević, S. Fratini, and M. Dressel. Quantum spin liquids unveil the genuine mott state. *Nature Materials*, 17(9):773–777, Sep 2018.

[7] G. R. Stewart. Non-fermi-liquid behavior in d- and f-electron metals. *Rev. Mod. Phys.*, 73:797–855, Oct 2001.

[8] Yuan Cao, Valla Fatemi, Ahmet Demir, Shiang Fang, Spencer L. Tomarken, Jason Y. Luo, Javier D. Sanchez-Yamagishi, Kenji Watanabe, Takashi Taniguchi, Efthimios Kaxiras, Ray C. Ashoori, and Pablo Jarillo-Herrero. Correlated insulator behaviour at half-filling in magic-angle graphene superlattices. *Nature*, 556(7699):80–84, Apr 2018.

[9] Yuan Cao, Valla Fatemi, Shiang Fang, Kenji Watanabe, Takashi Taniguchi, Efthimios Kaxiras, and Pablo Jarillo-Herrero. Unconventional superconductivity in magic-angle graphene superlattices. *Nature*, 556(7699):43–50, Apr 2018.

[10] J H de Boer and E J W Verwey. Semi-conductors with partially and with completely filled 3d-lattice bands. *Proceedings of the Physical Society*, 49(4S):59, aug 1937.

[11] N F Mott. The basis of the electron theory of metals, with special reference to the transition metals. *Proceedings of the Physical Society. Section A*, 62(7):416–422, jul 1949.

[12] N. F. Mott. On the transition to metallic conduction in semiconductors. *Canadian Journal of Physics*, 34(12A):1356–1368, 1956.

[13] N. F. Mott. The transition to the metallic state. *The Philosophical Magazine: A Journal of Theoretical Experimental and Applied Physics*, 6(62):287–309, 1961.

[14] Takayuki Isono, Taichi Terashima, Kazuya Miyagawa, Kazushi Kanoda, and Shinya Uji. Quantum criticality in an organic spin-liquid insulator κ-(bedt-ttf)2cu2(cn)3. *Nature Communications*, 7(1):13494, Nov 2016.

[15] P. W. Anderson. New approach to the theory of superexchange interactions. *Phys. Rev.*, 115:2–13, Jul 1959.

[16] Junjiro Kanamori. Electron Correlation and Ferromagnetism of Transition Metals. *Progress of Theoretical Physics*, 30(3):275–289, 09 1963.

[17] Martin C. Gutzwiller. Effect of correlation on the ferromagnetism of transition metals. *Phys. Rev. Lett.*, 10:159–162, Mar 1963.

[18] Antoine Georges, Gabriel Kotliar, Werner Krauth, and Marcelo J Rozenberg. Dynamical mean-field theory of strongly correlated fermion systems and the limit of infinite dimensions. *Reviews of Modern Physics*, 68(1):13, 1996.

[19] W. F. Brinkman and T. M. Rice. Application of gutzwiller's variational method to the metal-insulator transition. *Phys. Rev. B*, 2:4302–4304, Nov 1970.

[20] Martin C. Gutzwiller. Correlation of electrons in a narrow s band. *Phys. Rev.*, 137:A1726–A1735, Mar 1965.

[21] Peter Prelovšek, Jure Kokalj, Zala Lenarčič, and Ross H. McKenzie. Holon-doublon binding as the mechanism for the mott transition. *Phys. Rev. B*, 92:235155, Dec 2015.

[22] Sen Zhou, Yupeng Wang, and Ziqiang Wang. Doublon-holon binding, mott transition, and fractionalized antiferromagnet in the hubbard model. *Phys. Rev. B*, 89:195119, May 2014.

[23] Tsutomu Watanabe, Hisatoshi Yokoyama, Yukio Tanaka, and Jun-ichiro Inoue. Superconductivity and a mott transition in a hubbard model on an anisotropic triangular lattice. *Journal of the Physical Society of Japan*, 75(7):074707, 2006.

[24] Philip Phillips. *Advanced solid state physics*. Cambridge University Press, 2012. Discussion regarding f-sum rule and dynamical spectral weight transfer in pages 394-395, section 12.4 on Cooper pairs and computation of effective two particle energy.

[25] G. Baskaran. An exactly solvable fermion model: spinons, holons and a non-fermi liquid phase. *Modern Physics Letters B*, 05(09):643–649, 1991.

[26] Yasuhiro Hatsugai and Mahito Kohmoto. Exactly solvable model of correlated lattice electrons in any dimensions. *Journal of the Physical Society of Japan*, 61(6):2056–2069, 1992.

[27] Richard M. Martin, Lucia Reining, and David M. Ceperley. *Interacting Electrons: Theory and Computational Approaches*. Cambridge University Press, 2016.

[28] Walter Metzner and Dieter Vollhardt. Correlated lattice fermions in $d = \infty$ dimensions. *Phys. Rev. Lett.*, 62:324–327, Jan 1989.

[29] Antoine Georges. Strongly correlated electron materials: Dynamical mean-field theory and electronic structure. *AIP Conference Proceedings*, 715(1):3–74, 2004.

[30] J. Zaanen, G. A. Sawatzky, and J. W. Allen. Band gaps and electronic structure of transition-metal compounds. *Phys. Rev. Lett.*, 55:418, 1985.

[31] Masatoshi Imada, Atsushi Fujimori, and Yoshinori Tokura. Metal-insulator transitions. *Rev. Mod. Phys.*, 70:1039–1263, 1998.

[32] D. B. McWhan, T. M. Rice, and J. P. Remeika. Mott transition in cr-doped $\mathrm{v_2o_3}$. *Phys. Rev. Lett.*, 23:1384–1387, Dec 1969.

[33] P. Limelette, A. Georges, D. Jérome, P. Wzietek, P. Metcalf, and J. M. Honig. Universality and critical behavior at the mott transition. *Science*, 302(5642):89–92, 2003.

[34] M. J. Rozenberg, G. Kotliar, H. Kajueter, G. A. Thomas, D. H. Rapkine, J. M. Honig, and P. Metcalf. Optical conductivity in mott-hubbard systems. *Phys. Rev. Lett.*, 75:105–108, Jul 1995.

[35] M. M. Qazilbash, M. Brehm, Byung-Gyu Chae, P.-C. Ho, G. O. Andreev, Bong-Jun Kim, Sun Jin Yun, A. V. Balatsky, M. B. Maple, F. Keilmann, Hyun-Tak Kim, and D. N. Basov. Mott transition in vo2 revealed by infrared spectroscopy and nano-imaging. *Science*, 318(5857):1750–1753, 2007.

[36] H. Takagi, B. Batlogg, H. L. Kao, J. Kwo, R. J. Cava, J. J. Krajewski, and W. F. Peck. Systematic evolution of temperature-dependent resistivity in $_{2-x}sr_{x4}$. *Phys. Rev. Lett.*, 69:2975–2978, Nov 1992.

[37] Yoichi Ando, Seiki Komiya, Kouji Segawa, S. Ono, and Y. Kurita. Electronic phase diagram of high-T_c cuprate superconductors from a mapping of the in-plane resistivity curvature. *Phys. Rev. Lett.*, 93:267001, Dec 2004.

[38] T. R. Lemberger, I. Hetel, A. Tsukada, M. Naito, and M. Randeria. Superconductor-to-metal quantum phase transition in overdoped $\mathrm{la}_{2-x}\mathrm{sr}_x\mathrm{cuo}_4$. *Phys. Rev. B*, 83:140507, Apr 2011.

[39] Andrea Damascelli, Zahid Hussain, and Zhi-Xun Shen. Angle-resolved photoemission studies of the cuprate superconductors. *Rev. Mod. Phys.*, 75:473–541, Apr 2003.

[40] Jonathan A. Sobota, Yu He, and Zhi-Xun Shen. Angle-resolved photoemission studies of quantum materials. *Rev. Mod. Phys.*, 93:025006, May 2021.

[41] I M Vishik. Photoemission perspective on pseudogap, superconducting fluctuations, and charge order in cuprates: a review of recent progress. *Reports on Progress in Physics*, 81(6):062501, apr 2018.

[42] María Luisa Medarde. Structural, magnetic and electronic properties of perovskites (r = rare earth). *Journal of Physics: Condensed Matter*, 9(8):1679, feb 1997.

[43] Yuxin Wang, Kun Jiang, Jianjun Ying, Tao Wu, Jinguang Cheng, Jiangping Hu, and Xianhui Chen. Recent progress in nickelate superconductors. *National Science Review*, 12(10):nwaf373, 09 2025.

[44] Pascal Puphal, Thomas Schäfer, Bernhard Keimer, and Matthias Hepting. Superconductivity in infinite-layer and ruddlesden–popper nickelates. *Nature Reviews Physics*, 8(2):70–85, Feb 2026.

[45] Sajedeh Manzeli, Dmitry Ovchinnikov, Diego Pasquier, Oleg V. Yazyev, and Andras Kis. 2d transition metal dichalcogenides. *Nature Reviews Materials*, 2(8):17033, Jun 2017.

[46] Saju Joseph, Jainy Mohan, Seetha Lakshmy, Simil Thomas, Brahmananda Chakraborty, Sabu Thomas, and Nandakumar Kalarikkal. A review of the synthesis, properties, and applications of 2d transition metal dichalcogenides and their heterostructures. *Materials Chemistry and Physics*, 297:127332, 2023.

[47] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. Charge-density waves in metallic, layered, transition-metal dichalcogenides. *Phys. Rev. Lett.*, 32:882–885, Apr 1974.

[48] J. A. Wilson, F. J. Di Salvo, and S. Mahajan. Charge-density waves and superlattices in the metallic layered transition metal dichalcogenides. *Advances in Physics*, 50(8):1171–1248, 2001.

[49] P. Fazekas and E. Tosatti. Electrical, structural and magnetic properties of pure and doped 1t-tas$_2$. *Philosophical Magazine B*, 39:229–244, 1979.

[50] L. Perfetti, T. A. Gloor, F. Mila, H. Berger, and M. Grioni. Unexpected periodicity in the quasi-two-dimensional mott insulator $1t-\mathrm{tas}_2$ revealed by angle-resolved photoemission. *Phys. Rev. B*, 71:153101, Apr 2005.

[51] B. Sipos, A. F. Kusmartseva, A. Akrap, H. Berger, L. Forró, and E. Tutiš. From mott state to superconductivity in 1t-tas2. *Nature Materials*, 7(12):960–965, Dec 2008.

[52] Doohee Cho, Sangmo Cheon, Ki-Seok Kim, Sung-Hoon Lee, Yong-Heum Cho, Sang-Wook Cheong, and Han Woong Yeom. Nanoscale manipulation of the mott insulating state coupled to charge order in 1t-tas2. *Nature Communications*, 7(1):10453, Jan 2016.

[53] S. V. Borisenko, A. A. Kordyuk, A. N. Yaresko, V. B. Zabolotnyy, D. S. Inosov, R. Schuster, B. Büchner, R. Weber, R. Follath, L. Patthey, and H. Berger. Pseudogap and charge density waves in two dimensions. *Phys. Rev. Lett.*, 100:196402, May 2008.

[54] S. V. Borisenko, A. A. Kordyuk, V. B. Zabolotnyy, D. S. Inosov, D. Evtushinsky, B. Büchner, A. N. Yaresko, A. Varykhalov, R. Follath, W. Eberhardt, L. Patthey, and H. Berger. Two energy gaps and fermi-surface "arcs" in $nbse_2$. *Phys. Rev. Lett.*, 102:166402, Apr 2009.

[55] V. Vescoli et al. Charge-density-wave correlations in the optical properties of layered transition-metal dichalcogenides. *Physical Review Letters*, 81:453–456, 1998.

[56] Lijun Li, Xiaoyu Deng, Zhen Wang, Yu Liu, Milinda Abeykoon, Eric Dooryhee, Aleksandra Tomic, Yanan Huang, John B. Warren, Emil S. Bozin, Simon J. L. Billinge, Yuping Sun, Yimei Zhu, Gabriel Kotliar, and Cedomir Petrovic. Superconducting order from disorder in 2h-tase2−xsx. *npj Quantum Materials*, 2(1):11, Feb 2017.

[57] Yoichi Kamihara, Takumi Watanabe, Masahiro Hirano, and Hideo Hosono. Iron-based layered superconductor la[o1-xfx]feas (x = 0.05−0.12) with tc = 26 k. *Journal of the American Chemical Society*, 130(11):3296–3297, 2008. PMID: 18293989.

[58] J. Paglione and R. L. Greene. High-temperature superconductivity in iron-based materials. *Nature Physics*, 6:645–658, 2010.

[59] P. J. Hirschfeld, M. M. Korshunov, and I. I. Mazin. Gap symmetry and structure of fe-based superconductors. *Reports on Progress in Physics*, 74:124508, 2011.

[60] G. R. Stewart. Superconductivity in iron compounds. *Rev. Mod. Phys.*, 83:1589–1652, Dec 2011.

[61] David C. Johnston. The puzzle of high temperature superconductivity in layered iron pnictides and chalcogenides. *Advances in Physics*, 59(6):803–1061, 2010.

[62] S. Kasahara, H. J. Shi, K. Hashimoto, S. Tonegawa, Y. Mizukami, T. Shibauchi, K. Sugimoto, T. Fukuda, T. Terashima, Andriy H. Nevidomskyy, and Y. Matsuda. Electronic nematicity above the structural and superconducting transition in bafe2(as1−xp x)2. *Nature*, 486(7403):382–385, Jun 2012.

[63] R. M. Fernandes, A. V. Chubukov, and J. Schmalian. What drives nematic order in iron-based superconductors? *Nature Physics*, 10(2):97–104, Feb 2014.

[64] James G. Analytis, H.-H. Kuo, Ross D. McDonald, Mark Wartenbe, P. M. C. Rourke, N. E. Hussey, and I. R. Fisher. Transport near a quantum critical point in bafe2(as1−xpx)2. *Nature Physics*, 10(3):194–197, Mar 2014.

[65] M. M. Qazilbash, J. J. Hamlin, R. E. Baumbach, Lijun Zhang, D. J. Singh, M. B. Maple, and D. N. Basov. Electronic correlations in the iron pnictides. *Nature Physics*, 5(9):647–650, Sep 2009.

[66] H. Shishido, A. F. Bangura, A. I. Coldea, S. Tonegawa, K. Hashimoto, S. Kasahara, P. M. C. Rourke, H. Ikeda, T. Terashima, R. Settai, Y. Ōnuki, D. Vignolles, C. Proust, B. Vignolle, A. McCollam, Y. Matsuda, T. Shibauchi, and A. Carrington. Evolution of the fermi surface

of bafe$_2$(as$_{1-x}$$\mathbf{p}_x$)$_2$ on entering the superconducting dome. *Phys. Rev. Lett.*, 104:057008, Feb 2010.

[67] K. Terashima, Y. Sekiba, J. H. Bowen, K. Nakayama, T. Kawahara, T. Sato, P. Richard, Y.-M. Xu, L. J. Li, G. H. Cao, Z.-A. Xu, H. Ding, and T. Takahashi. Fermi surface nesting induced strong pairing in iron-based superconductors. *Proceedings of the National Academy of Sciences*, 106(18):7330–7333, 2009.

[68] J.-H. Chu, H.-H. Kuo, J. G. Analytis, and I. R. Fisher. Divergent nematic susceptibility in an iron arsenide superconductor. *Science*, 337:710–712, 2012.

[69] Hsueh-Hui Kuo, Jiun-Haw Chu, Johanna C. Palmstrom, Steven A. Kivelson, and Ian R. Fisher. Ubiquitous signatures of nematic quantum criticality in optimally doped fe-based superconductors. *Science*, 352(6288):958–962, 2016.

[70] D. S. Inosov, J. T. Park, P. Bourges, D. L. Sun, Y. Sidis, A. Schneidewind, K. Hradil, D. Haug, C. T. Lin, B. Keimer, and V. Hinkov. Normal-state spin dynamics and temperature-dependent spin-resonance energy in optimally doped $BaFe_{1.85}Co_{0.15}As_2$. *Nature Physics*, 6:178–181, 2010.

[71] Andrey Chubukov. Pairing mechanism in fe-based superconductors. *Annual Review of Condensed Matter Physics*, 3(Volume 3, 2012):57–92, 2012.

[72] G. R. Stewart. Heavy-fermion systems. *Reviews of Modern Physics*, 56:755–787, 1984.

[73] Steffen Wirth and Frank Steglich. Exploring heavy fermions from macroscopic to microscopic length scales. *Nature Reviews Materials*, 1(10):16051, Aug 2016.

[74] A. C. Hewson. *The Kondo Problem to Heavy Fermions*. Cambridge University Press, 1993.

[75] Hilbert v. Löhneysen, Achim Rosch, Matthias Vojta, and Peter Wölfle. Fermi-liquid instabilities at magnetic quantum phase transitions. *Rev. Mod. Phys.*, 79:1015–1075, Aug 2007.

[76] Philipp Gegenwart, Qimiao Si, and Frank Steglich. Quantum criticality in heavy-fermion metals. *Nature Physics*, 4(3):186–197, Mar 2008.

[77] J. Custers, P. Gegenwart, H. Wilhelm, K. Neumaier, Y. Tokiwa, O. Trovarelli, C. Geibel, F. Steglich, C. Pépin, and P. Coleman. The break-up of heavy electrons at a quantum critical point. *Nature*, 424(6948):524–527, Jul 2003.

[78] S. Paschen, T. Lühmann, S. Wirth, P. Gegenwart, O. Trovarelli, C. Geibel, F. Steglich, P. Coleman, and Q. Si. Hall-effect evolution across a heavy-fermion quantum critical point. *Nature*, 432(7019):881–885, Dec 2004.

[79] A. Schröder, G. Aeppli, R. Coldea, M. Adams, O. Stockert, H.v. Löhneysen, E. Bucher, R. Ramazashvili, and P. Coleman. Onset of antiferromagnetism in heavy-fermion metals. *Nature*, 407(6802):351–355, Sep 2000.

[80] O. Stockert, J. Arndt, E. Faulhaber, C. Geibel, H. S. Jeevan, S. Kirchner, M. Loewenhaupt, K. Schmalzl, W. Schmidt, Q. Si, and F. Steglich. Magnetically driven superconductivity in cecu2si2. *Nature Physics*, 7(2):119–124, Feb 2011.

[81] Hiroaki Shishido, Rikio Settai, Hisatomo Harima, and Yoshichika Ōnuki. A drastic change of the fermi surface at a critical pressure in cerhin5: dhva study under pressure. *Journal of the Physical Society of Japan*, 74(4):1103–1106, 2005.

[82] Qimiao Si and Frank Steglich. Heavy fermions and quantum phase transitions. *Science*, 329(5996):1161–1166, 2010.

[83] Jun Kondo. Resistance minimum in dilute magnetic alloys. *Progress of theoretical physics*, 32(1):37–49, 1964.

[84] JR Schrieffer and P.A.Wolff. Relation between the anderson and kondo hamiltonians. *Phys. Rev.*, 149:491, 1966.

[85] Kenneth G Wilson. The renormalization group: Critical phenomena and the kondo problem. *Reviews of Modern Physics*, 47(4):773, 1975.

[86] H. R. Krishna-murthy, J. W. Wilkins, and K. G. Wilson. Renormalization-group approach to the anderson model of dilute magnetic alloys. i. static properties for the symmetric case. *Phys. Rev. B*, 21:1003–1043, Feb 1980.

[87] S Nishimoto and E Jeckelmann. Density-matrix renormalization group approach to quantum impurity problems. *Journal of Physics: Condensed Matter*, 16(4):613, jan 2004.

[88] N. Andrei. Diagonalization of the kondo hamiltonian. *Phys. Rev. Lett.*, 45:379–382, Aug 1980.

[89] P B Wiegmann. Exact solution of the s-d exchange model (kondo problem). *Journal of Physics C: Solid State Physics*, 14(10):1463–1478, apr 1981.

[90] P Nozieres. A "fermi-liquid" description of the kondo problem at low temperatures. *Journal of Low Temperature Physics*, 17:31, 1974.

[91] Ian Affleck. Conformal field theory approach to the kondo effect. *Acta Phys.Polon.B*, 26(1869), 1995.

[92] T.A. Costi and A.C. Hewson. A new approach to the calculation of spectra for strongly correlated systems. *Physica B: Condensed Matter*, 163(1):179–181, 1990.

[93] T.A. Costi and A.C. Hewson. Static and dynamic properties of the anderson model with conduction electron screening. *Physica C: Superconductivity*, 185-189:2649–2650, 1991.

[94] Ryu Takayama and Osamu Sakai. Excitation spectra of the anderson model with coulomb interaction between localized and conduction electrons. *Physica B: Condensed Matter*, 186-188:915–917, 1993.

[95] Wanda C. Oliveira and Luiz N. Oliveira. Generalized numerical renormalization-group method to calculate the thermodynamical properties of impurities in metals. *Phys. Rev. B*, 49:11986–11994, May 1994.

[96] David C. Langreth. Friedel sum rule for anderson's model of localized impurity states. *Phys. Rev.*, 150:516–518, Oct 1966.

[97] JM Luttinger. Fermi surface and some simple equilibrium properties of a system of interacting fermions. *Physical Review*, 119(4):1153, 1960.

[98] Joaquin Mazdak Luttinger and John Clive Ward. Ground-state energy of a many-fermion system. ii. *Physical Review*, 118(5):1417, 1960.

[99] Masaki Oshikawa. Topological approach to luttinger's theorem and the fermi surface of a kondo lattice. *Physical Review Letters*, 84(15):3370, 2000.

[100] Kazuhiro Seki and Seiji Yunoki. Topological interpretation of the luttinger theorem. *Physical Review B*, 96(8):085124, 2017.

[101] Nozières, Ph. and Blandin, A. Kondo effect in real metals. *J. Phys. France*, 41(3):"193–211", 1980.

[102] J Gan. On the multichannel kondo model. *Journal of Physics: Condensed Matter*, 6(24):4547, jun 1994.

[103] Ian Affleck and Andreas W.W. Ludwig. Critical theory of overscreened kondo fixed points. *Nuclear Physics B*, 360(2):641–696, 1991.

[104] Andreas W. W. Ludwig and Ian Affleck. Exact, asymptotic, three-dimensional, space- and time-dependent, green's functions in the multichannel kondo effect. *Phys. Rev. Lett.*, 67:3160–3163, Nov 1991.

[105] Ian Affleck, Andreas W. W. Ludwig, H.-B. Pang, and D. L. Cox. Relevance of anisotropy in the multichannel kondo effect: Comparison of conformal field theory and numerical renormalization-group results. *Phys. Rev. B*, 45:7918–7935, Apr 1992.

[106] Ian Affleck and Andreas WW Ludwig. Exact conformal-field-theory results on the multichannel kondo effect: Single-fermion green's function, self-energy, and resistivity. *Physical Review B*, 48(10):7297, 1993.

[107] Olivier Parcollet and Antoine Georges. Transition from overscreening to underscreening in the multichannel kondo model: Exact solution at large N. *Phys. Rev. Lett.*, 79:4665–4668, Dec 1997.

[108] Ian Affleck. Non-fermi liquid behavior in kondo models. *Journal of the Physical Society of Japan*, 74(1):59–66, 2005.

[109] V. J. Emery and S. Kivelson. Mapping of the two-channel kondo problem to a resonant-level model. *Phys. Rev. B*, 46:10812–10817, Nov 1992.

[110] David G. Clarke, Thierry Giamarchi, and Boris I. Shraiman. Curie and non-curie behavior of impurity spins in quantum antiferromagnets. *Phys. Rev. B*, 48:7070–7076, Sep 1993.

[111] Gergely Zaránd and Jan von Delft. Analytical calculation of the finite-size crossover spectrum of the anisotropic two-channel kondo model. *Phys. Rev. B*, 61:6918–6933, Mar 2000.

[112] Jan von Delft, Gergely Zaránd, and Michele Fabrizio. Finite-size bosonization of 2-channel kondo model: A bridge between numerical renormalization group and conformal field theory. *Phys. Rev. Lett.*, 81:196–199, Jul 1998.

[113] A. J. Schofield. Bosonization in the two-channel kondo model. *Phys. Rev. B*, 55:5627–5630, Mar 1997.

[114] R Bulla, T Costi, and T Pruschke. Numerical renormalisation group method for quantum impurity systems. *Rev. Mod. Phys.*, 80:395, 2008.

[115] H. B. Pang and D. L. Cox. Stability of the fixed point of the two-channel kondo hamiltonian. *Phys. Rev. B*, 44:9454–9457, Nov 1991.

[116] N. Andrei and C. Destri. Solution of the multichannel kondo problem. *Phys. Rev. Lett.*, 52:364–367, Jan 1984.

[117] A. M. Tsvelick and P. B. Wiegmann. Solution of then-channel kondo problem (scaling and integrability). *Zeitschrift für Physik B Condensed Matter*, 54(3):201–206, Sep 1984.

[118] A M Tsvelick. The thermodynamics of multichannel kondo problem. *Journal of Physics C: Solid State Physics*, 18(1):159–170, jan 1985.

[119] Natan Andrei and Andrés Jerez. Fermi- and non-fermi-liquid behavior in the anisotropic multichannel kondo model: Bethe ansatz solution. *Phys. Rev. Lett.*, 74:4507–4510, May 1995.

[120] Gergely Zaránd, Theo Costi, Andres Jerez, and Natan Andrei. Thermodynamics of the anisotropic two-channel kondo problem. *Phys. Rev. B*, 65:134416, Mar 2002.

[121] A. V. Rozhkov. Impurity entropy for the two-channel kondo model. *International Journal of Modern Physics B*, 12(32):3457–3463, 1998.

[122] Ru Zheng, Rong-Qiang He, and Zhong-Yi Lu. Order parameter for the multichannel kondo model at quantum criticality. *Phys. Rev. B*, 103:045111, Jan 2021.

[123] D. Withoff and E. Fradkin. Phase transitions in gapless fermi systems with magnetic impurities. *Phys. Rev. Lett.*, 64:1835–1838, 1990.

[124] C. Gonzalez-Buxton and K. Ingersent. Renormalization-group study of anderson and kondo impurities in gapless fermi systems. *Phys. Rev. B*, 57:14254–14269, 1998.

[125] R. Bulla, T. Pruschke, and A. C. Hewson. Anderson impurity in pseudogap fermi systems. *J. Phys.: Condens. Matter*, 9:10463–10472, 1997.

[126] K. Ingersent. Behavior of magnetic impurities in gapless fermi systems. *Phys. Rev. B*, 54:11936–11953, 1996.

[127] K. Ingersent and Q. Si. Critical local-moment fluctuations in the pseudogap anderson model. *Phys. Rev. Lett.*, 89:076403, 2002.

[128] L. Fritz and M. Vojta. Phase transitions in the pseudogap anderson and kondo models. *Phys. Rev. B*, 70:214427, 2004.

[129] O. Parcollet and A. Georges. Non-fermi-liquid regime of a doped mott insulator. *Phys. Rev. B*, 59:5341–5360, 1999.

[130] Qimiao Si, Silvio Rabello, Kevin Ingersent, and J. Lleweilun Smith. Locally critical quantum phase transitions in strongly correlated metals. *Nature*, 413(6858):804–808, Oct 2001.

[131] R. Bulla, M. Vojta, and W. Hofstetter. Numerical renormalization group for quantum impurities in a pseudogap. *Phys. Rev. B*, 62:R6115–R6118, 2000.

[132] B. A. Jones and C. M. Varma. Study of two magnetic impurities in a fermi gas. *Phys. Rev. Lett.*, 58:843–846, 1987.

[133] B. A. Jones, C. M. Varma, and J. W. Wilkins. Low-temperature properties of the two-impurity kondo hamiltonian. *Phys. Rev. Lett.*, 61:125–128, 1988.

[134] I. Affleck, A. W. W. Ludwig, and B. A. Jones. Conformal-field-theory approach to the two-impurity kondo problem. *Phys. Rev. B*, 52:9528–9546, 1995.

[135] I. Affleck and A. W. W. Ludwig. Exact conformal-field-theory results on the multichannel kondo effect. *Phys. Rev. B*, 48:7297–7321, 1993.

[136] Krzysztof P. Wójcik and Johann Kroha. Quantum spin liquid in an rkky-coupled two-impurity kondo system. *Phys. Rev. B*, 107:L121111, Mar 2023.

[137] Elio J. König, Piers Coleman, and Yashar Komijani. Frustrated kondo impurity triangle: A simple model of deconfinement. *Phys. Rev. B*, 104:115103, Sep 2021.

[138] A. Spinelli, M. Gerrits, R. Toskovic, B. Bryant, M. Ternes, and A. F. Otte. Exploring the phase diagram of the two-impurity kondo problem. *Nature Communications*, 6(1):10046, Nov 2015.

[139] Hal Tasaki. The hubbard model - an introduction and selected rigorous results. *Journal of Physics: Condensed Matter*, 10(20):4353, may 1998.

[140] Daniel P. Arovas, Erez Berg, Steven A. Kivelson, and Srinivas Raghu. The hubbard model. *Annual Review of Condensed Matter Physics*, 13(Volume 13, 2022):239–274, 2022.

[141] Mingpu Qin, Thomas Schäfer, Sabine Andergassen, Philippe Corboz, and Emanuel Gull. The hubbard model: A computational perspective. *Annual Review of Condensed Matter Physics*, 13(Volume 13, 2022):275–302, 2022.

[142] Elliott H Lieb and F Yu Wu. Absence of mott transition in an exact solution of the short-range, one-band model in one dimension. *Physical Review Letters*, 20(25):1445, 1968.

[143] Antoine Georges and Gabriel Kotliar. Hubbard model in infinite dimensions. *Phys. Rev. B*, 45:6479–6483, Mar 1992.

[144] V. Janis and D. Vollhardt. Comprehensive mean field theory for the hubbard model. *International Journal of Modern Physics B*, 06(05n06):731–747, 1992.

[145] M. J. Rozenberg, X. Y. Zhang, and G. Kotliar. Mott-hubbard transition in infinite dimensions. *Phys. Rev. Lett.*, 69:1236–1239, Aug 1992.

[146] Marcelo J. Rozenberg, R. Chitra, and Gabriel Kotliar. Finite temperature mott transition in the hubbard model in infinite dimensions. *Phys. Rev. Lett.*, 83:3498–3501, Oct 1999.

[147] R. M. Noack and F. Gebhard. Mott-hubbard transition in infinite dimensions. *Phys. Rev. Lett.*, 82:1915–1918, Mar 1999.

[148] P. Werner and A. J. Millis. Doping-driven mott transition in the one-band hubbard model. *Phys. Rev. B*, 75:085108, 2007.

[149] Nozières, Ph. Some comments on kondo lattices and the mott transition. *Eur. Phys. J. B*, 6(4):”447–457”, ”” 1998.

[150] G. Kotliar, Sahana Murthy, and M. J. Rozenberg. Compressibility divergence and the finite temperature mott transition. *Phys. Rev. Lett.*, 89:046401, Jul 2002.

[151] Kotliar, G. Landau theory of the mott transition in the fully frustrated hubbard model in infinite dimensions. *Eur. Phys. J. B*, 11(1):27–39, 1999.

[152] Xiaoyu Deng, Jernej Mravlje, Rok Žitko, Michel Ferrero, Gabriel Kotliar, and Antoine Georges. How bad metals turn good: Spectroscopic signatures of resilient quasiparticles. *Phys. Rev. Lett.*, 110:086401, Feb 2013.

[153] R. Žitko, D. Hansen, E. Perepelitsky, J. Mravlje, A. Georges, and B. S. Shastry. Extremely correlated fermi liquid theory meets dynamical mean-field theory: Analytical insights into the doping-driven mott transition. *Phys. Rev. B*, 88:235132, Dec 2013.

[154] J. Vučičević, D. Tanasković, M. J. Rozenberg, and V. Dobrosavljević. Bad-metal behavior reveals mott quantum criticality in doped hubbard models. *Phys. Rev. Lett.*, 114:246402, Jun 2015.

[155] Nagamalleswararao Dasari, N. S. Vidhyadhiraja, Mark Jarrell, and Ross H. McKenzie. Quantum critical local spin dynamics near the mott metal-insulator transition in infinite dimensions. *Phys. Rev. B*, 95:165105, Apr 2017.

[156] Heike Eisenlohr, Seung-Sup B. Lee, and Matthias Vojta. Mott quantum criticality in the one-band hubbard model: Dynamical mean-field theory, power-law spectra, and scaling. *Phys. Rev. B*, 100:155152, Oct 2019.

[157] David E Logan and Martin R Galpin. Mott insulators and the doping-induced mott transition within DMFT: exact results for the one-band hubbard model. *Journal of Physics: Condensed Matter*, 28(2):025601, dec 2015.

[158] Y. Ōno, R. Bulla, and A. C. Hewson. Phase diagram of the mott transition in a two-band hubbard model in infinite dimensions. *The European Physical Journal B - Condensed Matter and Complex Systems*, 19(3):375–384, Feb 2001.

[159] Marcelo J Rozenberg. Integer-filling metal-insulator transitions in the degenerate hubbard model. *Physical Review B*, 55(8):R4855, 1997.

[160] FC Zhang and TM Rice. Effective hamiltonian for the superconducting cu oxides. *Physical Review B*, 37(7):3759, 1988.

[161] Motoharu Kitatani, Liang Si, Oleg Janson, Ryotaro Arita, Zhicheng Zhong, and Karsten Held. Nickelate superconductors - a renaissance of the one-band hubbard model. *npj Quantum Materials*, 5(1):59, Aug 2020.

[162] B J Powell and Ross H McKenzie. Strong electronic correlations in superconducting organic charge transfer salts. *Journal of Physics: Condensed Matter*, 18(45):R827, oct 2006.

[163] J. E. Hirsch. Two-dimensional hubbard model: Numerical simulation study. *Phys. Rev. B*, 31:4403–4419, Apr 1985.

[164] Steven R White, Douglas J Scalapino, Robert L Sugar, EY Loh, James E Gubernatis, and Richard T Scalettar. Numerical study of the two-dimensional hubbard model. *Physical Review B*, 40(1):506, 1989.

[165] T. Schäfer, F. Geles, D. Rost, G. Rohringer, E. Arrigoni, K. Held, N. Blümer, M. Aichhorn, and A. Toschi. Fate of the false mott-hubbard transition in two dimensions. *Phys. Rev. B*, 91:125109, Mar 2015.

[166] Ettore Vitali, Hao Shi, Mingpu Qin, and Shiwei Zhang. Computation of dynamical correlation functions for many-fermion systems with auxiliary-field quantum monte carlo. *Phys. Rev. B*, 94:085140, Aug 2016.

[167] Fedor Šimkovic, J. P. F. LeBlanc, Aaram J. Kim, Youjin Deng, N. V. Prokof'ev, B. V. Svistunov, and Evgeny Kozik. Extended crossover from a fermi liquid to a quasiantiferromagnet in the half-filled 2d hubbard model. *Phys. Rev. Lett.*, 124:017003, Jan 2020.

[168] K. Borejsza and N. Dupuis. Antiferromagnetism and single-particle properties in the two-dimensional half-filled hubbard model: A nonlinear sigma model approach. *Phys. Rev. B*, 69:085119, Feb 2004.

[169] Aaram J. Kim, Fedor Simkovic, and Evgeny Kozik. Spin and charge correlations across the metal-to-insulator crossover in the half-filled 2d hubbard model. *Phys. Rev. Lett.*, 124:117602, Mar 2020.

[170] Marcel Klett, Nils Wentzell, Thomas Schäfer, Fedor Simkovic, Olivier Parcollet, Sabine Andergassen, and Philipp Hansmann. Real-space cluster dynamical mean-field theory: Center-focused extrapolation on the one- and two particle-levels. *Phys. Rev. Res.*, 2:033476, Sep 2020.

[171] HJ Schulz. Incommensurate antiferromagnetism in the two-dimensional hubbard model. *Physical review letters*, 64(12):1445, 1990.

[172] R. Frésard, M. Dzierzawa, and P. Wölfle. Slave-boson approach to spiral magnetic order in the hubbard model. *Europhysics Letters*, 15(3):325, jun 1991.

[173] Walter Metzner, Manfred Salmhofer, Carsten Honerkamp, Volker Meden, and Kurt Schönhammer. Functional renormalization group approach to correlated fermion systems. *Reviews of Modern Physics*, 84(1):299, 2012.

[174] Hiroyuki Yamase, Andreas Eberlein, and Walter Metzner. Coexistence of incommensurate magnetism and superconductivity in the two-dimensional hubbard model. *Physical review letters*, 116(9):096402, 2016.

[175] S Raghu, SA Kivelson, and DJ Scalapino. Superconductivity in the repulsive hubbard model: An asymptotically exact weak-coupling solution. *Physical Review B*, 81(22):224505, 2010.

[176] Fedor Šimkovic, Youjin Deng, and Evgeny Kozik. Superfluid ground state phase diagram of the two-dimensional hubbard model in the emergent bardeen-cooper-schrieffer regime. *Phys. Rev. B*, 104:L020507, Jul 2021.

[177] Jing Wang, Andreas Eberlein, and Walter Metzner. Competing order in correlated electron systems made simple: Consistent fusion of functional renormalization and mean-field theory. *Phys. Rev. B*, 89:121116, Mar 2014.

[178] M Zegrodnik, A Biborski, M Fidrysiak, and J Spałek. Superconductivity in the three-band model of cuprates: nodal direction characteristics and influence of intersite interactions. *Journal of Physics: Condensed Matter*, 33(41):415601, aug 2021.

[179] C. Weber, C. Yee, K. Haule, and G. Kotliar. Scaling of the transition temperature of hole-doped cuprate superconductors with the charge-transfer energy. *Europhysics Letters*, 100(3):37001, nov 2012.

[180] Steven R. White and D. J. Scalapino. Doping asymmetry and striping in a three-orbital cuo$_2$ hubbard model. *Phys. Rev. B*, 92:205112, Nov 2015.

[181] Cornelia Hille, Daniel Rohe, Carsten Honerkamp, and Sabine Andergassen. Pseudogap opening in the two-dimensional hubbard model: a functional renormalization group analysis. *arXiv preprint arXiv:2003.01447*, 2020.

[182] Wei Wu, Mathias S. Scheurer, Shubhayu Chatterjee, Subir Sachdev, Antoine Georges, and Michel Ferrero. Pseudogap and fermi-surface topology in the two-dimensional hubbard model. *Phys. Rev. X*, 8:021048, May 2018.

[183] Bo-Xiao Zheng, Chia-Min Chung, Philippe Corboz, Georg Ehlers, Ming-Pu Qin, Reinhard M. Noack, Hao Shi, Steven R. White, Shiwei Zhang, and Garnet Kin-Lic Chan. Stripe order in the underdoped region of the two-dimensional hubbard model. *Science*, 358(6367):1155–1160, 2017.

[184] Luca F. Tocchio, Arianna Montorsi, and Federico Becca. Metallic and insulating stripes and their relation with superconductivity in the doped Hubbard model. *SciPost Phys.*, 7:021, 2019.

[185] Yi-Fan Jiang, Jan Zaanen, Thomas P. Devereaux, and Hong-Chen Jiang. Ground state phase diagram of the doped hubbard model on the four-leg cylinder. *Phys. Rev. Res.*, 2:033073, Jul 2020.

[186] Mingpu Qin, Chia-Min Chung, Hao Shi, Ettore Vitali, Claudius Hubig, Ulrich Schollwöck, Steven R. White, and Shiwei Zhang. Absence of superconductivity in the pure two-dimensional hubbard model. *Phys. Rev. X*, 10:031016, Jul 2020.

[187] Th. Maier, M. Jarrell, Th. Pruschke, and J. Keller. A non-crossing approximation for the study of intersite correlations. *The European Physical Journal B - Condensed Matter and Complex Systems*, 13(4):613–624, Feb 2000.

[188] C. Huscroft, M. Jarrell, Th. Maier, S. Moukouri, and A. N. Tahvildarzadeh. Pseudogaps in the 2d hubbard model. *Phys. Rev. Lett.*, 86:139–142, Jan 2001.

[189] M. Civelli, M. Capone, S. S. Kancharla, O. Parcollet, and G. Kotliar. Dynamical breakup of the fermi surface in a doped mott insulator. *Phys. Rev. Lett.*, 95:106402, Sep 2005.

[190] O. Parcollet, G. Biroli, and G. Kotliar. Cluster dynamical mean field analysis of the mott transition. *Phys. Rev. Lett.*, 92:226402, Jun 2004.

[191] O. Gunnarsson, T. Schäfer, J. P. F. LeBlanc, E. Gull, J. Merino, G. Sangiovanni, G. Rohringer, and A. Toschi. Fluctuation diagnostics of the electron self-energy: Origin of the pseudogap physics. *Phys. Rev. Lett.*, 114:236402, Jun 2015.

[192] Ansgar Liebsch and Ning-Hua Tong. Finite-temperature exact diagonalization cluster dynamical mean-field study of the two-dimensional hubbard model: Pseudogap, non-fermi-liquid behavior, and particle-hole asymmetry. *Phys. Rev. B*, 80:165126, Oct 2009.

[193] Emanuel Gull, Olivier Parcollet, Philipp Werner, and Andrew J. Millis. Momentum-sector-selective metal-insulator transition in the eight-site dynamical mean-field approximation to the hubbard model in two dimensions. *Phys. Rev. B*, 80:245102, Dec 2009.

[194] Wei Wu, Mathias S. Scheurer, Michel Ferrero, and Antoine Georges. Effect of van hove singularities in the onset of pseudogap states in mott insulators. *Phys. Rev. Res.*, 2:033067, Jul 2020.

[195] Youjin Deng, Evgeny Kozik, Nikolay V. Prokof'ev, and Boris V. Svistunov. Emergent bcs regime of the two-dimensional fermionic hubbard model: Ground-state phase diagram. *Europhysics Letters*, 110(5):57001, jun 2015.

[196] Xi Chen, J. P. F. LeBlanc, and Emanuel Gull. Superconducting fluctuations in the normal state of the two-dimensional hubbard model. *Phys. Rev. Lett.*, 115:116402, Sep 2015.

[197] E. Gull and A. J. Millis. Pairing glue in the two-dimensional hubbard model. *Phys. Rev. B*, 90:041110, Jul 2014.

[198] Peter Cha, Aavishkar A. Patel, Emanuel Gull, and Eun-Ah Kim. Slope invariant t-linear resistivity from local self-energy. *Phys. Rev. Res.*, 2:033434, Sep 2020.

[199] Edwin W. Huang, Ryan Sheppard, Brian Moritz, and Thomas P. Devereaux. Strange metallicity in the doped hubbard model. *Science*, 366(6468):987–990, 2019.

[200] J. J. Vučičević, J. Kokalj, R. Žitko, N. Wentzell, D. Tanasković, and J. Mravlje. Conductivity in the square lattice hubbard model at high temperatures: Importance of vertex corrections. *Phys. Rev. Lett.*, 123:036601, Jul 2019.

[201] Barbara M. Terhal, Michael M. Wolf, and Andrew C. Doherty. Quantum entanglement: A modern perspective. *Physics Today*, 56(4):46–52, 2003.

[202] Ryszard Horodecki, Paweł Horodecki, Michał Horodecki, and Karol Horodecki. Quantum entanglement. *Reviews of modern physics*, 81(2):865, 2009.

[203] J. Eisert, M. Cramer, and M. B. Plenio. Colloquium: Area laws for the entanglement entropy. *Rev. Mod. Phys.*, 82:277–306, Feb 2010.

[204] Nicolas Laflorencie. Quantum entanglement in condensed matter systems. *Physics Reports*, 646:1–59, 2016.

[205] Bei Zeng, Xie Chen, Duan-Lu Zhou, and Xiao-Gang Wen. *Quantum information meets quantum matter*. Springer, 2019.

[206] Manuel Erhard, Mario Krenn, and Anton Zeilinger. Advances in high-dimensional quantum entanglement. *Nature Reviews Physics*, 2(7):365–381, Jul 2020.

[207] X. G. Wen. Topological orders in rigid states. *International Journal of Modern Physics B*, 04(02):239–271, 1990.

[208] X. G. Wen and Q. Niu. Ground-state degeneracy of the fractional quantum hall states in the presence of a random potential and on high-genus riemann surfaces. *Phys. Rev. B*, 41:9377–9396, May 1990.

[209] X.G. Wen and A. Zee. Quantum statistics and superconductivity in two spatial dimensions. *Nuclear Physics B - Proceedings Supplements*, 15:135–156, 1990.

[210] Xiao-Gang Wen. Topological orders and chern-simons theory in strongly correlated quantum liquid. *International Journal of Modern Physics B*, 05(10):1641–1648, 1991.

[211] Xiao-Gang Wen. Topological orders and edge excitations in fractional quantum hall states. *Advances in Physics*, 44(5):405–473, 1995.

[212] Xiao-Gang Wen. Topological order: From long-range entangled quantum matter to a unified origin of light and electrons. *ISRN Condensed Matter Physics*, 2013:198710, Mar 2013.

[213] Xiao-Gang Wen. Colloquium: Zoo of quantum-topological phases of matter. *Reviews of Modern Physics*, 89(4):041004, 2017.

[214] R. Tao and Yong-Shi Wu. Gauge invariance and fractional quantum hall effect. *Phys. Rev. B*, 30:1097–1098, Jul 1984.

[215] Qian Niu, Ds J Thouless, and Yong-Shi Wu. Quantized hall conductance as a topological invariant. *Physical Review B*, 31(6):3372, 1985.

[216] X. G. Wen. Vacuum degeneracy of chiral spin states in compactified space. *Phys. Rev. B*, 40:7387–7390, Oct 1989.

[217] Hui Li and F Duncan M Haldane. Entanglement spectrum as a generalization of entanglement entropy: Identification of topological order in non-abelian fractional quantum hall effect states. *Physical review letters*, 101(1):010504, 2008.

[218] Xiao-Liang Qi, Hosho Katsura, and Andreas W. W. Ludwig. General relationship between the entanglement spectrum and the edge state spectrum of topological quantum states. *Phys. Rev. Lett.*, 108:196402, May 2012.

[219] Frank Pollmann, Ari M. Turner, Erez Berg, and Masaki Oshikawa. Entanglement spectrum of a topological phase in one dimension. *Phys. Rev. B*, 81:064439, Feb 2010.

[220] Hong Yao and Xiao-Liang Qi. Entanglement entropy and entanglement spectrum of the kitaev model. *Phys. Rev. Lett.*, 105:080501, Aug 2010.

[221] Zhao Liu, Hong-Li Guo, Vlatko Vedral, and Heng Fan. Entanglement spectrum: Identification of the transition from vortex-liquid to vortex-lattice state in a weakly interacting rotating bose-einstein condensate. *Phys. Rev. A*, 83:013620, Jan 2011.

[222] Pasquale Calabrese and Alexandre Lefevre. Entanglement spectrum in one-dimensional systems. *Phys. Rev. A*, 78:032329, Sep 2008.

[223] Andreas M. Läuchli, Emil J. Bergholtz, Juha Suorsa, and Masudul Haque. Disentangling entanglement spectra of fractional quantum hall states on torus geometries. *Phys. Rev. Lett.*, 104:156404, Apr 2010.

[224] John Schliemann. Entanglement spectrum and entanglement thermodynamics of quantum hall bilayers at $\nu = 1$. *Phys. Rev. B*, 83:115322, Mar 2011.

[225] B. I. Halperin. Quantized hall conductance, current-carrying edge states, and the existence of extended states in a two-dimensional disordered potential. *Phys. Rev. B*, 25:2185–2190, Feb 1982.

[226] Xiao-Liang Qi, Hosho Katsura, and Andreas W. W. Ludwig. General relationship between the entanglement spectrum and the edge state spectrum of topological quantum states. *Phys. Rev. Lett.*, 108:196402, May 2012.

[227] Nicolas Laflorencie. Quantum entanglement in condensed matter systems. *Physics Reports*, 646:1–59, 2016. Quantum entanglement in condensed matter systems.

[228] Saurya Das and S. Shankaranarayanan. How robust is the entanglement entropy-area relation? *Phys. Rev. D*, 73:121701, Jun 2006.

[229] M. Cramer, J. Eisert, M. B. Plenio, and J. Dreißig. Entanglement-area law for general bosonic harmonic lattice systems. *Phys. Rev. A*, 73:012309, Jan 2006.

[230] B. Blok and X. G. Wen. Effective theories of the fractional quantum hall effect: Hierarchy construction. *Phys. Rev. B*, 42:8145–8156, Nov 1990.

[231] N. Read. Excitation structure of the hierarchy scheme in the fractional quantum hall effect. *Phys. Rev. Lett.*, 65:1502–1505, Sep 1990.

[232] Daniel S. Rokhsar and Steven A. Kivelson. Superconductivity and the quantum hard-core dimer gas. *Phys. Rev. Lett.*, 61:2376–2379, Nov 1988.

[233] N. Read and B. Chakraborty. Statistics of the excitations of the resonating-valence-bond state. *Phys. Rev. B*, 40:7133–7140, Oct 1989.

[234] R. Moessner and S. L. Sondhi. Resonating valence bond phase in the triangular lattice quantum dimer model. *Phys. Rev. Lett.*, 86:1881–1884, Feb 2001.

[235] Eddy Ardonne, Paul Fendley, and Eduardo Fradkin. Topological order and conformal quantum critical points. *Annals of Physics*, 310(2):493–551, 2004.

[236] Michael A. Levin and Xiao-Gang Wen. String-net condensation: A physical mechanism for topological phases. *Phys. Rev. B*, 71:045110, Jan 2005.

[237] Alexei Kitaev and John Preskill. Topological entanglement entropy. *Physical review letters*, 96(11):110404, 2006.

[238] Hong-Chen Jiang, Zhenghan Wang, and Leon Balents. Identifying topological order by entanglement entropy. *Nature Physics*, 8(12):902–905, Dec 2012.

[239] Alioscia Hamma, Radu Ionicioiu, and Paolo Zanardi. Bipartite entanglement and entropic boundary law in lattice spin systems. *Phys. Rev. A*, 71:022315, Feb 2005.

[240] Michael Levin and Xiao-Gang Wen. Detecting topological order in a ground state wave function. *Phys. Rev. Lett.*, 96:110405, Mar 2006.

[241] Alexei Kitaev and John Preskill. Topological entanglement entropy. *Phys. Rev. Lett.*, 96:110404, Mar 2006.

[242] Yi Zhang, Tarun Grover, Ari Turner, Masaki Oshikawa, and Ashvin Vishwanath. Quasiparticle statistics and braiding from ground-state entanglement. *Phys. Rev. B*, 85:235151, Jun 2012.

[243] Tarun Grover, Ari M. Turner, and Ashvin Vishwanath. Entanglement entropy of gapped phases and topological order in three dimensions. *Phys. Rev. B*, 84:195120, Nov 2011.

[244] Alioscia Hamma and Daniel A. Lidar. Adiabatic preparation of topological order. *Phys. Rev. Lett.*, 100:030502, Jan 2008.

[245] Tian Lan and Xiao-Gang Wen. Topological quasiparticles and the holographic bulk-edge relation in $(2 + 1)$-dimensional string-net models. *Phys. Rev. B*, 90:115119, Sep 2014.

[246] Paul Fendley, Sergei V. Isakov, and Matthias Troyer. Fibonacci topological order from quantum nets. *Phys. Rev. Lett.*, 110:260408, Jun 2013.

[247] Zheng-Cheng Gu, Michael Levin, Brian Swingle, and Xiao-Gang Wen. Tensor-product representations for string-net condensed states. *Phys. Rev. B*, 79:085118, Feb 2009.

[248] Kevin Slagle, David Aasen, and Dominic Williamson. Foliated Field Theory and String-Membrane-Net Condensation Picture of Fracton Order. *SciPost Phys.*, 6:43, 2019.

[249] Xie Chen, Zheng-Cheng Gu, and Xiao-Gang Wen. Local unitary transformation, long-range quantum entanglement, wave function renormalization, and topological order. *Physical review b*, 82(15):155138, 2010.

[250] Zheng-Cheng Gu and Xiao-Gang Wen. Tensor-entanglement-filtering renormalization approach and symmetry-protected topological order. *Physical Review B*, 80(15):155131, 2009.

[251] Oskar Vafek and Ashvin Vishwanath. Dirac fermions in solids: From high-tc cuprates and graphene to topological insulators and weyl semimetals. *Annual Review of Condensed Matter Physics*, 5(1):83–112, 2014.

[252] B. Andrei Bernevig and Taylor L. Hughes. *Topological Insulators and Topological Superconductors*. Princeton University Press, 2013.

[253] Jérôme Cayssol. Introduction to dirac materials and topological insulators. *Comptes Rendus Physique*, 14(9):760–778, 2013. Topological insulators / Isolants topologiques.

[254] Roderich Moessner and Joel E. Moore. *Topological Phases of Matter*. Cambridge University Press, 2021.

[255] Stephan Rachel. Interacting topological insulators: a review. *Reports on Progress in Physics*, 81(11):116501, oct 2018.

[256] Jinying Wang, Shibin Deng, Zhongfan Liu, and Zhirong Liu. ”The rare two-dimensional materials with Dirac cones”. *National Science Review*, 2(1):22–39, 01 2015.

[257] Menghao Wu, Zhijun Wang, Junwei Liu, Wenbin Li, Huahua Fu, Lei Sun, Xin Liu, Minghu Pan, Hongming Weng, Mircea Dincă, Liang Fu, and Ju Li. Conetronics in 2d metal-organic frameworks: double/half dirac cones and quantum anomalous hall effect. *2D Materials*, 4(1):015015, nov 2016.

[258] Xiaojuan Ni, Huaqing Huang, and Jean-Luc Brédas. Emergence of a two-dimensional topological dirac semimetal phase in a phthalocyanine-based covalent organic framework. *Chemistry of Materials*, 34(7):3178–3184, 2022.

[259] Shi-Liang Zhu, Baigeng Wang, and L.-M. Duan. Simulation and detection of dirac fermions with cold atoms in an optical lattice. *Phys. Rev. Lett.*, 98:260402, Jun 2007.

[260] L. Lepori, G. Mussardo, and A. Trombettoni. (3+1) massive dirac fermions with ultracold atoms in frustrated cubic optical lattices. *EPL (Europhysics Letters)*, 92(5):50003, dec 2010.

[261] O Boada, A Celi, J I Latorre, and M Lewenstein. Dirac equation for cold atoms in artificial curved spacetimes. *New Journal of Physics*, 13(3):035002, mar 2011.

[262] Yoshihito Kuno, Ikuo Ichinose, and Yoshiro Takahashi. Generalized lattice wilson–dirac fermions in $(1 + 1)$ dimensions for atomic quantum simulation and topological phases. *Scientific Reports*, 8(1):10699, Jul 2018.

[263] Kenjiro K. Gomes, Warren Mar, Wonhee Ko, Francisco Guinea, and Hari C. Manoharan. Designer dirac fermions and topological phases in molecular graphene. *Nature*, 483(7389):306–310, Mar 2012.

[264] Xiang Yuan, Cheng Zhang, Yanwen Liu, Awadhesh Narayan, Chaoyu Song, Shoudong Shen, Xing Sui, Jie Xu, Haochi Yu, Zhenghua An, Jun Zhao, Stefano Sanvito, Hugen Yan, and Faxian Xiu. Observation of quasi-two-dimensional dirac fermions in zrte5. *NPG Asia Materials*, 8(11):e325–e325, Nov 2016.

[265] Matthieu Bellec, Ulrich Kuhl, Gilles Montambaux, and Fabrice Mortessagne. Topological transition of dirac points in a microwave experiment. *Phys. Rev. Lett.*, 110:033902, Jan 2013.

[266] Guifre Vidal, José Ignacio Latorre, Enrique Rico, and Alexei Kitaev. Entanglement in quantum critical phenomena. *Physical review letters*, 90(22):227902, 2003.

[267] J. I. Latorre, E. Rico, and G. Vidal. Ground state entanglement in quantum spin chains, 2003.

[268] J. I. Latorre, C. A. Lütken, E. Rico, and G. Vidal. Fine-grained entanglement loss along renormalization-group flows. *Phys. Rev. A*, 71:034301, Mar 2005.

[269] B.-Q. Jin and V. E. Korepin. Quantum spin chain, toeplitz determinants and the fisher—hartwig conjecture. *Journal of Statistical Physics*, 116(1):79–95, Aug 2004.

[270] Neill Lambert, Clive Emary, and Tobias Brandes. Entanglement and the phase transition in single-mode superradiance. *Phys. Rev. Lett.*, 92:073602, Feb 2004.

[271] Christoph Holzhey, Finn Larsen, and Frank Wilczek. Geometric and renormalized entropy in conformal field theory. *Nuclear Physics B*, 424(3):443–467, 1994.

[272] H. Casini and M. Huerta. A finite entanglement entropy and the c-theorem. *Physics Letters B*, 600(1):142–150, 2004.

[273] Pasquale Calabrese and John Cardy. Entanglement entropy and quantum field theory. *Journal of Statistical Mechanics: Theory and Experiment*, 2004(06):P06002, 2004.

[274] Tobias J Osborne and Michael A Nielsen. Entanglement in a simple quantum phase transition. *Physical Review A*, 66(3):032110, 2002.

[275] Andreas Osterloh, Luigi Amico, Giuseppe Falci, and Rosario Fazio. Scaling of entanglement close to a quantum phase transition. *Nature*, 416(6881):608–610, 2002.

[276] H Casini, C D Fosco, and M Huerta. Entanglement and alpha entropies for a massive dirac field in two dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2005(07):P07007–P07007, jul 2005.

[277] V. E. Korepin. Universality of entropy scaling in one dimensional gapless models. *Phys. Rev. Lett.*, 92:096402, Mar 2004.

[278] J. L. Cardy, O. A. Castro-Alvaredo, and B. Doyon. Form factors of branch-point twist fields in quantum integrable models and entanglement entropy. *Journal of Statistical Physics*, 130(1):129–168, Jan 2008.

[279] H Casini and M Huerta. Reduced density matrix and internal dynamics for multicomponent regions. *Classical and Quantum Gravity*, 26(18):185005, sep 2009.

[280] Benjamin Doyon. Bipartite entanglement entropy in massive two-dimensional quantum field theory. *Phys. Rev. Lett.*, 102:031602, Jan 2009.

[281] H Casini and M Huerta. Entanglement entropy in free quantum field theory. *Journal of Physics A: Mathematical and Theoretical*, 42(50):504007, dec 2009.

[282] Dimitri Gioev and Israel Klich. Entanglement entropy of fermions in any dimension and the widom conjecture. *Physical review letters*, 96(10):100503, 2006.

[283] Michael M. Wolf. Violation of the entropic area law for fermions. *Phys. Rev. Lett.*, 96:010404, Jan 2006.

[284] Weifei Li, Letian Ding, Rong Yu, Tommaso Roscilde, and Stephan Haas. Scaling behavior of entanglement in two- and three-dimensional free-fermion systems. *Phys. Rev. B*, 74:073103, Aug 2006.

[285] T. Barthel, M.-C. Chung, and U. Schollwöck. Entanglement scaling in critical two-dimensional fermionic and bosonic systems. *Phys. Rev. A*, 74:022329, Aug 2006.

[286] S. Farkas and Z. Zimborás. The von neumann entropy asymptotics in multidimensional fermionic systems. *Journal of Mathematical Physics*, 48(10):102110, 2007.

[287] Michael M Wolf, Frank Verstraete, Matthew B Hastings, and J Ignacio Cirac. Area laws in quantum systems: mutual information and correlations. *Physical review letters*, 100(7):070502, 2008.

[288] Olalla A Castro-Alvaredo and Benjamin Doyon. Bi-partite entanglement entropy in massive (1+1)-dimensional quantum field theories. *Journal of Physics A: Mathematical and Theoretical*, 42(50):504006, dec 2009.

[289] Brian Swingle. Entanglement entropy and the fermi surface. *Physical review letters*, 105(5):050502, 2010.

[290] Robert Helling, Hajo Leschke, and Wolfgang Spitzer. "A Special Case of a Conjecture by Widom with Implications to Fermionic Entanglement Entropy". *International Mathematics Research Notices*, 2011(7):1451–1482, 06 2010.

[291] P. Calabrese, M. Mintchev, and E. Vicari. Entanglement entropies in free-fermion gases for arbitrary dimension. *EPL (Europhysics Letters)*, 97(2):20009, jan 2012.

[292] Brian Swingle. Rényi entropy, mutual information, and fluctuation properties of fermi liquids. *Phys. Rev. B*, 86:045109, Jul 2012.

[293] Pasquale Calabrese, Mihail Mintchev, and Ettore Vicari. Exact relations between particle fluctuations and entanglement in fermi gases. *EPL (Europhysics Letters)*, 98(2):20003, apr 2012.

[294] Mao Tian Tan and Shinsei Ryu. Particle number fluctuations, renyi entropy, and symmetry-resolved entanglement entropy in a two-dimensional fermi gas from multidimensional bosonization. *Phys. Rev. B*, 101:235169, Jun 2020.

[295] Sara Murciano, Paola Ruggiero, and Pasquale Calabrese. Symmetry resolved entanglement in two-dimensional systems via dimensional reduction. *Journal of Statistical Mechanics: Theory and Experiment*, 2020(8):083102, aug 2020.

[296] Xiao Chen, William Witczak-Krempa, Thomas Faulkner, and Eduardo Fradkin. Two-cylinder entanglement entropy under a twist. *Journal of Statistical Mechanics: Theory and Experiment*, 2017(4):043104, apr 2017.

[297] Pablo Bueno and William Witczak-Krempa. Holographic torus entanglement and its renormalization group flow. *Phys. Rev. D*, 95:066007, Mar 2017.

[298] Max A. Metlitski, Carlos A. Fuertes, and Subir Sachdev. Entanglement entropy in the $o(n)$ model. *Phys. Rev. B*, 80:115122, Sep 2009.

[299] Raúl E Arias, David D Blanco, and Horacio Casini. Entanglement entropy as a witness of the aharonov–bohm effect in QFT. *Journal of Physics A: Mathematical and Theoretical*, 48(14):145401, mar 2015.

[300] Seth Whitsitt, William Witczak-Krempa, and Subir Sachdev. Entanglement entropy of large-n wilson-fisher conformal field theory. *Phys. Rev. B*, 95:045148, Jan 2017.

[301] Wei Zhu, Xiao Chen, Yin-Chen He, and William Witczak-Krempa. Entanglement signatures of emergent dirac fermions: Kagome spin liquid and quantum criticality. *Science Advances*, 4(11):eaat5535, 2018.

[302] Shijie Hu, W. Zhu, Sebastian Eggert, and Yin-Chen He. Dirac spin liquid on the spin-$1/2$ triangular heisenberg antiferromagnet. *Phys. Rev. Lett.*, 123:207203, Nov 2019.

[303] Valentin Crépel, Anna Hackenbroich, Nicolas Regnault, and Benoit Estienne. Universal signatures of dirac fermions in entanglement and charge fluctuations. *Phys. Rev. B*, 103:235108, Jun 2021.

[304] Joshuah T Heath and Kevin S Bedell. Necessary and sufficient conditions for the validity of luttinger's theorem. *New Journal of Physics*, 22(6):063011, jun 2020.

[305] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-i: a tensor network approach. *Nuclear Physics B*, 960:115170, 2020.

[306] Anirban Mukherjee and Siddhartha Lal. Holographic unitary renormalisation group for correlated electrons-ii: insights on fermionic criticality. *Nuclear Physics B*, 960:115163, 2020.

[307] G. 't Hooft. Dimensional reduction in quantum gravity. *arXiv:gr-qc/9310026*, 1993.

[308] Erik Verlinde. On the origin of gravity and the laws of newton. *Journal of High Energy Physics*, 2011(4):29, Apr 2011.

[309] Hsu-Wen Chiang, Yao-Chieh Hu, and Pisin Chen. Quantization of spacetime based on a spacetime interval operator. *Phys. Rev. D*, 93:084043, Apr 2016.

[310] Mark Van Raamsdonk. Building up spacetime with quantum entanglement. *General Relativity and Gravitation*, 42(10):2323–2329, 2010.

[311] Mark Van Raamsdonk. *Lectures on Gravity and Entanglement*, chapter Chapter 5, pages 297–351. World Scientific, 2017.

[312] Xiao-Liang Qi. Exact holographic mapping and emergent space-time geometry. *arXiv preprint arXiv:1309.6282*, 2013.

[313] Ching Hua Lee and Xiao-Liang Qi. Exact holographic mapping in free fermion systems. *Physical Review B*, 93(3):035112, 2016.

[314] Raphael Bousso. The holographic principle. *Rev. Mod. Phys.*, 74:825–874, Aug 2002.

[315] Joseph Katz and Jian Sheng. Applications of holography in fluid mechanics and particle dynamics. *Annual Review of Fluid Mechanics*, 42(1):531–555, 2010.

[316] A.G. Green. An introduction to gauge-gravity duality and its application in condensed matter. *Contemporary Physics*, 54(1):33–48, 2013.

[317] Sean A Hartnoll. Lectures on holographic methods for condensed matter physics. *Classical and Quantum Gravity*, 26(22):224002, oct 2009.

[318] Subir Sachdev. What can gauge-gravity duality teach us about condensed matter physics? *Annual Review of Condensed Matter Physics*, 3(1):9–33, 2012.

[319] Jan Zaanen, Yan Liu, Ya-Wen Sun, and Koenraad Schalm. *Holographic Duality in Condensed Matter Physics*. Cambridge University Press, 2015.

[320] Anosh Joseph. Review of lattice supersymmetry and gauge-gravity duality. *International Journal of Modern Physics A*, 30(27):1530054, 2015.

[321] Horaţiu Năstase. *Introduction to the AdS/CFT Correspondence*. Cambridge University Press, 2015.

[322] Johanna Erdmenger. Introduction to gauge/gravity duality (tasi lectures 2017), 2018.

[323] G.'t Hooft. A planar diagram theory for strong interactions. *Nuclear Physics B*, 72(3):461–473, 1974.

[324] A.M. Polyakov. Thermal properties of gauge fields and quark liberation. *Physics Letters B*, 72(4):477–480, 1978.

[325] A.M. Polyakov. *Gauge Fields and Strings (1st ed.)*. Routledge, 1987.

[326] David J. Gross and Washington Taylor. Two-dimensional qcd is a string theory. *Nuclear Physics B*, 400(1):181–208, 1993.

[327] E.T. Akhmedov. A remark on the ads/cft correspondence and the renormalization group flow. *Physics Letters B*, 442(1):152–158, 1998.

[328] Enrique Álvarez and César Gómez. Geometric holography, the renormalization group and the c-theorem. *Nuclear Physics B*, 541(1):441–460, 1999.

[329] D. Z. Freedman, S. S. Gubser, K. Pilch, and N. P. Warner. Renormalization group flows from holography–supersymmetry and a c-theorem. *Advances in Theoretical and Mathematical Physics*, 3(2):363–417, 1999.

[330] M. Porrati and A. Starinets. Rg fixed points in supergravity duals of 4-d field theory and asymptotically ads spaces. *Physics Letters B*, 454(1):77–83, 1999.

[331] Shamit Kachru, Michael Schulz, and Eva Silverstein. Self-tuning flat domain walls in 5d gravity and string theory. *Phys. Rev. D*, 62:045021, Jul 2000.

[332] L. Girardello, M. Petrini, M. Porrati, and A. Zaffaroni. The supergravity dual of n=1 super yang–mills theory. *Nuclear Physics B*, 569(1):451–469, 2000.

[333] Jan De Boer, Erik Verlinde, and Herman Verlinde. On the holographic renormalization group. *Journal of High Energy Physics*, 2000(08):003, 2000.

[334] Jan de Boer. The holographic renormalization group. *Fortschritte der Physik*, 49(4-6):339–358, 2001.

[335] Clifford V Johnson, Kenneth J Lovis, and David C Page. Probing some N = 1 AdS/CFT RG flows. *Journal of High Energy Physics*, 2001(05):036–036, may 2001.

[336] Satoshi Yamaguchi. AdS branes corresponding to superconformal defects. *Journal of High Energy Physics*, 2003(06):002–002, jun 2003.

[337] Juan Maldacena. The large-n limit of superconformal field theories and supergravity. *International journal of theoretical physics*, 38(4):1113–1133, 1999.

[338] S.S. Gubser, I.R. Klebanov, and A.M. Polyakov. Gauge theory correlators from non-critical string theory. *Physics Letters B*, 428(1):105–114, 1998.

[339] Edward Witten. Anti de sitter space and holography. *arXiv preprint hep-th/9802150*, 2:253–291, 1998.

[340] Jacob D. Bekenstein. Black holes and entropy. *Phys. Rev. D*, 7:2333–2346, Apr 1973.

[341] S. W. Hawking. Black hole explosions? *Nature*, 248(5443):30–31, Mar 1974.

[342] S. W. Hawking. Particle creation by black holes. *Communications in Mathematical Physics*, 43(3):199–220, Aug 1975.

[343] Shinsei Ryu and Tadashi Takayanagi. Holographic derivation of entanglement entropy from the anti–de sitter space/conformal field theory correspondence. *Physical review letters*, 96(18):181602, 2006.

[344] Shinsei Ryu and Tadashi Takayanagi. Aspects of holographic entanglement entropy. *Journal of High Energy Physics*, 2006(08):045, 2006.

[345] P. K. Kovtun, D. T. Son, and A. O. Starinets. Viscosity in strongly interacting quantum field theories from black hole physics. *Phys. Rev. Lett.*, 94:111601, Mar 2005.

[346] Kedar Damle and Subir Sachdev. Nonzero-temperature transport near quantum critical points. *Phys. Rev. B*, 56:8714–8733, Oct 1997.

[347] Sean A. Hartnoll, Pavel K. Kovtun, Markus Müller, and Subir Sachdev. Theory of the nernst effect near quantum phase transitions in condensed matter and in dyonic black holes. *Phys. Rev. B*, 76:144502, Oct 2007.

[348] Sung-Sik Lee. Non-fermi liquid from a charged black hole: A critical fermi ball. *Phys. Rev. D*, 79:086006, Apr 2009.

[349] Hong Liu, John McGreevy, and David Vegh. Non-fermi liquids from holography. *Phys. Rev. D*, 83:065029, Mar 2011.

[350] Thomas Faulkner, Hong Liu, John McGreevy, and David Vegh. Emergent quantum criticality, fermi surfaces, and ads$_2$. *Phys. Rev. D*, 83:125002, Jun 2011.

[351] Mihailo Čubrović, Jan Zaanen, and Koenraad Schalm. String theory, quantum phase transitions, and the emergent fermi liquid. *Science*, 325(5939):439–444, 2009.

[352] Steven S. Gubser, Silviu S. Pufu, and Fabio D. Rocha. Quantum critical superconductors in string theory and m-theory. *Physics Letters B*, 683(2):201–204, 2010.

[353] Jerome P. Gauntlett, Julian Sonner, and Toby Wiseman. Holographic superconductivity in m theory. *Phys. Rev. Lett.*, 103:151601, Oct 2009.

[354] Sean A. Hartnoll, Christopher P. Herzog, and Gary T. Horowitz. Building a holographic superconductor. *Phys. Rev. Lett.*, 101:031601, Jul 2008.

[355] Sean A Hartnoll, Christopher P Herzog, and Gary T Horowitz. Holographic superconductors. *Journal of High Energy Physics*, 2008(12):015–015, dec 2008.

[356] RongGen Cai, Li Li, LiFang Li, and RunQiu Yang. Introduction to holographic superconductor models. *Science China Physics, Mechanics & Astronomy*, 58(6):1–46, Jun 2015.

[357] Chunyan Wang, Dan Zhang, Guoyang Fu, and Jian-Pin Wu. Analytical study of the holographic superconductor from higher derivative theory. *Advances in High Energy Physics*, 2020:5902473, Jan 2020.

[358] Aristomenis Donos and Jerome P. Gauntlett. Holographic striped phases. *Journal of High Energy Physics*, 2011(8):140, Aug 2011.

[359] Aristomenis Donos. Striped phases from holography. *Journal of High Energy Physics*, 2013(5):59, May 2013.

[360] Sera Cremonini, Li Li, and Jie Ren. Holographic fermions in striped phases. *Journal of High Energy Physics*, 2018(12):80, Dec 2018.

[361] Shamit Kachru, Xiao Liu, and Michael Mulligan. Gravity duals of lifshitz-like fixed points. *Phys. Rev. D*, 78:106005, Nov 2008.

[362] Mihailo Čubrović, Yan Liu, Koenraad Schalm, Ya-Wen Sun, and Jan Zaanen. Spectral probes of the holographic fermi ground state: Dialing between the electron star and ads dirac hair. *Phys. Rev. D*, 84:086002, Oct 2011.

[363] T. Senthil, Subir Sachdev, and Matthias Vojta. Fractionalized fermi liquids. *Phys. Rev. Lett.*, 90:216403, May 2003.

[364] Subir Sachdev. Model of a fermi liquid using gauge-gravity duality. *Phys. Rev. D*, 84:066009, Sep 2011.

[365] Sean A. "Hartnoll, Diego M. Hofman, and David" Vegh. "Stellar spectroscopy: Fermions and holographic Lifshitz criticality". *"JHEP"*, "08":"096", "2011".

[366] Liza Huijse and Subir Sachdev. Fermi surfaces and gauge-gravity duality. *Phys. Rev. D*, 84:026001, Jul 2011.

[367] Francesco Aprile, Diederik Roest, and Jorge G. Russo. Holographic superconductors from gauged supergravity. *Journal of High Energy Physics*, 2011(6):40, Jun 2011.

[368] Aristomenis Donos and Jerome P. Gauntlett. Superfluid black branes in ads4 × s7. *Journal of High Energy Physics*, 2011(6):53, Jun 2011.

[369] Hiroaki Matsueda. Multiscale entanglement renormalization ansatz for kondo problem. *arXiv:1208.2872*, 2012.

[370] Hiroaki Matsueda, Masafumi Ishihara, and Yoichiro Hashizume. Tensor network and a black hole. *Phys. Rev. D*, 87:066002, Mar 2013.

[371] Hiroaki Matsueda. Emergent general relativity from fisher information metric. *arXiv:1310.1831*, 2013.

[372] Hiroaki Matsueda. Geometry and dynamics of emergent spacetime from entanglement spectrum, 2014.

[373] Bartłomiej Czech. Einstein equations from varying complexity. *Phys. Rev. Lett.*, 120:031601, Jan 2018.

[374] Thomas Faulkner, Felix M. Haehl, Eliot Hijano, Onkar Parrikar, Charles Rabideau, and Mark Van Raamsdonk. Nonlinear gravity from entanglement in conformal field theories. *Journal of High Energy Physics*, 2017(8):57, Aug 2017.

[375] Thomas Faulkner, Monica Guica, Thomas Hartman, Robert C. Myers, and Mark Van Raamsdonk. Gravitation from entanglement in holographic cfts. *Journal of High Energy Physics*, 2014(3):51, Mar 2014.

[376] Yingfei Gu, Ching Hua Lee, Xueda Wen, Gil Young Cho, Shinsei Ryu, and Xiao-Liang Qi. Holographic duality between $(2+1)$-dimensional quantum anomalous hall state and $(3+1)$-dimensional topological insulators. *Phys. Rev. B*, 94:125107, Sep 2016.

[377] Xueda Wen, Gil Young Cho, Pedro L. S. Lopes, Yingfei Gu, Xiao-Liang Qi, and Shinsei Ryu. Holographic entanglement renormalization of topological insulators. *Phys. Rev. B*, 94:075124, Aug 2016.

[378] Robert G. Leigh, Onkar Parrikar, and Alexander B. Weiss. Exact renormalization group and higher-spin holography. *Phys. Rev. D*, 91:026002, Jan 2015.

[379] Robert G. Leigh, Onkar Parrikar, and Alexander B. Weiss. Holographic geometry of the renormalization group and higher spin symmetries. *Phys. Rev. D*, 89:106012, May 2014.

[380] Michael R. Douglas, Luca Mazzucato, and Shlomo S. Razamat. Holographic dual of free field theory. *Phys. Rev. D*, 83:071701, Apr 2011.

[381] Robert de Mello Koch, Antal Jevicki, Kewang Jin, and João P. Rodrigues. $\mathrm{ads}_4/\mathrm{cft}_3$ construction from collective fields. *Phys. Rev. D*, 83:025006, Jan 2011.

[382] Sung-Sik Lee. Holographic description of quantum field theory. *Nuclear Physics B*, 832(3):567–585, 2010.

[383] Sung-Sik Lee. Tasi lectures on emergence of supersymmetry, gauge theory and string in condensed matter systems. *arXiv:1009.5127*, 2010.

[384] Sung-Sik Lee. Background independent holographic description: from matrix field theory to quantum gravity. *Journal of High Energy Physics*, 2012(10):160, Oct 2012.

[385] Sung-Sik Lee. Quantum renormalization group and holography. *Journal of High Energy Physics*, 2014(1):76, 2014.

[386] Guifre Vidal. Entanglement renormalization. *Physical review letters*, 99(22):220405, 2007.

[387] Guifré Vidal. Class of quantum many-body states that can be efficiently simulated. *Physical review letters*, 101(11):110501, 2008.

[388] Brian Swingle. Entanglement renormalization and holography. *Physical Review D*, 86(6):065007, 2012.

[389] G. Evenbly and G. Vidal. Scaling of entanglement entropy in the (branching) multiscale entanglement renormalization ansatz. *Phys. Rev. B*, 89:235113, Jun 2014.

[390] Ashley Milsted and Guifre Vidal. Geometric interpretation of the multi-scale entanglement renormalization ansatz, 2018.

[391] Glen Evenbly and Guifré Vidal. Tensor network states and geometry. *Journal of Statistical Physics*, 145(4):891–918, 2011.

[392] Justin Reyes and Miles Stoudenmire. A multi-scale tensor network architecture for classification and regression. *arXiv:2001.08286*, 2020.

[393] Nathan A. McMahon, Sukhbinder Singh, and Gavin K. Brennen. A holographic duality from lifted tensor networks. *npj Quantum Information*, 6(1):36, Apr 2020.

[394] Anirban Mukherjee and Siddhartha Lal. Superconductivity from repulsion in the doped 2d electronic hubbard model: an entanglement perspective. *Journal of Physics: Condensed Matter*, 34(27):275601, apr 2022.

[395] Ki-Seok Kim, Suk Bum Chung, Chanyong Park, and Jae-Ho Han. *Phys. Rev. D*, 99:105012, 2019.

[396] Ki-Seok Kim, Miok Park, Jaeyoon Cho, and Chanyong Park. Emergent geometric description for a topological phase transition in the kitaev superconductor model. *Phys. Rev. D*, 96:086015, Oct 2017.

[397] Ki-Seok Kim and Chanyong Park. Emergent geometry from field theory: Wilson's renormalization group revisited. *Phys. Rev. D*, 93:121702, Jun 2016.

[398] Ki-Seok Kim. Emergent dual holographic description for interacting dirac fermions in the large n limit. *Phys. Rev. D*, 102:086014, Oct 2020.

[399] Ki-Seok Kim. Geometric encoding of renormalization group β-functions in an emergent holographic dual description. *Phys. Rev. D*, 102:026022, Jul 2020.

[400] Ki-Seok Kim. Emergent geometry in recursive renormalization group transformations. *Nuclear Physics B*, 959:115144, 2020.

[401] Ki-Seok Kim. Emergent dual holographic description for interacting dirac fermions in the large n limit. *Phys. Rev. D*, 102:086014, Oct 2020.

[402] Ki-Seok Kim and Shinsei Ryu. Entanglement transfer from quantum matter to classical geometry in an emergent holographic dual description of a scalar field theory. *Journal of High Energy Physics*, 2021(5):260, May 2021.

[403] Kenneth G. Wilson. The renormalization group: Critical phenomena and the kondo problem. *Rev. Mod. Phys.*, 47:773–840, Oct 1975.

[404] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions: I. t = 0 phase diagram of the 1/2-filled hubbard model. *New Journal of Physics*, 22(6):063007, jun 2020.

[405] Anirban Mukherjee and Siddhartha Lal. Scaling theory for mott–hubbard transitions-II: quantum criticality of the doped mott insulator. *New Journal of Physics*, 22(6):063008, jun 2020.

[406] Anirban Mukherjee and Siddhartha Lal. Superconductivity from repulsion in the doped 2d electronic hubbard model: an entanglement perspective. *Journal of Physics: Condensed Matter*, 34(27):275601, apr 2022.

[407] Kenji Suzuki and Ryoji Okamoto. General structure of effective interaction in degenerate perturbation theory. *Progress of Theoretical Physics*, 71(6):1221–1238, 1984.

[408] PW Anderson. A poor man's derivation of scaling laws for the kondo problem. *Journal of Physics C: Solid State Physics*, 3(12):2436, 1970.

[409] FDM Haldane. Scaling theory of the asymmetric anderson model. *Physical Review Letters*, 40(6):416, 1978.

[410] Siddhartha Patra, Abhirup Mukherjee, Anirban Mukherjee, N S Vidhyadhiraja, A Taraphder, and Siddhartha Lal. Frustration shapes multi-channel kondo physics: a star graph perspective. *Journal of Physics: Condensed Matter*, 35(31):315601, may 2023.

[411] Stefan Kehrein. *The flow equation approach to many-particle systems*, volume 217. Springer, 2007.

[412] Franz Wegner. Flow-equations for hamiltonians. *Annalen der physik*, 506(2):77–91, 1994.

[413] Samuel L Braunstein and Carlton M Caves. Statistical distance and the geometry of quantum states. *Physical Review Letters*, 72(22):3439, 1994.

[414] Philipp Hauke, Markus Heyl, Luca Tagliacozzo, and Peter Zoller. Measuring multipartite entanglement through dynamic susceptibilities. *Nature Physics*, 12(8):778–782, Aug 2016.